

# Calibration-Aware Reinforcement Learning for Large Language Models: A Survey of Objectives, Optimization, and Decision-Making

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## Abstract

Large language models increasingly emit confidence reports, predictive distributions, and typed decisions that determine whether a system answers, abstains, retrieves evidence, or spends more computation. We survey calibration-aware reinforcement learning (RL), in which a reported probability is scored by the reward, consumed by the policy’s actions, or both. Such a probability means something only relative to an event, the reporter’s information, and a population or policy. Unlike post-hoc calibration, RL can change the answers being assessed, the incentive actually optimized, and the behavior that consumes the number. We organize the literature around three gaps. Between reporting and capability, proper-score geometry shows that a joint answer–confidence reward decomposes into terms for mean accuracy, the distribution of success across inputs, and reporting error; so it can prefer a less accurate policy even under truthful reporting. Between the stated objective and the implemented update, group standardization, the treatment of parameter-dependent rewards, and finite-ensemble scoring can change what training optimizes. Between ranking and decision value, a score can keep its ordering while losing the numerical meaning that a cost-derived threshold requires. Exact constructions make each gap concrete, and a method landscape traces representative protocols to primary sources. We then give a comparator guide; an evaluation contract that separates reporting gains from changes in the answer policy, effects of the implemented update, and operating-point selection; and open questions. The practical conclusion is a comparator rule: post-hoc fitting is the comparator to beat when predictions can remain fixed, and direct supervision when the desired report has a tractable differentiable loss; RL is a distinct intervention only when reasoning, retrieval, answering, abstention, or information acquisition must adapt through outcome feedback.

## 1. Introduction and scope

A language model’s confidence becomes consequential when another component acts on it. A reported correctness probability of 0.9 may trigger automatic acceptance, whereas one of 0.5 may trigger retrieval; a numerical forecast may determine resource allocation. Typed-decision services such as TypeSafe’s Jev expose decisions and probability fields directly to downstream software [1]. These probability fields must carry the numerical meaning that the consuming decision assumes; fluent justification or valid formatting is not enough. Reinforcement learning (RL) complicates this requirement because it can change both the report and the behavior that the report describes.

**From feedback sources to calibrated decisions.** Reinforcement learning from human feedback (RLHF) typically optimizes a policy against a reward model fitted to human comparisons, so that preferences that resist exact rules can guide training [2]. Reinforcement learning with verifiable rewards (RLVR) instead takes feedback from an explicit check, such as answer verification or executable tests [3]. Both labels primarily identify the *source of feedback*; neither says what a reported confidence means. TypeSafe’s reinforcement learning *for* calibrated decisions (RLCD) names the target instead: typed decisions with calibrated probabilities [1], [4]. Its public materials do not disclose a complete training recipe, so we treat RLCD as an industrial interface case rather than as an established algorithm. We use *calibration-aware RL* for the broader class of training procedures in which a probability claim is part of what is optimized: the reward scores a reported probability, the policy acts on one, or both. The claim may be a textual confidence, a class-probability vector, a predictive distribution, or a probability field attached to a typed decision.

Labels that name a feedback source and labels that name a target are complementary. Future-as-Label, for example, combines verifiable event outcomes with proper scoring rules for probability reports [5], showing that the same feedback source can support either a correctness reward or a probability-scoring objective. Table 1 compares RLHF, RLVR, and RLCD as entry points. Sections 2–5 then follow a probability claim from its target through the scoring incentive and the implemented update to the decision that consumes it.

**Table 1. RLHF, RLVR, and RLCD as entry points.** The labels emphasize different aspects of learning; they do not name three mutually exclusive algorithms or imply a performance ranking. RLCD is TypeSafe’s term; this survey covers calibration-aware RL more broadly.

| Entry point                       | Defining emphasis and signal                                                                                                  | Main opportunity                                                                                                        | Limitation relevant to calibration                                                                                                                                                                |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RLHF</b><br>Human feedback     | <b>Feedback source.</b> Human judgments typically train a reward model for policy optimization [2].                           | Captures preferences and response qualities that resist exact rule-based specification.                                 | Depends on annotation costs, annotator populations, and reward proxies. Preferred expressions of certainty need not identify correctness probability [6].                                         |
| <b>RLVR</b><br>Verifiable rewards | <b>Feedback source.</b> Checked answers, tests, or resolved outcomes provide rewards [3], [5].                                | Scales automatic feedback where adequate verification is available and can also support probability-scoring objectives. | Depends on verifier validity and coverage. Rewarding correctness alone does not ensure calibrated reports [7].                                                                                    |
| <b>RLCD</b><br>(TypeSafe)         | <b>Objective and interface.</b> Public materials describe training for typed decisions and calibrated probabilities [1], [4]. | Exposes decisions and probability fields to software for routing, thresholding, and escalation.                         | The complete training recipe is undisclosed. Probability semantics and task-level performance require validation; the interface alone does not establish an advantage over other training routes. |

The literature supplies apparently conflicting observations. The GPT-4 technical report shows that, on the same MMLU subset, answer-option calibration is worse after post-training than before [8], whereas a study of reasoning models finds their discretized verbal confidence better calibrated than that of non-reasoning counterparts, with task- and interface-dependent exceptions [9]. Another comparison reports lower average expected calibration error (ECE) after group relative policy optimization (GRPO) [10] or direct preference optimization (DPO) [11] than at initialization, while a supervised baseline attains lower ECE than either [12]. Across the checkpoints studied, distillation and RL generally improve verbalized calibration on reasoning tasks while reducing “I don’t know” responses on factuality benchmarks [13]. Other studies report a similar decline in abstention: reinforcement fine-tuning can reduce refusal on unanswerable questions, which the Hallucination Tax study evaluates as answer/refusal behavior rather than numerical calibration [14], and models tuned for reasoning by distillation or supervised fine-tuning can gain accuracy on answerable questions while losing abstention recall on unanswerable ones [15, Table 4]. Results also differ by training pipeline: among released reasoning models, those trained by supervised fine-tuning followed by RL scored higher on factual question answering (QA) than their non-reasoning comparison models in all four pairs examined, whereas RL-only models scored lower on TriviaQA than their starting models in all three [16, Tables 1–2]. Finally, changing only an answer’s chat-role framing can change confidence [17]. These results need not conflict: they concern different probability readouts, supervision, post-training pipelines, and evaluation populations. Interpreting them therefore requires separating the training intervention from the probability interface and from the evaluation population.

Three gaps separate an improved confidence metric from a claim that downstream software can use, and they organize this survey. **The reporting-capability gap** (Section 3): a reward that elicits honest confidence can also prefer one distribution of success across questions over another; in an exact two-policy example, maximizing a correctness-plus-Brier reward under truthful reporting selects the policy with lower mean accuracy. **The objective-update gap** (Section 4): normalization, token masks, finite rollouts, and direct reward derivatives determine whether training follows the stated objective; in one reported comparison, GRPO with division by the group standard deviation reached ECE 0.292, against 0.036 without it [18]. **The ranking-decision gap** (Section 5): a score can keep its ordering, and hence its area under the receiver operating characteristic (ROC) curve (AUROC), while losing the numerical meaning that a cost-derived threshold requires. Each gap calls for a different comparison and potentially a different remedy; a single “better confidence” label obscures these choices.

Our thesis is that calibration-aware RL must preserve the meaning of a probability claim—its event, information, and population—from target construction through objective design and implemented learning to decision use. The three gaps mark where along that path the meaning can be lost. Method comparisons must therefore distinguish fixed-outcome reporting, joint answer–confidence objectives, and action rewards, rather than group methods by optimizer or output format.

The same lens answers a central question for method choice: when does RL add something that post-hoc calibration or direct supervision cannot? The answer depends on what has to change. If the predictions can remain fixed and only the reported number must move, post-hoc fitting is the comparator to beat; if the desired report has a tractable differentiable loss, direct supervision is; RL offers something distinct only when the model’s own behavior—how it reasons, retrieves, answers, abstains, or acquires information—must adapt through outcome feedback (Section 6.1). What the downstream decision needs is a separate question, and its answer need not favor the design with the best average metric or the richest output: lower average accuracy can accompany a larger safely answerable subset, ranking can suffice for fixed coverage, and direct action learning can suit a single cost regime. Probability estimation earns its additional complexity when numerical meaning is needed across decisions, information states, or operating points (Section 5).

## 1.1. Contributions and relation to existing surveys

Existing surveys cover confidence estimation and calibration [19], reward modeling and RL objectives for reasoning [20], RL for large reasoning models, from reward design to policy optimization and sampling [21], uncertainty as an active signal in reasoning and agents [22], uncertainty quantification and its applications [23], and abstention [24]. Together, these surveys establish the broader landscape of uncertainty estimation, calibration, and decision control. More closely related work examines how confidence signals are used in learning and decision-making. *Confidence as Control* [25] organizes confidence-related signals by source, decision unit, and control role across training and deployment, including preference and policy optimization. *Beyond Natural-Language Interfaces: Decisions, Scores, and Predictive Distributions* studies how predictive reports are produced, transformed, and delivered to downstream consumers, with particular attention to supervision, access, computation, and evaluation.

In contrast, the present survey is organized around the three gaps introduced above, which connect probability-learning incentives, implemented optimization updates, and downstream decision losses. This perspective complements prior treatments of confidence, uncertainty, and predictive interfaces by emphasizing how probabilistic semantics interact with training objectives and decision-theoretic consequences. A common way to narrate this literature is as a pipeline that first estimates uncertainty, then calibrates it, and finally uses it for control; Section 2.4 traces the lineage behind that reading (Figure 3). We do not organize the survey by these stages, because each gap arises where a later stage changes what an earlier one established: training can change the answers whose confidence was estimated, the implemented update can depart from the calibration objective, and a decision can need numerical meaning that an estimate validated only by its ranking does not supply. Where the broad survey of RL for reasoning treats model confidence mainly as an intrinsic reward [21, pp. 17–18], the present survey treats the reported probability as a claim whose event, information, and population must be preserved through training. Appendix E provides a detailed comparison of scope and coverage.

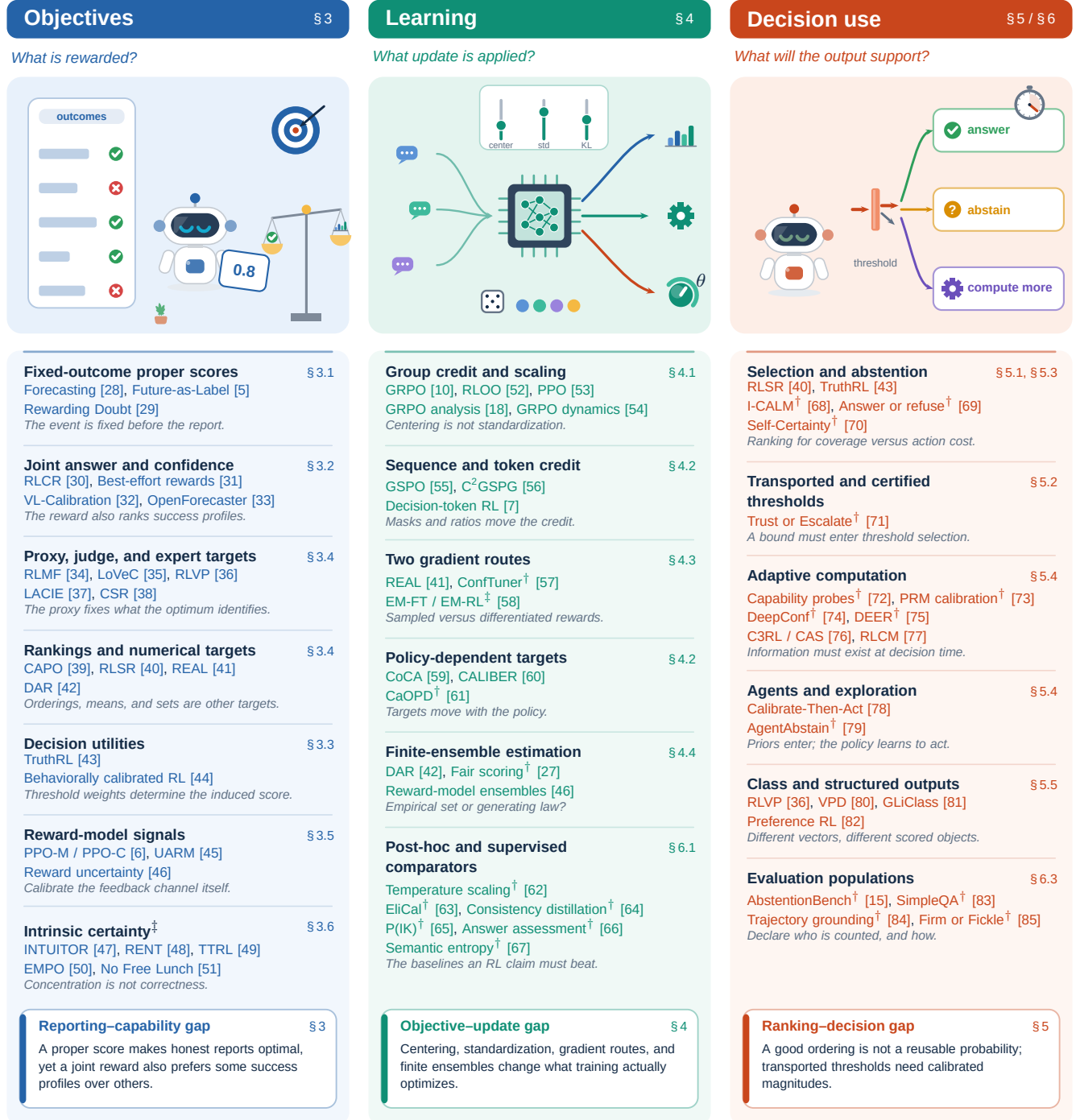
The survey makes four contributions. It compares methods by the probability claim their output makes and the interface that carries it (Section 2; Figure 1; Table 3; Appendix B). It makes each of the three gaps precise with exact constructions and the corresponding evidence (Sections 3–5). It turns these distinctions into a comparator guide and an evaluation contract (Section 6; Tables 5–7). Finally, it states four open questions with the evidence that would distinguish competing explanations (Section 7). The identities behind these constructions draw on established statistics [26], [27]; our contribution is to integrate them with concrete learning protocols and decision requirements.

## 1.2. Scope and method

We review calibration-aware RL together with supervised, preference-based, distillation, and post-hoc comparators, and we draw on foundational results, post-training diagnostics, and relevant decision interfaces. A source enters the collection if it trains, fits, or evaluates a probability report, a correctness or capability estimate, or an action that consumes one, or if it supplies an optimizer, comparator, theory, diagnostic, or decision interface that the comparisons use. Selection combines targeted primary-source searches and citation tracing. We verify source identity and check

# Calibration-aware reinforcement learning

A map of objectives, learning mechanisms, and decision uses. Method names link to references; section marks link to the discussion.



<sup>†</sup> comparator without online rollout-reward training, or classical theory   <sup>‡</sup> boundary family: capability training without a probability target   Methods may recur across views.

Jev<sup>†</sup> [1]   § 5.6   \* Industrial interface case: typed decisions and probability fields for software; the full RL training recipe is undisclosed.

**Figure 1.** Faceted taxonomy of calibration-aware RL and its comparators. Three complementary views organize objectives, learning mechanisms, and decision uses; a method may recur across views. The drawings are conceptual illustrations, not empirical results or a sequence of training stages. Method names and section marks are clickable links to bibliography entries and to the relevant discussion. Detailed distinctions appear in Sections 3–6 and Table 3. Jev is an industrial interface case, not an established algorithm.

claim-bearing methods, equations, experimental tables, and implementation details; the source collection records the version and reading range of each source (Appendix F).

The collection contains 140 academic records and eight primary web sources. Targeted searches for 2025–2026 work on confidence rewards, label-free certainty rewards, reward-model uncertainty, adaptive computation, abstention decisions, and evaluation extend its core of calibration-aware training protocols, as do the primary sources of the optimizers the survey names. A second extension adds the sources needed to trace the lineage in Section 2.4: two semantic-entropy studies, a self-certainty selection score, a semantic-entropy reward, and a multi-turn consistency study. Table 3 and Appendix B organize the learning and comparator protocols within the collection. We report model results under their original study populations and protocols, and we label analytic constructions and public-code observations separately.

## 2. Targets and comparison framework

A probability claim begins with an event and the information available when its probability is reported. The same “0.8” can describe the correctness of an answer, the success rate of a sampling policy, or agreement within an annotator population, and each requires different supervision and supports different decisions. A probability claim therefore needs three specifications alongside the number itself: its event, the reporter’s information, and the population or policy over which frequencies are taken. Post-hoc calibration can usually hold these three fixed; RL often cannot, because training can change the answers being assessed and the reasoning or evidence on which the report conditions. We compare methods by the claim their output makes and then trace that claim through the objective, the implemented update, and the decision that uses it (Figure 2).

### 2.1. Events, information, and reports

Let  $X$  denote an input,  $A$  a generated answer or action, and  $Z \in \{0, 1\}$  a declared binary outcome. For answer confidence,  $Z$  may indicate whether  $A$  passes a specified correctness test. For forecasting, it may indicate whether an external event occurs. Write  $\mathcal{I}$  for the declared statistical conditioning information and

$$p_{\mathcal{I}} = \mathbb{E}[Z \mid \mathcal{I}], \quad C \in [0, 1]$$

for the corresponding conditional probability and its report. Conditional scoring arguments require the report to use only the declared information: formally, it must be measurable with respect to  $\mathcal{I}$ , so any information the report can depend on must be included in  $\mathcal{I}$ . The protocol must separately record the evidence actually available to the model and must not supply feedback that arrives after the decision being evaluated.

For a deterministic correctness rule, conditioning on the complete prompt and answer makes the label probability zero or one, whatever the computational difficulty. A nondegenerate target therefore needs a stated source of uncertainty: coarser conditioning, such as on a report or a restricted representation; uncertainty over latent or stochastic labels; or a stochastic answering procedure. Population calibration given a score remains meaningful even for deterministic benchmark labels. A post-answer reporter can inspect the realized answer and reasoning, whereas a pre-generation capability estimator cannot, so the claim must state its report timing as well as its statistical conditioning. Access to reasoning does not show that a reporter uses it. A vision–language study measures trajectory grounding as the drop in expected confidence when attention to the model’s own trajectory is removed; at 8B, RL-based calibration training improved outcome-level calibration while trajectory grounding fell below zero (Table 7) [84, Tables 2 and 5]. At 4B the post-training score was not distinguishable from zero, but the 8B result shows that a report calibrated at the outcome level can behave more like a question-level estimate than an assessment of the realized answer.

For a fixed generation and decoding policy  $\pi$ , capability can instead be written

$$p_{\pi}(X) = \Pr_{\pi}(Z = 1 \mid X),$$

where the probability averages over the declared sampling procedure and any outcome randomness. Repeated rollouts estimate this policy-specific success rate; they do not directly label the correctness of an answer already selected. Because the rate is specific to the policy and its sampling procedure, capability estimates for repeated-attempt budgets depend on the decoder and on assumptions about repeated samples [72]. Changing the policy changes this target even

when the benchmark questions stay fixed. Comparisons between RLVR-trained models and their base models make this concrete: the RL-trained models lead at small  $k$  but are matched or overtaken at large  $k$ , and on AIME24 at  $k = 1024$ , 13.3% of problems were solved only by the base model and none only by the RL-trained model [86, Table 2]. Under independent draws, expected  $\text{pass}@k$  equals  $\mathbb{E}[1 - (1 - p_\pi(X))^k]$ , so  $\text{pass}@1$  and large- $k$   $\text{pass}@k$  summarize different features of one distribution of success probabilities.

## 2.2. The principal objects

Table 2 separates output form from statistical meaning. A vector of numbers does not establish full-distribution supervision, and a generated numeral is not necessarily an event probability.

**Table 2. Probability objects and what defines them.** A probability-valued output is interpretable only together with its event, the information it conditions on, and, where relevant, the generation policy or population behind it. The last group describes behavior rather than belief.

| Object                                            | Quantity and conditioning information                                                                                        | Representative setting                                                                                    |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Events, labels, and realized answers</b>       |                                                                                                                              |                                                                                                           |
| External binary event                             | $\Pr(Z = 1 \mid \mathcal{J})$ , for an outcome mechanism held fixed during the reporting comparison                          | Stochastic-outcome scoring experiments [18]                                                               |
| Class-label posterior                             | $Q_k$ intended to describe $\Pr(Y = k \mid \mathcal{J})$ , either for the complete vector or only for its selected component | Decision-token calibration [7]; verbalized vectors in RLVP [36]                                           |
| Realized-answer correctness                       | $\Pr(A \text{ is correct} \mid \mathcal{J}, A)$ , possibly including the answer’s reasoning trace                            | $P(\text{True})$ evaluation of a proposed answer [65]; joint answer–confidence learning [30]              |
| <b>Success rates of a named generation policy</b> |                                                                                                                              |                                                                                                           |
| Pre-generation capability                         | $\Pr_\pi(Z = 1 \mid X)$ , under a named generation and decoding policy                                                       | $P(\text{IK})$ under a specified sampling policy [65]; inference-budget prediction [72]                   |
| Continuation success                              | $\Pr_\pi(Z = 1 \mid X, A_{1:t})$ , success of continuing a partial trace under a named decoder                               | Calibrated process reward models (PRMs) for adaptive sampling [73]; distributional success estimates [87] |
| <b>Populations and numerical targets</b>          |                                                                                                                              |                                                                                                           |
| Population preference                             | $\Pr(B \succ A \mid X, \text{annotator process})$ , allowing persistent disagreement                                         | Preference-distribution learning [82]                                                                     |
| Numerical outcome statistic                       | A conditional mean or another declared functional of a numerical target                                                      | REAL’s mean-oriented objective [41]; sentence-level judge ratings in long-form generation [35]            |
| Numerical outcome distribution                    | $F(t \mid \mathcal{J}) = \Pr(Y \leq t \mid \mathcal{J})$ , beyond a point prediction                                         | Distributional regression with the empirical continuous ranked probability score (CRPS) [42]              |
| <b>Behavior rather than belief</b>                |                                                                                                                              |                                                                                                           |
| Action-selection probability                      | $\pi(a \mid s)$ , describing how often the policy selects an action                                                          | Correct/abstain/wrong action rewards [43]                                                                 |

Kadavath et al. established the useful  $P(\text{True})/P(\text{IK})$  distinction: evaluating a supplied answer and predicting success before an answer is supplied require different conditioning [65]. Their  $P(\text{IK})$  training, which updates the language model and a value head from sampled correctness labels, is a supervised counterpart to later RL approaches for capability estimation. Verbalized uncertainty training [88] and elicitation comparisons [89] add another axis: confidence in the same answer can be read from label logits, a numerical report, or a prompted self-evaluation. Sections 4 and 6.1 compare the learning choices for these interfaces.

The last row of the table, action-selection probability, is particularly important for decision-oriented large language models (LLMs). Increasing the probability of selecting a useful action does not, by itself, make that probability an esti-

mate of the action’s correctness. Similarly, a preference distribution measures the specified population’s choices, rather than uncertainty about a uniquely correct winner. In the principal experiments of [82], that population is simulated through persona-conditioned annotators. The target remains meaningful, but its population must accompany the claim.

The same separation applies inside a reporting interface. A distribution over candidate confidence numerals, such as the one ConFTuner trains [57, Eq. 1] (Section 4.3), is a mechanism for producing a report, not the answer’s correctness posterior. Likewise, emitting a full class vector does not mean every component was scored: a selected-answer reward can improve multiclass metrics empirically [80, Table 6] without being strictly proper for an unrestricted vector, as Section 5.5 discusses.

### 2.3. Calibration is one property among several

For binary correctness, we use **score calibration** to mean

$$\mathbb{E}[Z \mid C] = C \quad \text{almost surely.}$$

This matches Guo et al.’s definition of confidence calibration, which conditions on confidence alone [62, Eq. 1]. If  $\hat{Y}$  is the predicted class, class-conditional top-label calibration additionally requires  $\mathbb{E}[Z \mid C, \hat{Y}] = C$ . Full-vector calibration requires  $\mathbb{E}[\mathbf{1}\{Y = k\} \mid Q] = Q_k$  for each class. When the prediction and its confidence are determined from that vector, the full-vector condition implies the corresponding selected-class conditions. The reverse implications generally fail.

None of these equalities makes the report equal to the probability of correctness conditional on all available input information. A constant report equal to the population success rate is score-calibrated but cannot distinguish difficult from easy cases. Resolution concerns such informativeness; discrimination concerns ordering correct above incorrect answers. A strictly increasing transformation preserves scalar rankings, and hence AUROC under consistent tie handling, while generally changing calibration. Equal AUROC values alone, however, do not establish equal rankings or identical operating curves.

For binary correctness  $Z$  and a scalar report  $C$ , the population Brier decomposition makes the distinction between calibration and resolution precise. Write  $m(C) = \mathbb{E}[Z \mid C]$  and  $\mu = \mathbb{E}Z$ . Under their fixed joint distribution [90],

$$\mathbb{E}(C - Z)^2 = \underbrace{\mu(1 - \mu)}_{\text{uncertainty}} - \underbrace{\text{Var}(m(C))}_{\text{resolution}} + \underbrace{\mathbb{E}(C - m(C))^2}_{\text{squared reliability error}}.$$

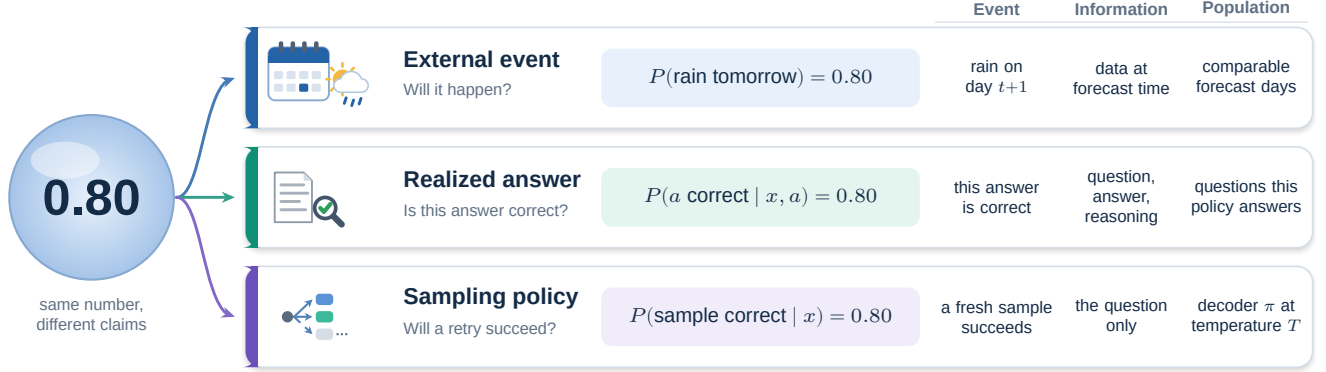
The identity conditions on the exact report; estimating its terms with bins introduces a separate approximation. Resolution measures information about correctness retained by the score; under calibration it becomes  $\text{Var}(C)$ . The decomposition also shows why RL needs separate treatment. With answers held fixed, the uncertainty term is constant and a Brier gain reflects resolution or reliability; when training changes the answer-generating policy, the base rate  $\mu$  moves as well, so part of a Brier change can reflect the outcome population rather than the report. Section 6 therefore separates fixed-answer from native-policy evaluation.

Sharpness describes the concentration of predictive distributions without reference to realized outcomes. Gneiting, Balabdaoui, and Raftery formulate maximizing sharpness subject to calibration as a guiding principle for probabilistic forecasting [91], which separates useful concentration from an unconditional preference for certainty. A scoring rule  $S(c, z)$  is *proper* if, for every outcome probability  $p$ , the expected score is optimized by reporting  $c = p$ , and *strictly proper* if that report is the unique optimum. Proper scores evaluate predictive quality more broadly than calibration does, so an improvement in them can reflect reduced reporting error, more informative predictions, or a changed outcome population. Strict propriety is defined within a specified report class [26], and it does not establish that finite neural optimization attains the truthful optimum.

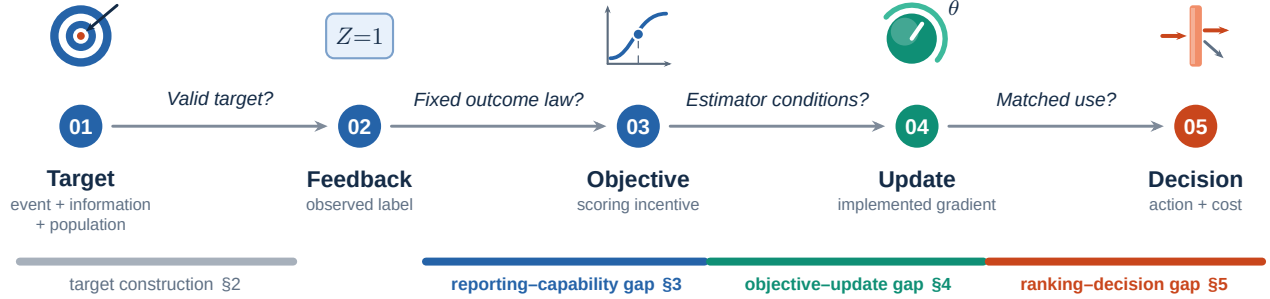
Coverage must also name its object. Selective-answering coverage is the fraction of queries answered, and selective risk is the error among the retained predictions [92]; risk-coverage curves and the area under them (AURC) summarize a selector across operating points, combining the underlying prediction errors with their ordering [93]. Predictive-set coverage is instead the probability that a set contains the target: conformal methods guarantee marginal coverage under specified sampling assumptions, and conditional or subgroup coverage requires stronger provisions [94].

A further use of the term *calibration* concerns a generative distribution over semantic facts. Kalai and Vempala’s hallucination lower bound applies to that generated fact distribution, under a sparse fact space and limited information about unseen arbitrary facts [95]. Transferring the bound to reported correctness confidence would require a model connecting the two objects.

(a) One reported number, three probability claims



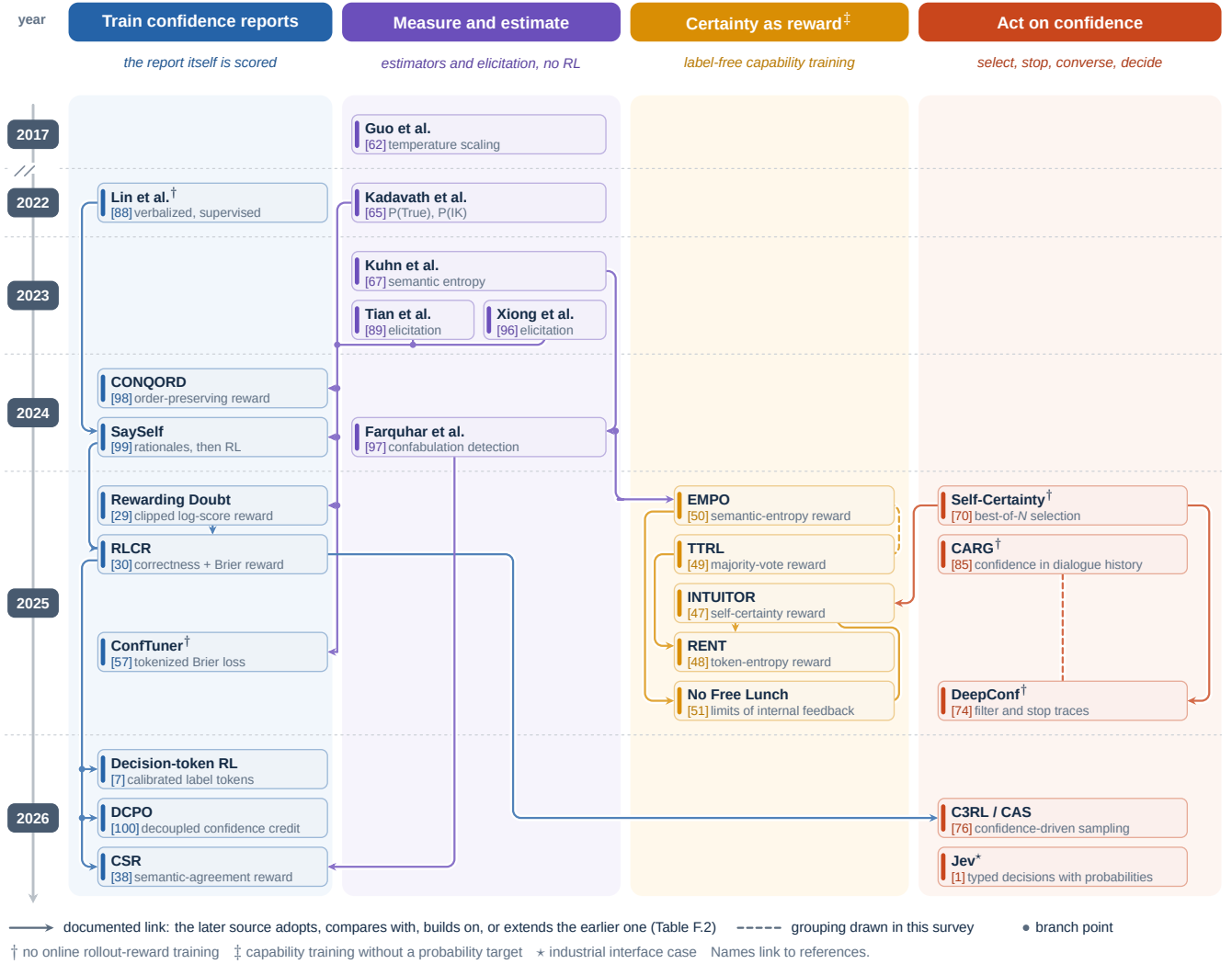
(b) The same claim must keep its meaning at every stage



**Figure 2.** A probability claim has a context. (a) The same reported number can state three different claims, about an external event, a realized answer, or the success of a sampling policy; each claim is fixed by its event, the reporter’s information, and a population. (b) The workflow connects the target to feedback, an objective, an implemented update, and a decision. Each connecting question identifies a condition that must be checked, not one that holds automatically, and the colored bars mark where the three gaps of Sections 3–5 arise. Section 2 uses this reading aid to organize established distinctions [22], [26], [65].

## 2.4. How the literature is organized

**How the lines developed.** Figure 3 traces four lines of work through the sources compared here. Estimation work showed how confidence can be measured or elicited without RL: post-hoc temperature scaling for neural classifiers [62], self-evaluation of a supplied answer and prediction of success before answering [65], prompted verbal elicitation [89], [96], and semantic entropy, the entropy of sampled answers grouped by meaning [67], [97]. Among the sources compared here, training of the reported confidence begins with supervised verbalized uncertainty [88] and, in 2024, turned to RL rewards on the report itself, among them SaySelf’s quadratic (Brier-type) reward, evaluated against those elicitation baselines [98], [99, p. 5]. In 2025, a clipped log-score reward was applied to confidence in a fixed answer [29], and RLCR combined a Brier term with a correctness reward, naming SaySelf and Rewarding Doubt as its closest related work [30, p. 11]; a supervised tokenized Brier loss offered a non-RL route [57]. Work in 2026 moved calibration-aware training beyond the verbal report: to decision-token probabilities [7], to separately credited confidence tokens [100], and to semantic agreement among sampled answers [38], each positioned against RLCR. Two further lines are represented here by 2025 work. Label-free methods rewarded certainty itself to train capability: semantic-entropy rewards [50], majority-vote rewards [49], self-certainty [47], and token entropy [48], followed by a controlled comparison of when such internal feedback helps [51]. Inference-time methods acted on confidence without training: self-certainty ranked candidates for best-of- $N$  selection [70] and supplied INTUITOR’s reward and, in a top- $k$  form, DeepConf’s global baseline score [47, p. 4], [74, p. 3]; DeepConf’s windowed confidence filtered and stopped parallel traces; and a token-probability confidence was written back into the dialogue history [85]. C3RL then joined the lines by training confidence



**Figure 3.** Lineage of the research lines compared in this survey. Columns group sources by how they treat confidence: training a stated confidence report, estimating or eliciting confidence without RL, rewarding certainty to train capability without a probability target, and acting on confidence at inference or in deployment. Solid arrows mark relations documented in the later source: it adopts the earlier method or statistic, uses it as a baseline or a stated contrast, names it as its closest related work or as the line it builds on, or, with shared authors, extends it. The purple line bundles links from three elicitation studies to the four methods that compare against one or more of them. Dashed links mark groupings drawn in this survey. The figure is a selection; Table F.2 lists each drawn link with its locator; together with some documented links not drawn here. Years refer to first public versions; within a year, entries in a column follow that date, but vertical positions are not comparable across columns. Read as a lineage, the columns suggest a pipeline from estimation through calibration to control; Section 1.1 explains why the survey is organized by gaps instead.

reports for an adaptive sampler [76]. Solid links in the figure mark relations that the later source documents; Table F.2 gives their locators.

Figure 1 organizes the literature into three complementary views rather than mutually exclusive families: a method can share an objective with one neighbor, an update with another, and a deployment use with a third. Each view corresponds mainly to one gap and one section, and Table 3 compares the families along the axes that expose those gaps: what is scored, what training changes, and what a downstream decision may infer. The table’s unit of comparison is a learning target rather than a paper or optimizer name; Appendix B gives the source-level inventory.

**Objectives: what signal should improve?** This view carries the reporting–capability gap (Section 3). Its families differ first in how much of the claim they hold fixed: external-event forecasting fixes the outcome while the report changes [5], [28], [33], [101], [102], fixed-answer confidence training fixes a supplied answer [29], and joint answer–confidence learning changes answer and report together [30], [31], [32], [99], [100], [103], which is where reporting and capability can pull apart. Other families change the target itself: reward-model interventions repair how a preference signal treats

expressed certainty [6], while distributional and conformal reward models expose the scorer’s own uncertainty [45], [46]; label-free certainty and consensus rewards train capability without any probability target [47], [48], [49], [58]. Expert guidance, judges, and consistency or verification feedback provide other targets for reports [34], [35], [36], [38], [104], [105]. Ranking objectives prioritize useful ordering [39], [40], [98]; numerical prediction separates a conditional mean from a distribution of outcomes [41], [42]; and action utilities and threshold-derived reporting scores tie feedback to a declared deployment loss [43], [44], [106].

**Learning: how does the signal change parameters?** This view carries the objective–update gap (Section 4). Group centering and standardization turn trajectory outcomes into credit [10], [18], [30], [54]; treating a whole response as one action and using sequence-level importance ratios change the unit of that credit [52], [55]; token masks, trajectory baselines, paired comparisons, and process feedback localize that credit [7], [56], [77], [107]; and a parameter-dependent reward adds a direct gradient route [41]. Online targets track a changing answer policy [59], [60], distillation can match confidence to the information available to the student [61], and finite-rollout objectives choose between scoring the realized empirical set and its generating law [27], [42]. Supervised confidence fitting and post-hoc transformations are the simpler comparators when answers or logits can stay fixed [57], [62], [66], [88], [108]; Section 6 treats them as the baselines that an RL claim must beat.

**Decisions: which property will be used?** This view carries the ranking–decision gap (Section 5). Selective answering and abstention consume ordering or numerical confidence through coverage and error costs [40], [43], [109], [110], and agents add an executed action whose gate can move independently of the stated probability [78], [79], [111]. Adaptive computation needs success estimates before additional work is purchased [65], [72], [73], [76], [87], whereas early exit and parallel-trace filtering read confidence inside or across traces [74], [75], [112], and a dialogue method feeds it back to the model across turns [85]. Class and structured outputs range from supervised label-token prediction to learned probability vectors [36], [80], [113], with guard models sensitive to the task’s safety labels and validation population [114]. Forecasting and judgment need external-event probabilities, outcome distributions, or a stated annotator population [28], [42], [82], [115], with selective judging escalating uncertain verdicts [71]. Jev is an industrial interface case, kept distinct from the public training algorithms [1].

The table’s value lies in what it refuses to merge. Holding answers fixed isolates reporting, whereas joint training also changes which outcomes are scored; the reward, not the output format, decides whether training seeks probability magnitudes, useful ordering, or actions at a specified cost; and a mean, a probability vector, and a sampled predictive distribution require different feedback and losses. These distinctions determine the appropriate baseline and evaluation—fixed-answer calibration, selective risk, cost-sensitive utility, or distributional scoring. They also explain why a shared optimizer, such as proximal policy optimization (PPO) [53] or GRPO [10], or a shared output format does not show that two methods solve the same statistical problem. The next section begins with the first of these distinctions: what a reward values once the policy also chooses the answer being scored.

### 3. Objectives and joint capability

A strictly proper score makes truthful reporting uniquely optimal in expectation when the outcome law is fixed. Joint answer–confidence training does not keep that law fixed: when the policy also chooses the answer, the reward that makes reports honest also expresses a preference over how success is distributed across questions. This section develops that reporting–capability gap: fixed-outcome versus joint training (Section 3.1), a decomposition of the joint objective (Section 3.2), and best-effort conditions together with the decision thresholds that a reward averages (Section 3.3). Sections 3.4–3.6 turn to objectives with other targets: rankings, proxies, distributions, preference-derived reward signals, and intrinsic certainty.

Throughout,  $p_\pi$  denotes a conditional success probability under policy  $\pi$  and a declared information set, and the report  $c$  is measurable with respect to that information. The probability may average over a randomized answering procedure or over uncertainty remaining after information is withheld (Section 2.1).

#### 3.1. Fixed outcomes, selected answers, and forecasting

For a fixed external event, changing a forecast ordinarily leaves the event itself unchanged, and a strictly proper score uniquely favors the true conditional probability when that report is available and expectations exist [26]. Outcome-based forecasting turns this idea into a training pipeline, and its variants differ mainly in what they hold fixed and what

**Table 3. What calibration-aware training learns.** Families are compared by the object actually scored, the learning intervention, and the resulting decision semantics. Examples are representative; Appendix B records 77 source protocols and four controlled analyses.

| Family / exemplars                                                                                                                                                     | Scored target and signal                                                                                                    | Learning intervention                                                                                         | Decision use and interpretation                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A. Probability reports: what event and information define the target?</b>                                                                                           |                                                                                                                             |                                                                                                               |                                                                                                                                                                                 |
| <b>External events</b><br>Forecasting [28],<br>outcome-based RL [101],<br>teacher or market targets<br>[102], [116]                                                    | Event probability; resolved<br>outcomes or soft targets from<br>teacher or market probabilities,<br>with Brier/log rewards. | Update the report policy;<br>the event definition stays<br>fixed.                                             | Forecast-based decisions; assess<br>reliability and proper scores on<br>held-out resolved events.                                                                               |
| <b>Fixed-answer<br/>confidence</b><br>Rewarding Doubt [29]                                                                                                             | Supplied-answer correctness;<br>clipped log-score reward.                                                                   | Train confidence<br>generation with the answer<br>held as input.                                              | Answer assessment; isolates<br>reporting from the choice of<br>answer during training.                                                                                          |
| <b>Joint answer and<br/>confidence</b><br>RLCR [30], Multi-RLCR<br>[103], Verbalized<br>Probability Distribution<br>[80], VL-Calibration [32]                          | Selected-answer confidence or<br>candidate marginals;<br>correctness plus report-error<br>rewards.                          | Update the answer<br>distribution and reports<br>together.                                                    | Answering and selection;<br>reporting and capability interact.<br>A vector output may receive<br>scalar supervision, and<br>decomposed components may<br>receive proxy targets. |
| <b>Prospective success</b><br>CoCA [59], CALIBER [60];<br>capability probes <sup>†</sup> [72],<br>EliCal <sup>†</sup> [63], calibrated<br>PRMs <sup>†</sup> [73], [87] | Policy success before the<br>outcome; rollout-frequency<br>targets.                                                         | Train report/answer<br>segments, or fit a separate<br>probe, head, or PRM<br>calibration layer <sup>†</sup> . | Routing and compute allocation;<br>estimates depend on the policy<br>and information available when<br>used.                                                                    |
| <b>Class probabilities</b><br>Decision-token RL [7],<br>RLVP [36]                                                                                                      | Class-token probabilities or a<br>generated vector; local class loss<br>or teacher-guided reward.                           | Apply masked RL plus a<br>local loss, or RL to<br>generated vectors.                                          | Classification; evaluate the full<br>vector. Normalization or teacher<br>alignment alone does not<br>establish propriety.                                                       |
| <b>Preference probabilities</b><br>Judging with Confidence<br>[82]                                                                                                     | Annotator preference<br>probability; Brier/log rewards<br>on votes.                                                         | Train a<br>probability-generating<br>judgment policy.                                                         | Preference-sensitive judgment;<br>the annotator population, not<br>factual correctness, defines the<br>event.                                                                   |
| <b>Utility-derived reports</b><br>Behaviorally calibrated RL<br>[44]                                                                                                   | Correctness confidence; a score<br>derived from utility across<br>thresholds.                                               | Train confidence/value<br>prediction; apply<br>thresholds externally.                                         | Cost-sensitive decisions;<br>weighting the thresholds<br>determines which reporting score<br>is induced.                                                                        |

**Reading the table.** A target is fixed if the reporting intervention does not change it; it need not be observed without noise. A strictly proper score uniquely identifies a conditional probability when that probability is admissible. Composite rewards, restricted reports, changing policies, and finite optimization require separate analysis (Sections 3–4). <sup>†</sup> denotes a comparator without online rollout-reward training.

*Continued on next page*

Table 3 (continued)

| Family / exemplars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Scored target and signal                                                                                                         | Learning intervention                                                                  | Decision use and interpretation                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>B. Other learning targets and the role of simpler comparators</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                  |                                                                                        |                                                                                                                                                                                |
| <b>Proxy / reader alignment</b><br>RLMF [34], LoVeC [35], linguistic calibration [117]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Confidence aligned with sampling consistency or a judge, or reader belief; proxy, judge, or reader scores.                       | Change verbalized uncertainty, numeric or linguistic, and generated text.              | Uncertainty communication; interpretation depends on the proxy, judge, or reader. Factual calibration needs separate evidence.                                                 |
| <b>Reward-model feedback</b><br>PPO-M/PPO-C [6], UARM [45]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Learned response quality, the reward model’s treatment of expressed confidence, or its uncertainty about the reward label.       | Adapt the reward model before policy RL, or transform rewards or advantages during it. | RLHF; calibration of the trained policy’s reports must be measured separately, and reward-model coverage or ranking does not transfer automatically to the policy.             |
| <b>Relative ordering</b><br>CAPO [39], RLSR [40]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Correctness order; pairwise or rank-weighted selective-risk rewards.                                                             | Change likelihood or confidence ordering.                                              | Fixed-coverage selection; good ordering leaves the probability scale unidentified.                                                                                             |
| <b>Selective actions</b><br>TruthRL [43], UCPO [110]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Answer/abstain behavior; ternary utilities or group-dependent action credit.                                                     | Update the action policy under the chosen reward.                                      | Selective answering; evaluate utility and coverage. Action likelihood is not correctness confidence.                                                                           |
| <b>Numerical means</b><br>REAL [41]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Expected numerical rating; mean squared error plus likelihood.                                                                   | Combine policy and direct prediction gradients, as the paper proposes.                 | Point prediction; a correct mean does not identify a predictive distribution.                                                                                                  |
| <b>Predictive distributions</b><br>DAR [42]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | A finite set of generated outcomes; empirical CRPS.                                                                              | Assign leave-one-out credit, then apply group-standardized RL.                         | Distribution-sensitive decisions; distinguish the scored finite set from its generating law.                                                                                   |
| <b>Intrinsic certainty</b> <sup>‡</sup><br>INTUITOR [47], RENT [48], TTRL [49], EMPO [50]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Concentration of the policy’s own token distribution, or agreement with its majority answer or answer cluster; no outcome label. | Reward the generation policy for concentration with GRPO-style updates.                | Label-free capability adaptation; the statistic is not a correctness probability, and gains under self-certainty and entropy rewards can reverse with continued training [51]. |
| <b>Supervised confidence</b> <sup>†</sup><br>ConfTuner [57], answer assessment [66], intermediate-answer probes [112]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Fixed-answer correctness; semantic report error or binary prediction loss.                                                       | Directly fit confidence tokens or a separate estimator.                                | Answer assessment and early exit; tests whether rollout RL adds value beyond supervised reporting.                                                                             |
| <b>Post-hoc calibration</b> <sup>†</sup><br>Temperature scaling [62], self-consistency distillation [64]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Fixed-logit class probabilities or response embeddings; held-out likelihood loss or unlabeled self-consistency targets.          | Fit a scalar temperature or an external regressor with the generator fixed.            | Probability rescaling; temperature scaling preserves the class argmax, and calibration data must be representative.                                                            |
| <b>Cross-family comparison.</b> Probability magnitudes, ordering, action utility, conditional means, and predictive distributions are different learning targets. Shared PPO/GRPO labels do not make them equivalent: group normalization, token masks, and direct report gradients change the implemented update (Section 4). <sup>†</sup> denotes a comparator without online rollout-reward training; <sup>‡</sup> denotes a boundary family that trains capability without a probability target. The table states what each objective targets; it does not guarantee that calibration is achieved. |                                                                                                                                  |                                                                                        |                                                                                                                                                                                |

they vary. Future-as-Label controls information: it restricts predictor inputs to pre-resolution material and rewards resolved historical events with the log score [5]. Outcome-based Reinforcement Learning to Predict the Future varies the policy update under negative Brier rewards [101]. Proper-score forecasting varies the score itself, comparing log, Brier, spherical, and Beta-family rewards under a shared training setup, and shows that scores with the same truthful population optimum can produce different finite-budget learning profiles [28]. Population propriety therefore does not

settle optimization. A further variant changes the label rather than the score: a GRPO study trains forecasters against prediction-market prices used as soft cross-entropy targets [102, pp. 7, 9–10, 16–17]. Because cross-entropy is linear in its target, a price that equals the event’s conditional probability given information containing the reporter’s preserves the conditional optimum while removing outcome noise. That guarantee depends on the market’s information and pricing, however: the study does not report when prices were recorded, one displayed example pairs a context summary written after the question’s deadline with a price of zero [102, pp. 21–23], and the price-trained models are evaluated by agreement with prices rather than by calibration on resolved events [102, Table 5.1].

A fixed-answer confidence task has a similar structure. Rewarding Doubt first generates an answer and then treats it as an input while training the confidence report with a log score [29], so the answer’s outcome law is held fixed during the reporting comparison. Joint answer–confidence training instead allows the policy to improve the answer as well as its report. RLCR addresses the resulting need to reward truthful uncertainty while retaining an incentive to answer correctly [30]. The distinction lies in what the policy can change, not in whether its output is text.

Open-ended forecasting sits between fixed-event forecasting and selected-answer confidence. OpenForecaster predicts a free-form future outcome and assigns a probability to that selected answer [33]: the world outcome is external, but the proposition being scored depends on the model’s choice. Its fixed binary questions use negative Brier loss, whereas its free-form questions need a joint reward (Section 3.3). Even within forecasting, a fixed-event probability and a selected-answer confidence call for different objectives.

Report timing also matters. CoCA and CALIBER distinguish capability confidence, a pre-generation estimate of policy success, from response confidence, a post-generation assessment of the realized answer and reasoning [59], [60]. Under a fixed policy, one binary outcome supplies valid squared-loss supervision for its conditional success probability; multiple independent outcomes reduce estimation variance. During joint training, the answering policy changes, and an early confidence token may itself become part of the answer context. Target construction must track these dependencies.

### 3.2. Separating capability, its distribution, and reporting regret

Let  $S(c, z)$  be a proper score in reward orientation for binary  $z$ , with Bayes reward

$$G(p) = pS(p, 1) + (1 - p)S(p, 0).$$

Define its reporting regret by

$$D_G(p, c) = G(p) - \{pS(c, 1) + (1 - p)S(c, 0)\} \geq 0.$$

Under the standard regularity conditions, this is the regret associated with the convex (Savage) representation of proper scores (Appendix A.2), and strict propriety on a nondegenerate binary probability interval gives strictly convex  $G$  [26].

Consider  $R = \alpha Z + \beta S(c, Z)$ , where  $\beta > 0$ , and let  $\mu = \mathbb{E}[p_\pi]$ . A fixed report-independent additive constant shifts all policies equally and is omitted. For common input weights and finite expectations,

$$\begin{aligned} J(\pi, c) &= \alpha\mu + \beta G(\mu) \\ &\quad + \beta\{\mathbb{E}[G(p_\pi)] - G(\mu)\} \\ &\quad - \beta\mathbb{E}[D_G(p_\pi, c)]. \end{aligned}$$

The first line depends on mean capability. The second is a Jensen gap describing the distribution of conditional success probabilities, which we call the success-rate profile. The last measures reporting error relative to the declared information; unlike binned ECE, it does not aggregate errors within confidence bins. The identity applies to arbitrary admissible reports, and truthful reports make its final term vanish. Define the *calibrated value*  $V(p) = \alpha p + \beta G(p)$ , the expected reward at success probability  $p$  under truthful reporting. The first two lines sum to  $\mathbb{E}[V(p_\pi)]$ , so truthful reports give  $J = \mathbb{E}[V(p_\pi)]$ .

For the correctness-plus-Brier reward used by RLCR,

$$R = Z - (c - Z)^2, \quad J = \mu^2 + \text{Var}(p_\pi) - \mathbb{E}[(p_\pi - c)^2].$$

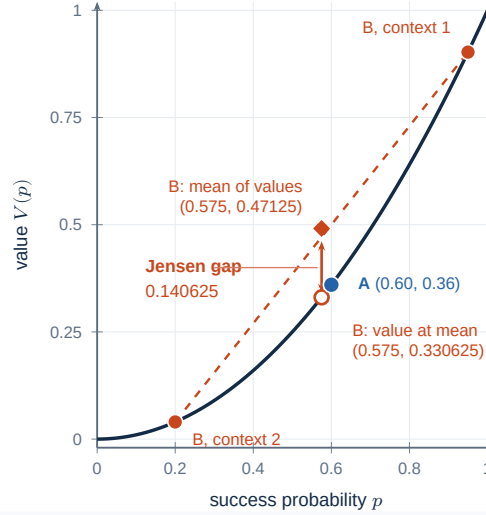
Here the calibrated value is  $V(p) = p^2$  [30]. Increased reward can therefore reflect a higher mean success rate, a more dispersed success-rate profile, or more accurate reporting. The variance term concerns actual conditional success probabilities across information states. Randomizing reports around one unchanged probability instead increases reporting regret.

### (a) Two policies

two equally weighted contexts; truthful reports



### (b) Truthful reports: value $V(p) = p^2$



### (c) Which policy wins?

|                                              | A     | B     | wins |
|----------------------------------------------|-------|-------|------|
| <b>Mean accuracy</b><br>$\mathbb{E}[p]$      | 0.600 | 0.575 | A    |
| <b>Joint reward</b><br>$R = Z - (C - Z)^2$   | 0.360 | 0.471 | B    |
| <b>Selective utility</b><br>$+1/-1/0$ at 0.5 | 0.200 | 0.450 | B    |
| <b>pass@2</b><br>$\mathbb{E}[1 - (1 - p)^2]$ | 0.840 | 0.679 | A    |

At threshold 0.5, A accepts both contexts; B accepts only its 0.95 context. Reward and selective utility favor B; mean accuracy and repeated attempts favor A.

$$J = \mathbb{E}[p^2] = \mu^2 + \text{Var}(p) \quad (\text{truthful reports})$$

$$\text{Jensen gap} = \mathbb{E}[V(p)] - V(\mathbb{E}p)$$

exact construction; equal weights

**Figure 4.** Classical proper-score geometry in an exact two-context construction. (a) Policy A has success rates (0.60, 0.60) and policy B (0.95, 0.20); contexts have equal weight and reports are truthful, so reporting regret vanishes and  $J = \mu^2 + \text{Var}(p) = \mathbb{E}[V(p)]$ , with  $V(p) = p^2$ . (b) B’s expected value, 0.47125, is the midpoint of the chord joining its two contexts and exceeds  $V(\mu_B) = 0.330625$  by the Jensen gap 0.140625; A’s contexts coincide at (0.60, 0.36). (c) A has higher mean accuracy and pass@2, whereas B has the higher correctness-plus-Brier reward  $R = Z - (C - Z)^2$  and higher selective utility at payoffs  $+1/-1/0$  with threshold 0.5. All plotted values are exact. The RLCR reward motivates the construction [30]; the geometry follows classical scoring theory [26].

Figure 4 makes the gap concrete with two equally likely contexts and a policy family restricted to A, with success probabilities (0.60, 0.60), and B, with (0.95, 0.20). With truthful reports, B’s expected value lies at the midpoint of the chord joining its two contexts on the convex curve, and the Jensen gap above  $V(\mu_B)$  lifts that value above A’s. Exact reward maximization therefore selects B although A has higher mean accuracy. The preference arises because the family couples performance across contexts; Appendix A.2 derives the construction and its assumptions.

The same allocation can be useful for selective decisions. With payoffs  $+1, -1, 0$  for correct answers, incorrect answers, and abstention, truthful thresholding at 0.5 gives A expected utility 0.20, whereas B answers only in its first context and obtains 0.45 (Figure 4, right, third row). A more uneven success-rate profile can therefore support more valuable selective decisions despite lower average accuracy; the deployment objective decides whether that allocation is beneficial.

The construction separates improving success wherever possible (the pointwise best effort that Section 3.3 formalizes) from aggregate capability allocation. A nondecreasing calibrated value favors a higher success probability wherever it can be raised at one information state without lowering another. Across coupled policies, however, maximizing  $\mathbb{E}[V(p_\pi)]$  also values their success-rate profiles, and a finite affine correctness bonus changes the slope of  $V$  but not its strict convexity (Appendix A.3 gives the restricted-family ordering result). Reward maximization and accuracy maximization agree under identifiable conditions. A pointwise-dominating feasible policy optimizes both when  $V$  is nondecreasing. With finite endpoint values and  $p_\pi \in \{0, 1\}$ , expected calibrated value is affine in mean accuracy, with slope equal to the difference between the endpoint values, and support on any two fixed probability values likewise reduces the comparison to an affine function of their mixture weight. When truthful reports are unattainable, policy-dependent reporting regret remains part of the comparison.

Accuracy weighting provides a quantitative control. For bounded  $G$ , attainable truthful reports, and exact optimization over the same feasible family,

$$0 \leq \text{Acc}(\pi_{\text{acc}}) - \text{Acc}(\pi_{\alpha}) \leq \frac{\beta \text{osc}(G)}{\alpha}, \quad \alpha > 0,$$

where  $\text{Acc}(\pi) = \mathbb{E}[p_{\pi}]$  is mean accuracy,  $\text{osc}(G) = \sup G - \inf G$  is the range of the Bayes reward (1/4 for the binary Brier score; Appendix A.3),  $\pi_{\text{acc}}$  maximizes accuracy, and  $\pi_{\alpha}$  maximizes the weighted reward. Under these conditions the bound doubles as a design rule: for any  $\varepsilon > 0$ , choosing  $\alpha \geq \beta \text{osc}(G)/\varepsilon$  ensures that the exact optimum gives up at most  $\varepsilon$  in accuracy. The bound, derived in Appendix A.3, concerns objective optima; whether neural optimization reaches them is the question of Section 4.

In practice, a joint reward’s gain should therefore be attributed before it is interpreted. Accuracy, a fixed-answer evaluation of calibration or reporting error, and an estimated success-rate profile, reported separately, show whether a gain came from capability, from reporting, or from allocation. Correctness-only RL already reshapes the profile: as GRPO training proceeded in one study, pass@1 on its training distribution rose from 26.1 to 42.5 while pass@256 fell from 66.3 to 64.3, and, in the same study’s base-versus-RL comparison on Minerva, per-question accuracies moved toward both one and zero [86, Table 4, Fig. 5]. An allocation effect of a joint reward should therefore be compared with a correctness-only RL control on the same decoder before it is attributed to the reporting term; Section 4.1 shows that the update’s normalization can itself shift the profile. Repeated-attempt objectives weight the same profile in the opposite direction: because  $1 - (1 - p)^k$  is concave in  $p$  for  $k \geq 2$ , policy A in Figure 4 has pass@2 0.84 whereas policy B has 0.679, although the correctness-plus-Brier reward prefers B.

### 3.3. Best-effort rewards and threshold-dependent utilities

Tan et al. describe joint confidence rewards through  $f(c)$  for a correct answer and  $g(c)$  for an incorrect one [31]. Their conditions combine proper reporting with a calibrated value that is nondecreasing over a specified probability interval; for differentiable rewards, the calibrated value’s derivative is  $f(p) - g(p)$ . Because the definition requires propriety but not strict propriety, it admits correctness-only rewards, which retain an accuracy incentive but leave the numerical report unidentified. Best effort is therefore weaker than eliciting a reusable probability.

For  $R = Z - k(c - Z)^2$  with nonnegative calibration weight  $k$ , the calibrated value is  $(1 - k)p + kp^2$ , which is nondecreasing on the full interval  $(0, 1)$  exactly when  $0 \leq k \leq 1$ . With a log-score component, the difference between the scores for success and failure is unbounded near an endpoint, so no fixed correctness bonus can keep the calibrated value nondecreasing over the whole interval [30], [31]. Log scoring remains proper for an external event or fixed answer; joint answer selection adds the best-effort requirement.

Published joint rewards show that these conditions bind in practice. VL-Calibration, which trains separate visual and reasoning confidences for vision–language reasoning, weights answer accuracy by 1 and a Brier penalty on their combined (holistic) confidence by 2 [32, Eq. 10, Table 5], which is  $k = 2$  in this notation. Setting aside its separate visual-certainty term, its calibrated value  $2p^2 - p$  decreases for success probabilities below 1/4. LoVeC, which attaches a 0–10 confidence to each sentence of a long-form answer, uses a released reward that clips, normalizes, and stretches a sentence-level log score and adds a small correctness bonus [35, p. 15]. For binary labels and truthful reports on its 11-point grid, our calculation from that code gives calibrated values of 31.62 at  $p = 0$ , 27.73 at  $p = 0.5$ , and 33.12 at  $p = 1$ . Free-form training can therefore prefer a statement known to be false, reported at zero confidence, over one with even odds of being true. Neither paper reports an aggregate loss of accuracy or factuality [32, Table 6], [35, Table 10]. Aggregate metrics can mask such incentives, however, whereas the separate measurements recommended at the end of Section 3.2 can expose their effects.

OpenForecaster provides a practical example of adjusting the balance between correctness and calibration weights [33]. Its free-form score is  $S' = 2cZ - c^2 = Z - (c - Z)^2$ , where  $Z$  indicates agreement with the resolved outcome; the final reward  $Z + S' = 2Z - (c - Z)^2$  strengthens the reward for a correct low-confidence guess. In the reported validation comparison, near-zero-confidence “Unknown” responses fell from approximately 40% with  $S'$  alone to 4% after adding the correctness term. This supports accuracy weighting as a way to improve exploration in that setting, alongside its population-level role in the accuracy bound.

Decision-based derivations explain which costs a reward averages. The bounded answer/abstain utilities in [44] assign  $+1$  or  $-1$  to an answered outcome and  $2t - 1$  to abstention at threshold  $t$ . Averaging these utilities under a uniform threshold density yields  $2cZ - c^2$ , exactly correctness-plus-Brier. Deriving these utilities from the original ones involves a  $t$ -dependent scale, so the uniform-averaging result is specific to that representation. The unscaled integral discussed in [31] instead produces  $f(c) = c$ ,  $g(c) = c + \log(1 - c)$ . Both constructions average over thresholds but weight cost regimes differently. Choosing a joint score therefore also chooses which operating thresholds training emphasizes, which links this section’s objectives to the decisions in Section 5.

At the other extreme, when one operating cost is stable, the reward can target the action directly instead of averaging over thresholds, as TruthRL does [43]. Probability reporting is the complementary interface when one model must serve several thresholds or changing costs; Section 5.3 compares the two.

### 3.4. Ranking, proxy targets, and distributional outputs

The remaining objectives change what the reward identifies, not only how strongly it rewards honesty. Each is best read by asking which statistical quantity its loss would recover at the optimum.

Ranking objectives are well matched to candidate selection. CAPO orders correct and incorrect trajectories, while CONQORD aligns confidence differences with preference-model quality differences [39], [98]. Their useful output is a relative score; assigning that score a numerical probability interpretation requires an additional outcome-based calibration step. A generated confidence numeral can serve either role; its training objective, not its format, determines which.

Proxy-based methods make feedback available when exact outcome supervision is costly. RLMF uses sampling consistency to supervise uncertainty expressions, and CSR (calibration with semantic reward) derives confidence from relations among generated answers [34], [38]. Their transfer to correctness depends on the relationship between the proxy and the deployment outcome. CSR, for instance, reads its confidence from discrete semantic entropy, computed from the frequencies of answer clusters among sampled responses [38, p. 13], [97, p. 8]. The semantic-entropy studies evaluate it, the later study also in its discrete variant, only by ranking metrics such as AUROC and rejection accuracy [67, p. 7], [97, p. 3], and the later study states that it does not detect errors that the model makes consistently [97, pp. 1, 4–5]. The loss selects a statistical quantity as well: absolute deviation elicits a median, whereas squared loss elicits a mean. The distinction matters for confidence-target methods: DCPO motivates its target with proper scoring, but its implemented reward is a negative absolute error to a mixed group and instance target, which elicits a conditional median rather than a mean (Appendix B) [100].

Label-free and judge-based targets make both points concrete. TTRL rewards each rollout for matching the majority answer of 64 samples on unlabeled test questions. On AIME 2024 the majority label was initially correct for 37% of questions, yet 92% of rollout rewards agreed with ground-truth rewards because wrong answers were scattered, an agreement that declined as training concentrated answers, although the authors report that it stayed above 75% [49, Figs. 9–10]; Section 3.6 returns to its effect on capability. A supervised analogue shows the dependence on labels: a confidence head trained only on self-consistency does not exceed the consistency baseline (in-domain average AUROC for Qwen2.5-7B, 71.19 versus 73.62) and reaches 84.36 only after calibration on 1k labeled questions [63, Table 2]. Self-consistency can also be distilled into a single-generation predictor without policy learning; such a predictor is a non-RL comparator for label-free confidence (Section 6.1) and can inherit the proxy’s consistent errors [64]. A judge-based target can change the estimated quantity itself: LoVeC scores sentence-level confidence against an LLM judge’s 0–10 veracity rating, so its stated loss recovers a conditional mean rating rather than a correctness probability unless the labels are binary [35, Eqs. 1–2, 6]; the clipping and stretching of its released reward (Section 3.3) need not preserve even that optimum. Across these protocols, the proxy fixes what the optimum identifies; its relation to correctness must be measured on held-out outcomes rather than assumed.

RLVP applies expert guidance to complete class vectors. It rewards a verbalized distribution  $p$  with

$$R_{\text{Prob}} = \exp\{-(1 - \lambda)D_{\text{KL}}(e_Y \| p) - \lambda D_{\text{KL}}(q \| p)\},$$

where KL denotes the Kullback–Leibler divergence,  $q$  is an expert’s soft distribution,  $e_Y$  the observed one-hot label, and  $\lambda$  anneals from 1 to 0, with added format and argmax-correctness rewards [36]. Expert matching supplies fine-grained comparisons among generated vectors and can improve learned predictions during the curriculum. The exponential transformation, however, changes the population incentive: at the hard-label endpoint  $R_{\text{Prob}} = p_Y$ , so for a report

issued before observing  $Y \sim r$  the expected reward is  $r^\top p$ , maximized by concentrating on a modal class. This analysis concerns the displayed probability component, and teacher-correct filtering further restricts the samples to which it applies; Appendix A.7 separates the teacher-matching incentive, the endpoint scoring property, and the limits of both for finite training.

Teacher information is another part of target design. CaOPD distills from a teacher that receives privileged demonstrations or feedback unavailable to the deployed student, replacing confidence targets with the current student’s empirical success rate while retaining a token-level reverse KL objective [61]. It is a calibration-oriented distillation comparator rather than a new proper-score RL reward, and its question-level target shows why teacher certainty, student capability, and confidence in a particular answer should be kept distinct. For expert-guided RL and distillation alike, the information behind a target defines what the student is asked to predict.

Numerical tasks further distinguish functionals from distributions. REAL’s squared-error argument identifies a conditional mean and connects it to population Pearson optimality [41]. DAR instead scores an empirical distribution of numerical rollouts [42]. The mean supports squared-error decisions, while a fuller distribution can support quantile or tail-sensitive uses; Section 4.4 examines how finite sampling affects DAR’s score.

Finally, a probability target can belong to a reader. Linguistic Calibration rewards the log probability assigned to the correct answer by a surrogate reader of generated text, with a normalization penalty and policy KL [117]. LACIE uses correctness and listener acceptance to construct preferences [37]. These methods shape how uncertainty is communicated and used, complementing objectives that directly score a speaker’s numeric report.

### 3.5. Calibrating preference-derived reward signals

RLHF introduces a learned quality model between an outcome and the policy update. Leng et al. show that reward models can prefer a response with a higher appended confidence even when its answer text is held constant, including for rejected responses [6]. Their intervention targets this feedback channel: reward-model judgments should respond to confidence in relation to response quality, rather than treating confidence itself as an unconditional advantage.

**PPO-M, calibrated reward modeling**, fine-tunes an existing reward model using within-response comparisons: for a chosen answer, high confidence is preferred to low confidence, and for a rejected answer the preference is reversed. This confidence–preference matching supplements the quality ranking learned by the original reward model, and PPO [53] then trains the answering policy on the modified scorer. It changes the reward model’s treatment of confidence rather than fitting a correctness posterior.

**PPO-C, calibrated reward calculation**, keeps the reward model fixed and first removes the explicit confidence report before scoring the answer. With  $\hat{r}$  the reward model’s quality score,  $c$  the reported confidence, and  $\Delta$  an exponential moving average of batch quality, the adjusted reward is

$$r = \hat{r} + w(\hat{r} - \Delta)(c - 0.5),$$

with  $w = 2$  in the main recipe [6, Eq. 3]. Above-average judged quality rewards higher confidence; below-average quality rewards lower confidence. For fixed quality and center, the adjustment is linear in  $c$ , so its optimum lies at an endpoint: it rewards aligning confidence with a quality proxy rather than identifying an interior truthful probability.

The methods provide evidence that modifying preference feedback can improve the calibration of the trained policy’s confidence. On Llama3-8B with chain-of-thought prompting on GSM8K, reported ECE is 0.2566 for PPO, 0.1909 for PPO-M, and 0.1546 for PPO-C [6, Table 1]. Their effects on ranking and broader instruction-following vary across settings. The family is therefore best understood as correcting a learnable bias in the reward signal, with probability calibration assessed on the resulting policy. It complements verifier-based proper scoring when preference data are the available supervision.

A second axis concerns the reward signal itself, namely what its values identify and how uncertain they are, rather than how it treats confidence expressed in a response. When preference optimization bypasses an explicit reward model, as DPO does, the implicit reward  $\beta \log[\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x)]$  is identified only up to a prompt-dependent shift, so only within-prompt differences carry Bradley–Terry preference-probability meaning [11, pp. 4–5]. If a reward model instead represents its uncertainty as a distribution  $p(r \mid x, y)$ , KL-regularized RLHF admits the effective reward  $\beta \log \mathbb{E}_p[e^{r/\beta}]$  under a Bayesian control-as-inference reading and  $-\beta \log \mathbb{E}_p[e^{-r/\beta}]$  under a KL-robust adversary. For a

Gaussian posterior, the latter equals  $\mu - \sigma^2 / (2\beta)$  when the adversary’s coefficient is tied to the policy’s, and mean, worst-case, and uncertainty-weighted ensemble rules appear as limits or truncations [46, Eqs. 4, 8, 10–12]. The construction helps only if  $\sigma^2$  tracks epistemic uncertainty [46, p. 4]. UARM instead calibrates the uncertainty estimate itself: it forms reward-model intervals by split conformal prediction and uses their width to down-weight GRPO advantages [45, Eqs. 6–11, 20–22], but its reported evidence concerns reward prediction rather than a trained policy [45, Table 1]. As with PPO-M and PPO-C, the calibration of the resulting policy’s confidence reports must be measured separately.

### 3.6. Intrinsic certainty and entropy as capability signals

A neighboring line uses confidence to improve reasoning itself. INTUITOR scores trajectories using average  $D_{\text{KL}}(U \| p_\theta)$ , where  $U$  is uniform over the token vocabulary, and uses this self-certainty statistic as an intrinsic GRPO reward [47, p. 4], adopting it from a best-of- $N$  method that ranks sampled candidates by the same statistic without training [70, Eqs. 5 and 9]. Other label-free protocols reward the mean negative next-token entropy [48], the negative summed token entropy or the sequence log-likelihood [58], agreement with a sampled majority answer [49], or the frequency of an output’s answer cluster among the group’s samples, which lowers semantic entropy [50, Eqs. 3–6]. Each reports gains on mathematical benchmarks without gold labels in the reward; TTRL, for example, raised Qwen2.5-Math-7B’s AIME 2024 pass@1 from 12.9 to 40.2 while training on the evaluation questions [49, Table 1]. Free-form tasks show a limit: at 7B, EMPO stayed below GRPO with gold-label rewards on MMLU-Pro and GPQA (50.1 versus 54.5 and 28.8 versus 33.8), although the gap narrows at 14B and reverses on GPQA at 3B [50, Table 2]. Positive-Advantage Reweighting instead regulates policy entropy by changing the contribution of positive-advantage tokens during correctness-reward training [118]. These mechanisms exploit or preserve properties of the generation policy, providing alternatives when gold-answer feedback is limited or exploration becomes concentrated.

Their uncertainty objects differ from a reported probability that an answer is correct: self-certainty is an unbounded token-distribution statistic whose level also falls with problem difficulty [70, Fig. 10], and the confidence analysis in the Positive-Advantage Reweighting study compares likelihoods of correct and incorrect responses. Such signals can support capability learning or discrimination, but calibrated risk decisions additionally require a validated mapping to outcomes. The capability benefit is itself conditional. A controlled comparison of self-certainty, token-entropy, and trajectory-entropy rewards finds that general base models mostly gain within the first 20 steps, in the model analyzed largely by eliminating format failures, and that later training often brings declines or collapse (a math-specialized base model kept improving), whereas an instruction-tuned model declines throughout, for example from 0.636 to 0.496 on MATH500 under self-certainty [51, Tables 2, 4–5]. None of these studies reports the calibration of a correctness probability: RENT’s description of its model as “well-calibrated overall” rests on a confidence–accuracy correlation and accuracy gains [48, p. 10], and “overconfidence” in the controlled comparison denotes shallow, premature reasoning rather than a measured calibration gap [51, pp. 9–10]. For EMPO, under independent sampling of  $G$  outputs and ignoring its entropy filter, the expected reward is  $1/G + (1 - 1/G) \sum_j p_j^2$ , where  $p_j$  is the probability of answer cluster  $j$  (our calculation); it therefore rises with the chance that two samples agree, and the authors note that training can converge on a confident but wrong majority cluster [50, p. 5]. Because each reward is maximized by a concentrated policy whether or not that policy is correct, the certainty statistic that served as the reward cannot be assumed to keep its relation to correctness after training.

The boundary between capability signals and calibrated reports suggests a constructive division of labor: intrinsic signals can guide exploration and reasoning, while outcome-based scores or held-out calibration establish the numerical meaning of reports. VL-Calibration builds both into one reward: answer correctness scores a holistic confidence, while a separate visual confidence is regressed onto a batch-standardized intrinsic certainty target, which gives that component relative rather than probabilistic meaning [32, Eqs. 5–11].

Across this section, objectives differ less in their optimizers than in what their optima identify. A strictly proper score makes truthful reporting optimal for whatever outcome law the policy induces. A joint reward also ranks outcome laws by how success is distributed, at an accuracy cost that Section 3.2 bounds at the exact optimum, and its score determines which decision thresholds training emphasizes. Ranking, proxy, distributional, and preference-model objectives identify other quantities altogether, and intrinsic-certainty rewards target no outcome probability. These are statements about population optima. Whether training reaches them depends on the implemented update (Section 4).

## 4. Implemented learning

Section 3 analyzed population objectives. Training optimizes an estimator of such an objective, shaped by baselines, normalization, token weighting, KL regularization, auxiliary gradients, and sampling, and these choices can change what is optimized. This potential mismatch between the population objective and the implemented update is the objective–update gap. Three mechanisms make it concrete: dividing by a group’s standard deviation can re-weight the outcomes on which proper elicitation depends and, under binary rewards, the prompts themselves according to their success probability (Section 4.1); treating a parameter-dependent reward as a stopped-gradient scalar removes one of two gradient routes (Section 4.3); and scoring a finite set of rollouts can move the optimum away from the generating distribution (Section 4.4). Section 4.2 maps where feedback enters the update in current protocols, which determines what a reproducible description must record.

### 4.1. Centering and standardization preserve different information

Consider independently sampled reports for one prompt, scored against a shared external binary outcome. In the setting analyzed by Bereket and Leskovec, subtracting the mean reward of a group of size  $G$  gives

$$\mathbb{E}[\hat{A}_{\text{center}}(x, c)] = \frac{G-1}{G} A(x, c),$$

where  $A(x, c)$  is the report’s population advantage, its expected reward minus the policy’s average expected reward at  $x$ . In expectation, centering therefore only rescales the population advantage; the proportionality is exact under the stated setup, before further transformations [18]. Group-mean centering is also a rescaled leave-one-out advantage:  $r_i - \bar{r} = \frac{G-1}{G}(r_i - \bar{r}_{-i})$ , where  $\bar{r}_{-i}$  averages the other rewards. GRPO’s centering [10, pp. 13–14] and the REINFORCE leave-one-out (RLOO) baseline [52, p. 6] therefore differ only by a constant factor, so a calibration-relevant difference between them must come from other components of the update, such as standardization. The leave-one-out justification requires that the baseline not depend on the scored sample’s own action. A shared external outcome drawn independently of the reports satisfies this requirement, whereas rewards that depend on other rollouts, such as ensemble or set scores, require a separate calculation.

Dividing by the group’s random standard deviation has a different effect. Reports within the group share a realized outcome, but their reward spread can differ according to whether that outcome is zero or one. Bereket and Leskovec’s large-group approximation shows that standardization assigns outcome-specific inverse scales that depend on the current reporting policy, so the update can change the outcome weighting on which proper elicitation depends. The approximation is not exact for finite groups and does not characterize every proper score.

For binary correctness rewards, where each rollout has its own outcome, standardization also re-weights prompts. Using population rather than group statistics, with success probability  $p$  under the sampling policy, it gives a success the advantage  $\sqrt{(1-p)/p}$  and a failure  $-\sqrt{p/(1-p)}$  [54, Eq. 3]. In an idealized analysis with KL regularization to a fixed reference and no clipping, each prompt’s success log-odds are shifted above their reference value by  $1/\beta$  under mean-only centering but by  $(p(1-p) + \varepsilon)^{-1/2}/\beta$  under standardization, where  $p$  is the previous iterate’s success probability and a small constant  $\varepsilon > 0$  stabilizes the division; whenever the reference success probability lies strictly between zero and one, every fixed point exceeds it [54, Theorems 2–3, Sec. 5]. The normalization choice is therefore itself a preference over success-rate profiles, the quantity that Section 3.2 separates from reporting.

In the stochastic-outcome setting, Bereket and Leskovec’s CRISPR experiment reports ECE 0.292 with GRPO and 0.036 without the standard-deviation division [18, Table 1]. Their public language-model launchers support a focused comparison by switching this normalization flag while retaining the other main settings; Appendix C.3 separates that evidence from a discrepancy in the released synthetic notebook’s clipping path.

Other studies retain standardization and report calibration improvements, including teacher-probability forecasting [116], separately credited capability and answer training [59], and decision-token training with an auxiliary direct loss [7]. Their targets and gradient paths differ from those in the stochastic-outcome setting. Conversely, RLCR combines its mean-centered update with a joint correctness/report reward, format incentives, and batch token normalization [30], so its gains cannot be attributed to centering alone. Uncertainty-aware variants change the scale again: UARM multiplies each advantage by a sample-specific reliability weight derived from reward-model intervals, so the group’s

advantages need not sum to zero [45, Eqs. 20–22]. Such differences motivate controlled estimator experiments with a common target, reward, and sampling budget.

## 4.2. Where feedback enters the update

Current protocols differ mainly in where feedback enters the update, and each placement changes what a reported improvement can mean.

**Segmented and staged learning.** DCPO and CoCA compute separate advantages for answer and confidence segments but construct targets differently: DCPO combines per-response correctness with a group success estimate [100], whereas CoCA predicts the current policy’s success before answering [59]. Decision-token RL instead zeroes the final token’s policy advantage and adds a local classification loss [7], and CURE separates calibration preferences from a later factuality stage that masks confidence-related tokens [104]. These methods allocate feedback to particular token losses within shared parameters, so their effect should be measured by both fixed-answer and native-policy evaluation.

Fixing the answer as an input takes a different route, as in Rewarding Doubt (Section 3.1) [29]. Entropy-target RL trains a separate uncertainty adapter against a calibrated sampling-entropy target while keeping answer generation fixed [119]. Because it amortizes an offline sampling-and-calibration pipeline, its target is question-level error under that pipeline, even though the reporter also sees a particular answer. Such designs help isolate assessment from answer selection and expose the data and sampling costs of confidence supervision.

**Sequence and group credit.** Group sequence policy optimization (GSPO) replaces token-level importance ratios with a length-normalized sequence ratio and clips whole responses while retaining group standardization [55, Eqs. 5–7]. C<sup>2</sup>GSPO likewise optimizes a sequence-level policy objective and couples it to binary cross-entropy on the geometric mean of token probabilities, which is the length-normalized sequence likelihood that also defines GSPO’s ratio. Its advantage divides the centered reward by one minus the old sequence score, introducing confidence-dependent weighting without group-standard-deviation division [56]. UCPO instead separates correct, incorrect, and abstaining outputs, adapts the abstention advantage to group composition, and filters groups lacking the required outcome types [110]. C<sup>2</sup>GSPO trains an internal likelihood score as a correctness proxy, whereas UCPO trains discrete uncertain behavior: their update rules respond to different objects despite a shared vocabulary of confidence calibration.

**Paired and process feedback.** RAC distributes confidence-ranking and clean-versus-corrupted evidence penalties across paired multimodal rollouts [107]. RLCM fits a probe to the success rate of answers forced from intermediate reasoning prefixes, training the probe with a supervised loss and the policy with final correctness plus an aggregate confidence-margin reward [77]. RefGRPO adds a decaying bonus for a binary self-assessment made after environment feedback [120]. These methods extend credit beyond a final numerical report—to evidence perturbations, reasoning prefixes, and post-action reflection—and their relevant evaluations are, respectively, directional behavior, prefix success prediction, and reflection accuracy.

**Feedback and target adaptation.** Some protocols change the feedback or the target rather than the credit. PPO-M and PPO-C modify how preference feedback treats expressed confidence, as Section 3.5 describes [6]. Success estimates in CoCA and CALIBER evolve with the answering policy [59], [60]. Refreshing such targets keeps them aligned with current behavior but adds sampling noise, and it creates dependence when the same rollouts supply both targets and advantages. CALIBER’s warmup illustrates a further issue: a statistically meaningful group target can have too little within-group variation to compete with other rewards.

Finally, reward magnitude interacts with KL penalties, format bonuses, and clipping: a positive affine transformation preserves a fixed report optimum but can change its weight against an unchanged auxiliary term. The KL term is itself a design choice with an anchor, a placement, and a coefficient. The KL-penalty variant of PPO penalizes divergence from the previous policy in the surrogate loss with an adaptive coefficient [53, Eq. 8]; PPO-based RLHF places a per-token penalty toward a reference model in the reward [10, Eq. 2]; and GRPO adds an estimated divergence from the reference model directly to the loss, with the reference reset to the current policy at each outer iteration of its iterative variant [10, Eqs. 3–4, Algorithm 1]. The anchor can change the limit of training, not only its path: in the idealized binary-reward dynamics of Section 4.1, anchoring the KL term only to the previous iterate drives every prompt that the reference policy sometimes, but not always, solves toward certain success, whereas an anchor that includes the reference model can leave an interior fixed point [54, Theorem 5, pp. 8–9]. Token aggregation adds another weighting choice. DAPO divides summed token losses by the observed total number of response tokens [121], whereas Dr. GRPO replaces response-length and question-level reward-standard-deviation normalization with fixed choices [122], and calibration protocols

add their own active-token reductions [7], [30]. A reproducible description therefore includes the baseline, scale, mask, auxiliary losses, reduction rule, importance-ratio unit and clipping range, and the anchor, placement, and coefficient of any KL term, not only the optimizer’s name. A component implied by an algorithm’s name may also be inactive: the main run in the original GRPO paper made a single policy update after each sampling stage [10, p. 15], so every gradient is evaluated at an importance ratio of one and the clipped surrogate never binds.

### 4.3. Differentiable rewards require a direct prediction gradient

Policy dependence has a second, analytically distinct meaning. A reward may depend differentiably on the current parameters even after a trajectory has been sampled. If

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [r_\theta(\tau)],$$

then, under the usual conditions for differentiating an expectation,

$$\nabla_\theta J = \mathbb{E}[r_\theta(\tau) \nabla_\theta \log \pi_\theta(\tau) + \nabla_\theta r_\theta(\tau)].$$

The first term changes the trajectory distribution; the second changes the evaluated prediction on that trajectory. Treating every reward as an external stopped-gradient scalar omits the latter. Periodically refreshed success targets (Section 4.2) create a different dependence: the target moves between updates, whereas a differentiable reward also responds within a single gradient step.

REAL makes this distinction explicit [41]. Its numerical prediction is a probability-weighted sum of score-token values, and its reward combines negative squared error with an auxiliary target-token log-likelihood, so the direct gradient can flow through both reward components; a parsed numerical string, by contrast, is scored only through its discrete generation action. The practical algorithm in REAL’s paper nevertheless departs from the exact identity: it standardizes and clips leave-one-out advantages and weights the direct prediction gradient by  $\beta$ , a weight distinct from reference-policy KL regularization; even  $\beta = 1$  retains the transformations of the trajectory term. Its ablation initialized by supervised fine-tuning (SFT) reports Pearson  $r = 0.679$  and root-mean-square error (RMSE) 0.968 at  $\beta = 0.01$ , versus  $r = 0.667$  and RMSE 0.935 at  $\beta = 1$  [41, Table 9], so correlation and numerical error favor different settings and evaluation should match the desired regression statistic.

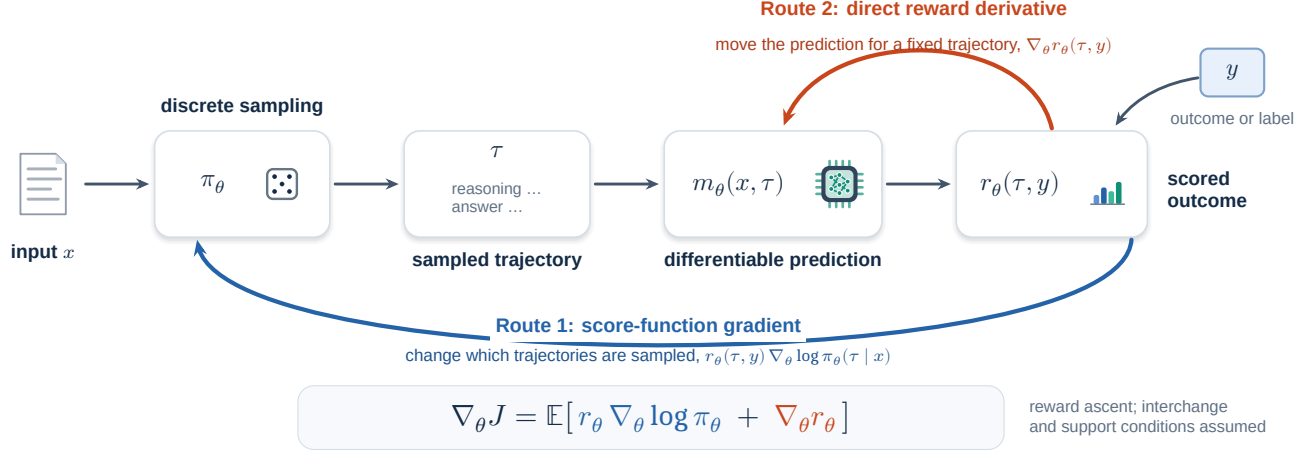
The public implementation shows why the active routes and transformations must be reported for each code version. At the audited commit, REAL’s example launchers set the direct-gradient and dedicated KL coefficients to zero, and the specialized actor normalizes negative mean-squared-error (MSE) and target-probability rewards separately rather than using the paper’s combined log-likelihood reward. The actor’s policy-gradient loss (GPG in the code) uses an unclipped score-function term, although the advantages themselves are clipped. These observations separate the paper’s formulation from a public execution path but do not reveal which settings produced the published checkpoints or ablation runs (Appendix C.4).

The two routes also sort existing protocols (Figure 5). A numeral parsed after generation, as in RLCR, receives only the score-function route [30]. ConfTuner uses only the direct route [57]: it assigns probabilities  $q_i$  to the tokens that denote the values  $i/N$ , through a softmax restricted to candidate confidence tokens, and minimizes

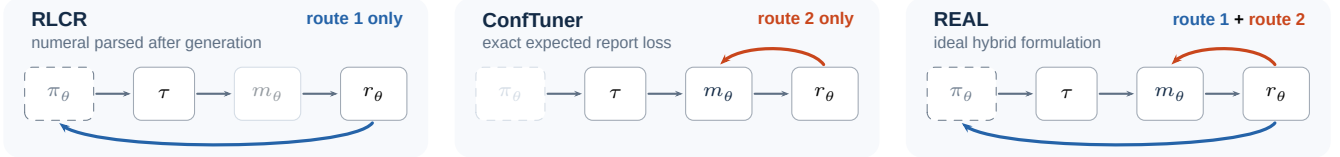
$$\ell(q, z) = \sum_{i=0}^N q_i (z - i/N)^2$$

by differentiating this expectation exactly. Summing over the finite reporting support removes the Monte Carlo variance of sampling the confidence action; label noise, the reporting grid, and distribution shift remain sources of error. For such an enumerable support, this direct derivative equals the exact expected score-function gradient for the report token, so relative to a sampled numeral ConfTuner removes sampling variance rather than a gradient term that RLCR omits. Entropy minimization gives a label-free instance of the same split: EM-FT takes only the direct route by differentiating next-token entropies along sampled outputs, whereas the token-level EM-RL variant takes only the score-function route by using their negative sum as the reward [58, Table 1]. REAL’s ideal hybrid formulation, by contrast, makes both routes available [41].

## (a) A reward that depends on parameters reaches them by two routes



## (b) Protocols use one route or both



**Figure 5.** Two routes from reward to learning. (a) A sampled trajectory  $\tau \sim \pi_{\theta}$  conditions a differentiable prediction  $m_{\theta}(x, \tau)$ , scored against an outcome  $y$ . Under the usual interchange and support conditions, the expectation gradient contains a score-function term that changes which trajectories are sampled (route 1) and a direct reward derivative that changes the prediction on a sampled trajectory (route 2); stopping the reward gradient removes route 2. (b) A numeral parsed after generation (e.g., RLCR [30]) uses route 1 only; a fixed answer with an exactly differentiated report loss (e.g., ConfTuner [57]) uses route 2 only; REAL motivates the ideal hybrid identity, which uses both [41]. Entropy minimization splits the same way: EM-FT takes route 2 and token-level EM-RL route 1 [58]. The diagram shows the ideal identity; it does not assert that every released run used both routes (Section 4.3).

#### 4.4. Finite ensembles change the scoring problem

Scoring several numerical rollouts changes the scoring problem even before optimization: a proper score for the issued empirical distribution need not be a fair score for the distribution that generated its members. “Fair” here has the ensemble-forecast meaning formalized by Ferro, rather than a demographic interpretation [27].

For independent  $X_1, \dots, X_K \sim F$ , a fixed observation  $y$ , and finite first moments, the usual empirical CRPS is

$$C_K = \frac{1}{K} \sum_i |X_i - y| - \frac{1}{2K^2} \sum_{i,j} |X_i - X_j|.$$

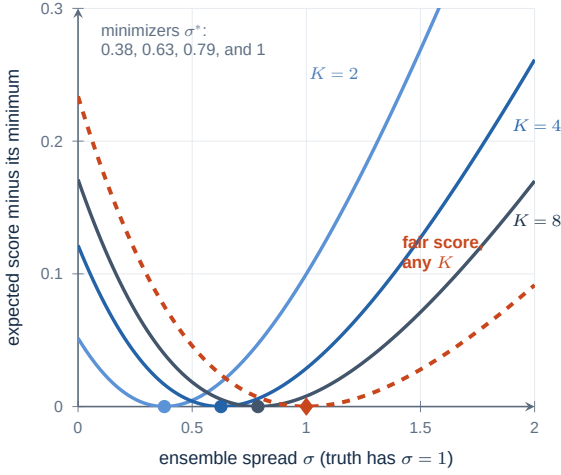
Its expectation satisfies

$$\mathbb{E}[C_K] = \text{CRPS}(F, y) + \frac{\mathbb{E}|X - X'|}{2K}.$$

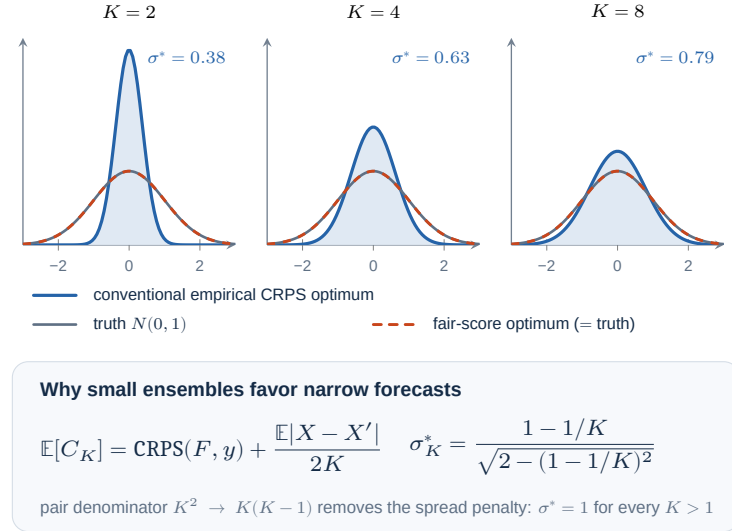
The extra term penalizes dispersion of the generating distribution. Replacing  $K^2$  in the pairwise denominator by  $K(K-1)$ , for  $K > 1$ , removes this term under independent sampling [27]. This classical adjustment shows that rollout count can change the expected training criterion, not only the variance of its estimate. In the Gaussian example of Appendix A.5, the conventional optimum has only about 0.38, 0.63, and 0.79 of the true spread for  $K = 2, 4, 8$ , whereas the fair score recovers the true spread for every  $K > 1$  (Figure 6). Reward-model ensembles raise an analogous issue in the feedback channel: with KL coefficient  $\beta$  and  $M$  reward-model samples  $R_1, \dots, R_M$ , the plug-in entropic reward  $-\beta \log[M^{-1} \sum_m e^{-R_m/\beta}]$  is consistent but biased toward less pessimism at finite  $M$  [46, Eq. 14].

DAR uses the empirical CRPS with the  $K^2$  denominator and assigns rollout  $i$  a leave-one-out contribution  $r_i = C_{K-1,-i} - C_K$  [42]. Its public reward manager implements these denominators, so the contribution retains the empirical-score denominators rather than implementing the fair-score correction. The resulting scalar rewards then undergo GRPO group centering and standardization; because each contribution depends on the other

(a) Expected score against forecast spread



(b) Optimal forecasts under each score



**Figure 6.** Classical finite-ensemble scoring example [27], computed from the analytic formulas in Appendix A.5. Truth is  $N(0, 1)$ ; the forecast generator is  $N(0, \sigma^2)$  with  $K$  iid members. (a) Expected conventional empirical CRPS, minus its minimum, as a function of  $\sigma$ : the minimizers  $\sigma_K^* = a/\sqrt{2 - a^2}$ , where  $a = 1 - 1/K$ , are approximately 0.38, 0.63, and 0.79 for  $K = 2, 4, 8$ , whereas the fair score, which replaces the pairwise denominator  $K^2$  by  $K(K-1)$ , is minimized at  $\sigma^* = 1$  for every  $K > 1$ . (b) The corresponding optimal densities against the truth. All curves are exact Gaussian calculations, not LLM measurements. The iid correction alone does not characterize DAR’s filtered, normalized update [42].

rollouts, this is the case that Section 4.1 flags as requiring a separate baseline calculation, and the effective  $K$  excludes invalid numerical parses, with fallback and penalty rules for exceptional groups. Defining a set score, assigning credit, and transforming the update are therefore three separate operations.

DAR’s reported CRPS improves over its MSE comparator, while SFT attains lower CRPS than DAR on the KBSS code task [42, Table 5]. The mixed result is a reason to report the intended distributional quantity and the actual estimator separately. Section 7.2 describes the controlled comparison that would attribute such effects; dependent generations, deterministic quantiles, and filtering fall outside the independent-draw analysis above.

The three mechanisms of the objective–update gap share a lesson: the name of the optimizer does not identify the objective being optimized. A calibration gain from PPO- or GRPO-style training can come from the population score, from how the estimator re-weights outcomes or prompts, from the anchor of its KL regularization, from which gradient routes are retained, or from how finite rollouts are scored. A method description therefore needs these choices alongside the reward. An objective that is optimized as intended still delivers value only through the decision that consumes its output, which Section 5 examines.

## 5. Decisions and interfaces

The decision that consumes a score determines which of the score’s statistical properties is useful; the ranking–decision gap arises from this dependence. A decision can consume an ordering, as when an auditor reviews a fixed quota; a numerical magnitude, as when a cost-derived threshold must remain valid as costs change; or the action itself, as when a service answers or abstains under one stable cost. Sections 5.1–5.3 match these uses to training targets, recalling from Section 3.3 that a joint score already weights some thresholds more than others. Sections 5.4 and 5.5 add two further questions—when the information behind a score becomes available and which object the decision reads—and Section 5.6 examines an industrial interface. In each case, the interface’s shape alone does not establish what training has learned.

## 5.1. Fixed costs and reusable probabilities

Suppose accepting an answer pays  $b > 0$  when correct and  $-\ell$  otherwise, while rejection pays zero. For a score  $C$  calibrated for correctness  $Z$ ,

$$\mathbb{E}[u(\text{accept}, Z) \mid C = c] = (b + \ell)c - \ell, \quad \tau = \frac{\ell}{b + \ell}.$$

Acceptance at  $c \geq \tau$  is therefore optimal among policies observing only that score, with either action optimal at equality. SimpleQA proposes this payoff structure as an evaluation metric: a correct answer earns 1, not attempting earns 0, and an incorrect answer costs  $\ell$ , so that with  $\ell = 9$  the score is positive only if at least 90% of attempted answers are correct [83, p. 6]. In the notation above,  $b = 1$  and  $\tau = \ell/(1 + \ell)$ . The optimality of the threshold rule assumes fixed payoffs and calibration under the deployment joint distribution, and it holds only among policies that observe the score alone. A policy with additional input information may do better; arbitrary contextual costs require that information to enter the decision. Appendix A.4 bounds the regret of thresholding an imperfect report; neither result is a guarantee about the trained generator (Section 3).

Probabilities are particularly useful when the cost ratio changes after training. The same report can support different thresholds for different services, provided its event and population remain appropriate. A score with good ordering but unknown probability semantics can support a threshold fitted on validation data; it cannot directly inherit a cost-derived numerical threshold. Behaviorally calibrated RL implements the reporting approach explicitly: its verbalized-confidence variant generates a report and applies the answering threshold externally [44]. Stating the payoffs to the model does not by itself implement the rule. When the payoffs made refusal optimal by construction, three frontier models still refused only 6–42% of GPQA questions; eliciting confidence and the answer-or-refuse decision in separate calls raised the six-model average normalized reward on MMLU at payoffs  $(1, -8)$  to 0.012, against  $-0.389$  when the two steps shared one prompt and  $-0.361$  when the prompt only stated the payoffs [69, Fig. 3a, Table 1]. Likewise, in the I-CALM study, raising the announced wrong-answer penalty from  $-1$  to  $-100$  did not lower one model’s PopQA coverage (0.694 with the larger penalty; 0.689 at  $-1$  with the payoffs alone), although by our calculation the implied expected-utility threshold rises from 0.70 to about 0.99 [68, Tables 4 and 10].

When the objective instead fixes the fraction of answers to accept, only ordering and tie handling determine the selected set. When all inputs share the same payoffs, ordering by conditional success is the appropriate target, and a selective-prediction objective is a useful comparator. RLSR’s HotpotQA-trained in-domain comparison shows that selective performance and calibration can move in opposite directions: relative to RLCR, the area under the risk–coverage curve (AURC) improves from 0.51 to 0.44 and accuracy at 10% coverage from 53.5% to 66.0%, while ECE worsens from 0.05 to 0.24 [40, Table 1]. These results motivate fixing and evaluating the operating objective before choosing a training loss. Safety detection illustrates the same need at the tail of the score distribution: Reasoning’s Razor reports settings where reasoning improves aggregate accuracy but reduces recall at a strict false-positive-rate threshold [123]. Such a result motivates reporting the relevant operating region alongside average calibration and discrimination.

## 5.2. The same ordering can support different transported thresholds

A threshold is *transported* when it is derived from costs and applied to a score without being refitted on labeled data from the deployment. Consider four equally likely contexts with correctness probabilities  $(0.1, 0.4, 0.6, 0.9)$ . Compare truthful reports with their monotone sharpening  $(0.001, 0.1, 0.9, 0.999)$  and the constant base-rate report 0.5. Acceptance pays  $+1$  for correctness and  $-4$  for error, so the probability threshold is 0.8. The results below are exact population calculations, rounded for display; Appendix A.6 gives the full numerical table.

Sharpening preserves ranking but moves the 0.6-success context across the transported 0.8 threshold and turns positive utility into negative utility (Figure 7). The constant report is calibrated, yet supplies no resolution between the contexts; rejecting everything is optimal given that report alone. For the metric comparison in Appendix A.6, reliability means  $\mathbb{E}[\mathbb{E}[Z \mid C] - C]$ , rather than the squared reliability term in the Brier decomposition [90].

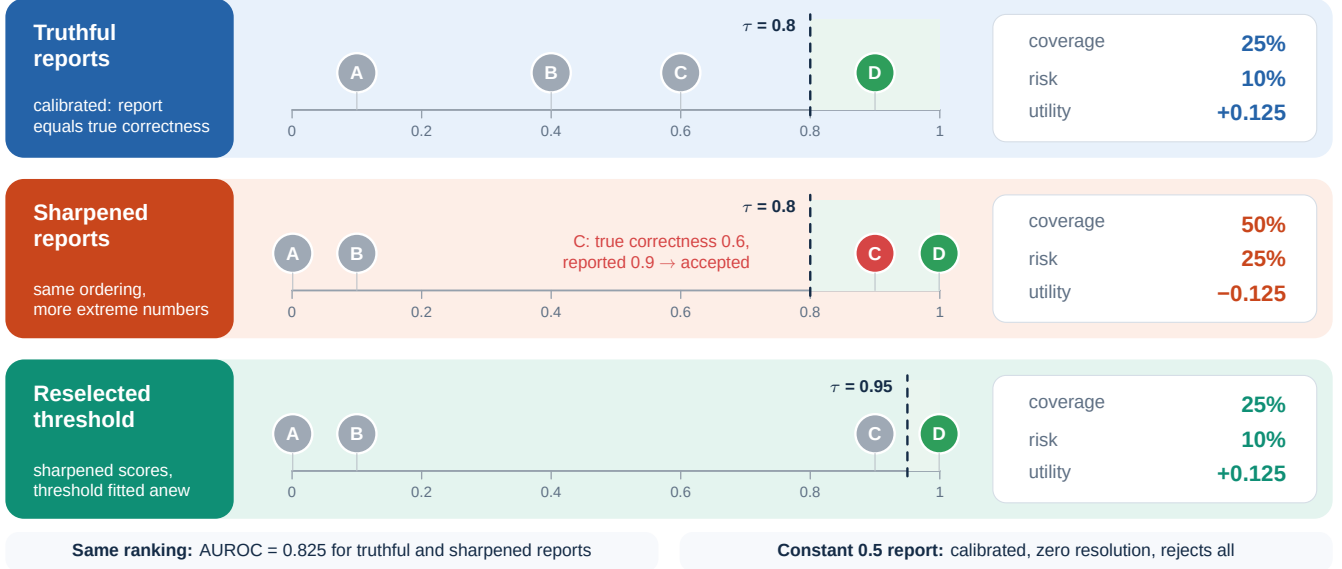
The truthful and sharpened reporters nevertheless have identical attainable threshold-based ROC and risk–coverage frontiers (Appendix A.6; Figure 7 illustrates the corresponding accepted sets). A sharpened-score threshold in  $(0.9, 0.999]$  recovers the truthful accepted set. Independent validation can select a replacement threshold, although finite samples do not guarantee recovery. Trust or Escalate, a cascade of LLM judges that escalates or abstains when a

### Same ordering, different decisions at a transported threshold

Four equally likely contexts (true correctness):

A 0.1    B 0.4    C 0.6    D 0.9

Payoffs  $+1/-4/0$  for a correct answer, an error, and an abstention give the cost-derived threshold  $\tau = 4/(1+4) = 0.8$ .



**Figure 7.** Exact four-context decision example with equal weights and correctness rates (0.1, 0.4, 0.6, 0.9). Each lane places the reports on  $[0, 1]$ , shades the accepted region, and colors accepted contexts by the sign of their expected utility. At the cost-derived threshold  $\tau = 0.8$ , truthful reports accept only context D, giving coverage 25%, risk 10%, and expected utility  $+0.125$  per input under payoffs  $+1/-4/0$ . Monotone sharpening to  $(0.001, 0.1, 0.9, 0.999)$  preserves ranking but also accepts context C, giving coverage 50%, risk 25%, and utility  $-0.125$ . Reselecting the sharpened-score threshold to 0.95 recovers the original accepted set and utility; this is threshold adaptation, not recalibration. Risk is the error rate among accepted answers; utility is averaged over all inputs. Both reporters have AUROC 0.825. The constant 0.5 report is calibrated with zero resolution and rejects all inputs at 0.8. Finite validation data determine how reliably an alternative threshold can be selected. Full metrics appear in Appendix A.6.

judge’s confidence is low (Section 5.5), compares the two routes: a threshold set equal to the target agreement level  $1 - \alpha$ , which presumes calibrated confidence, met a 0.9 target in 42.0% of 1,000 random splits with GPT-4 as the only judge and in none when applied to the cascade, whereas thresholds chosen on 500 labeled examples with an exact binomial bound met it in 90.8% of splits, at 55.7% coverage [71, Table 2]. The four-context example thus establishes the value of probability semantics for transporting thresholds, without ruling out thresholds learned from labeled validation data.

The invariance of the attainable frontiers relies on a strictly increasing transformation of the same scalar score, fixed answers and outcomes, and consistent treatment of ties (Section 2.3); it does not extend automatically to maps that merge distinct scores into ties, to a changed answer policy, or to general multiclass temperature scaling followed by maximum-confidence readout. Comparing cost-derived with independently selected thresholds therefore reveals whether an improvement comes from numerical probability meaning or from the attainable selection frontier, an evaluation design that Section 6.3 makes part of the contract.

### 5.3. Learning the action when the operating objective is fixed

For a stable answer-or-abstain service, learning the desired action can be sufficient. TruthRL assigns  $+1$ ,  $0$ , and  $-1$  to correct answers, abstentions, and incorrect answers [43]. For a fixed candidate with success probability  $p$ , answering has expected reward  $2p - 1$ , giving the ideal threshold 0.5. The training feedback identifies which action is desirable without uniquely identifying a numerical probability. This makes direct-action RL a relevant baseline when deployment uses that same utility and retraining can accommodate changes.

Direct-action RL should be judged by both the errors it removes and the correct answers it retains. On CRAG with retrieval, TruthRL reports accuracy 56.6% and hallucination 19.4% for Llama3.1-8B-Instruct, versus 60.3% and 39.5% for binary-reward RL [43, Table 1]. These rates use all questions as their denominator. The result shows a useful change

in operating behavior, at the cost of fewer correct answers than that comparator; it does not measure calibration of an unreported probability. The converse dissociation, in which the action changes while the stated probability barely moves, has been reported for agents. On forty questions whose outcomes cannot be known in advance, escalating an evidence panel raised the rate at which twelve frontier models committed to an answer from 6.5% to 54.0%, while the mean distance of their stated probabilities from 50% moved only from 4.7 to 7.7 points; the committed probabilities also scored worse than a base-rate forecast [111, pp. 9, 11]. Auditing the probability field would not reveal this change in action.

Action learning also introduces an optimization question. The abstention-collapse analysis studies a shared, bounded answer gate under error-penalized rewards [109]: both the task gradient and the policy-to-reference KL gradient contain a saturation factor, so a predominantly low-success training population can suppress answering even on easier inputs. Its controlled gate and constructed QA-mixture experiments motivate comparing direct abstention with mandatory-answer probability training followed by an external threshold. Because the analysis relies on the shared parameterization and stated training conditions, it complements TruthRL’s operating-point evidence rather than predicting collapse for every answer-or-abstain policy. An answer-or-abstain gate can also be learned through supervision: fine-tuning a 3B model on 540 synthetic cases removed commitment on the unknowable questions, but the learned gate weakened under an output format without a reasoning slot [111, Table 5, pp. 18–19], so evaluations of a learned answer-or-abstain policy should include changes to the output interface.

Evaluation incentives also shape which strategy is rewarded. *Why Language Models Hallucinate* explains how binary correct/incorrect grading can reward guessing relative to abstention, and motivates grading rules with an explicit confidence threshold [124]. A benchmark’s grading rule thus determines which answer-or-abstain behavior it rewards, and hence which behavior training aimed at that benchmark will favor. Announcing the payoff to the model is not a substitute, because stated payoffs need not change its behavior (Section 5.1). Whether RL or a simpler comparator is the right tool for learning answer-or-abstain behavior is the subject of Section 6.1.

## 5.4. Adaptive computation depends on when the score is available

Deciding whether to accept an existing answer differs from deciding whether another generation is worth its cost. The  $P(\text{IK})/P(\text{True})$  distinction of Section 2.2 applies directly [65]: capability confidence concerns the expected success of a specified stochastic decoder on a query, whereas response confidence concerns an already sampled answer [72]. A pre-generation router needs the former target for each available model, whereas a post-generation acceptance gate needs evidence about the actual candidate. A score acquired after execution or external verification conditions on further information and should be assessed at that stage. The rest of this subsection examines gates within one trace, rules across parallel traces, and budgets set before or between generations, and then the assumptions on which budget allocation rests.

Estimates that become available during generation allow early exit, which places an acceptance gate inside a single trace. DEER induces a trial answer at reasoning transitions and stops reasoning when the geometric mean of the trial answer’s maximum token probabilities exceeds 0.95 [75, Eq. 4, pp. 4–5]; it uses no training and reports no calibration metric, and across three models its outputs are 19.1–42.9% shorter than without early exit [75, Table 1]. A probe of reasoning-model hidden states instead predicts whether an intermediate answer is correct: its in-distribution ECE was 0.01–0.14, rising to 0.25–0.31 when probes for one model were transferred to AIME [112, Tables 1–2]. Thresholding the probe at 0.85 yielded 88.2% accuracy on MATH, against 88.6% without early exit, with about 24% fewer tokens [112, Fig. 5]. Both thresholds were examined on the evaluation benchmarks themselves, so these results describe operating points that were neither transported from costs (Section 5.2) nor validated on held-out data. Stopping also determines which answer is delivered, so the relevant comparison is between the current candidate’s success probability and that of the answer continued reasoning would deliver, net of the cost of continuing; both gates estimate only the former.

Parallel sampling adds decisions across traces. DeepConf filters, weights, and terminates traces using windowed averages of negative top- $k$  token log-probabilities, again without training [74, Eqs. 2–7]. DeepConf proposes these local measures as alternatives to a trace-level average of its own top- $k$  token confidence, which it calls self-certainty after the selection study [70], [74, pp. 2–3]. A trace stops when the confidence of its current window falls below a percentile of the warm-up traces’ confidences for the same problem, and sampling stops once the confidence-weighted vote share reaches 0.95 [74, Algorithm 2; p. 24, Table 12]. These rules need orderings within a problem and a confidence-weighted agreement statistic, but not probability semantics. At a 512-trace budget, DeepConf’s aggressive variant used 43.8–

84.7% fewer tokens than majority voting, with accuracy changes from  $-0.9$  to  $+7.8$  points [74, Table 2], whereas the corresponding offline filter lowered accuracy relative to majority voting in 8 of 20 settings [74, App. C]. RL has also been used to learn these signals: RLCM combines a supervised early-exit probe with a GRPO reward that includes a confidence margin [77], and C3RL trains the 1–10 confidence reports that its adaptive sampler sums to decide when to stop [76]. The training-free and supervised rules above are the comparators such controllers must beat (Section 6.1).

Under independent trials with the same query-level success probability  $p$ , the probability of at least one success after  $k$  samples and the marginal gain from another sample are

$$P(\text{success by } k) = 1 - (1 - p)^k, \quad \Delta_k = p(1 - p)^k.$$

Probability magnitudes, not only orderings, enter both quantities and hence any allocation rule built on them [72]. Instance-adaptive scaling inverts the first: it takes the smallest  $k$  with  $1 - (1 - p)^k \geq \gamma$  for a coverage target  $\gamma$ , capped at a maximum budget, and substitutes for  $p$  a lower quantile of the success probability predicted by a process reward model fine-tuned on one generator’s rollout frequencies [73, pp. 5–7, Prop. 1]. Calibration changed the allocation itself. The off-the-shelf reward model gave all six generators the same 1.6% of a 64-sample budget on MATH500, with accuracy close to single-sample accuracy, whereas calibrated estimates gave generator-specific budgets of 23–64%, with accuracy 0.7–2.2 points below fixed 64-sample best-of- $N$  [73, Table 2].

The same magnitudes can serve different allocation objectives. With a value assigned to success and a sampling cost, they support allocating computation where the expected gain is largest. A coverage target instead assigns the largest budgets to the smallest  $p$ : instance-adaptive scaling spent the full 64 samples on a 1B model’s AIME questions at 0.0167 accuracy [73, Table 2], even though the marginal gain  $p(1 - p)^k$  stays small there at every sample count. The same value–cost principle suggests comparing model-specific expected benefits and costs for routing, rather than routing on the entropy of a label distribution whose event is not the candidate model’s success. When an agent decides whether to retrieve, for example, supplying recalibrated success probabilities in its prompt raised its agreement with the cost-optimal choice from 56–72% to 89–99% across cost settings [78, Fig. 3].

These allocation rules rest on assumptions that delimit their use. The decoder must remain the one whose capability was estimated; correlated or adaptively changed samples need a different gain model. If uncertainty about  $p$  is represented by a predictive distribution, averaging  $1 - (1 - p)^k$  over it gives the gain model for samples that are independent given  $p$ ; because this function is concave in  $p$ , the uncertainty cannot lower the required budget relative to plugging in the mean. Distributional reward-model calibration takes this average over quantiles from a conditional optimal-transport map [87, Eq. 5], and its held-out calibration gains depended on the underlying reward model [87, Tables 1 and 3].

Pass@ $k$  further assumes that a successful candidate can be recognized; otherwise it measures an oracle event, not the performance of the deployed selector. The instance-adaptive guarantee states this assumption explicitly [73, Prop. 1], and with a weaker reward model as selector, fixed best-of-64 fell below single-sample accuracy for three of six generators on MATH500 (e.g., 0.628 versus 0.756 for R1-Distill-Qwen-7B) [73, Table 7]. The capability-prediction study likewise scores repeated-sample success rather than a deployed selector: it reports Qwen3-8B pass@64 simulation MSE 0.0205 for its capability probe versus 0.1486 for an oracle response-calibration comparator on MATH500 [72, Table 3]. This is a useful target-matching result, not evidence that capability confidence should replace response confidence when accepting an answer. Routing and allocation evaluations should record the information available before each decision and charge for obtaining it. Reported budgets differ in what they charge: DeepConf counts its warm-up traces [74], whereas instance-adaptive scaling does not count the per-generator calibration rollouts [73, Table 2; App. C.1–C.2]. Table 4 summarizes what each decision needs from an estimate.

## 5.5. Richer interfaces must expose the object the decision uses

Direct decision interfaces have several learning routes. SALSA maps classes to existing label tokens and fits their renormalized logits by supervised cross-entropy [113]; GLiClass uses encoder-based text–label representations, including independent sigmoid scores for multilabel tasks, with a PPO stage in its broader pipeline [81]; and BinaryPPO frames binary label prediction as policy optimization [125]. Their classification results call for separate proper-score and reliability evaluation when scores will be consumed as probabilities. In guard models, label-token readouts are already operational, and post-hoc calibration depends on the calibration-set distribution [114].

**Table 4. What each adaptive-computation decision needs from an uncertainty estimate.** Fixed stopping thresholds need validation on the deployment population; selection and relative termination need only orderings within a problem; budget allocation needs calibrated magnitudes for the named decoder. Unless marked RL, the estimators are training-free or fitted with the generator fixed; they serve as comparators for RL controllers of the same decisions (Section 6.1).

| Decision                                                                | Event the estimate must address                                                                       | Property the decision uses                                                                                           | Reviewed evidence     |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Within one trace: a fixed threshold gates acceptance</b>             |                                                                                                       |                                                                                                                      |                       |
| Stop reasoning and accept the current answer                            | Correctness of the current candidate; the alternative is the answer continued reasoning would deliver | A threshold validated on the deployment population; calibration only to derive it from costs or to transport it      | [75], [112]; RL: [77] |
| <b>Across parallel traces: orderings within a problem often suffice</b> |                                                                                                       |                                                                                                                      |                       |
| Terminate one of several traces                                         | Trace quality relative to the same problem’s warm-up traces                                           | Ordering within the problem                                                                                          | [74]                  |
| Filter traces or select an answer                                       | Correctness of each candidate’s answer                                                                | Ordering within the problem; weighted voting also uses score magnitudes, without probability meaning                 | [73], [74]            |
| Stop sampling further traces                                            | Correctness of the current majority answer, left implicit                                             | A threshold on a vote-share or stopping statistic; its probability meaning is not established                        | [74]; RL: [76]        |
| <b>Before or between generations: magnitudes enter a budget rule</b>    |                                                                                                       |                                                                                                                      |                       |
| Buy samples before generation                                           | Success of one sample from a named decoder, given the query                                           | Calibrated magnitude for that decoder; a selector that recognizes success; a stated coverage or value–cost objective | [72], [73], [87]      |
| Expand or prune search prefixes                                         | Success of continuing a prefix under the same decoder                                                 | Calibrated magnitude for each prefix                                                                                 | [73]                  |
| Retrieve or test before committing                                      | Success of committing now versus after acquiring information                                          | Calibrated magnitudes combined with an explicit cost or discount rule                                                | [78]; prompted and RL |

An output vector is valuable when downstream costs distinguish among several labels, but its training target still requires inspection. Verbalized Probability Distribution emits candidate probabilities and selects the maximum, yet inherits a scalar correctness-plus-Brier reward for that selected answer, plus a penalty for an invalid probability sum [80]. It nonetheless reports improved full-vector Brier scores after RLCR training, from 0.387 to 0.347 on MedQA for Qwen3-4B-Instruct [80, Table 6]: scalar supervision can improve the richer output empirically without identifying every nonselected probability through the reward. A downstream cost matrix that uses the full vector therefore deserves full-vector evaluation.

Inference alone can also preserve useful information: aggregating a judge’s rating-token distribution instead of taking its modal score improves the evaluated judgments in many settings, which makes it a comparator for learning a new judgment distribution [115]. A frozen judge’s confidence can likewise drive a decision without retraining. Trust or Escalate accepts a pairwise judgment when an ensemble of simulated annotators is confident enough and otherwise escalates to a stronger judge; it chooses thresholds on labeled calibration data so that agreement with human majority labels among accepted items is at least  $1 - \alpha$  with probability at least  $1 - \delta$  [71, Eqs. 1–5]. Because any nearly monotone confidence satisfies the guarantee, better confidence buys coverage rather than validity [71, p. 3].

A model’s confidence can also be returned to the model itself. Firm or Fickle, a multi-turn study that shares four authors with this survey, follows up initially correct multiple-choice answers with eight kinds of challenge, including expressions of doubt or disagreement, appeals to authority or consensus, and misleading suggestions; across six models, mean follow-up accuracy ranged from 0.416 to 0.713 [85, Tables 1–2]. Its training-free CARG method computes the geometric mean of the token probabilities of the answer statement “The correct answer: X”, appends this value to each turn of the dialogue history, and lets the model condition on it without a threshold or decision rule; the authors report mean follow-up accuracy of 0.748 for CARG against 0.713 for GPT-4o with its default prompt [85, pp. 5, 8–9; Table 6]. The appended number is a sequence-likelihood statistic whose calibration was not evaluated, and because every follow-up challenges

a correct answer and none supplies correct information, keeping every initial answer would maximize the score [85, pp. 4, 9]. The study therefore measures resistance to pressure rather than the quality of revision, a distinction that Section 7.4 takes up.

Preference judgments add another distinction. Judging with Confidence targets the probability that a specified annotator population, mainly persona-conditioned Gemini, prefers B to A, using Brier or log rewards on binary observations [82]. Its evaluation compares soft preference proportions by MSE but binarizes them into tie-excluded majority labels for ECE and Brier. For Gemma, RL-Brier improves soft-target MSE over temperature scaling (0.0764 versus 0.0839), while temperature scaling has lower reported ECE (0.0827 versus 0.0879) [82, Table 1]. Modeling disagreement and predicting the majority winner are distinct tasks, and the evaluation should match the intended one. The same principle applies to numerical decisions: the interface should expose, and evaluation should assess, the statistic that the decision’s loss needs (Section 3.4).

RLVP, whose incentives Section 3.4 analyzes, also raises interface questions [36]. Its XGBoost expert is fitted to each task’s training split and predicts soft targets on that same split, with teacher-correct filtering, so these targets are not independently observed posteriors, although the setup does not indicate test-label leakage. Matched supervised hard- and soft-target comparators, cross-fitted teacher targets, and untransformed log-score rewards are therefore useful comparisons. Its JSON class probabilities remain autoregressive text rather than a token-free decision head, and the chain of thought generated alongside them needs a separate faithfulness evaluation. RLVP’s training-set expert and CaOPD’s privileged-context teacher (Section 3.4) [61] are different protocols, but both show why teacher access, target construction, and the deployment information set belong in a method description (Appendix D.2).

The decoding procedure also determines which distribution a structured output represents. For multistep structured outputs, grammar-constrained decoding adds another distributional choice. Park et al. distinguish local token masking from sampling the language model conditioned on eventual satisfaction of the grammar [126]. The latter depends on future valid-continuation mass, and their ASAP procedure approaches that conditional distribution through additional sampling. This result concerns a sequence distribution conditional on a grammar, rather than correctness calibration. A single-step softmax over fixed candidate labels is a different case; its numerical meaning, and hence what a decision reading it may infer, depends on the candidate set and outcome supervision.

## 5.6. Jev as an industrial interface case

TypeSafe presents Jev as a system for typed decisions and probability outputs, generated in parallel for use inside software [1]; its training is described as RLCD [4]. Classification, routing, and extraction are plausible beneficiaries: applications can consume bounded outputs directly and retain probabilities for later cost-sensitive decisions. The public materials establish that design direction but do not specify the training recipe fully enough to identify it with any open academic algorithm.

The API also illustrates a semantic issue that matters at deployment. Its documentation distinguishes a probability vector from an additional confidence statistic derived from distribution shape [127]. Before applying the cost threshold in Section 5.1, an application must establish that its chosen field estimates the relevant event. A useful default statistic can support validation-based gating without thereby becoming a correctness probability.

Community artifacts implement different parts of this design. The Qwen-2.5-1B-RLCD readout engine, `jev-on-a-laptop`, and JEVfire explore constrained candidate scoring and output assembly with existing models [128], [129], [130], whereas AlexWortega/`openjev` describes a supervised sequence classifier trained with cross-entropy on entailment-style labels [131]. They are practical comparators for interface efficiency and classifier training, but their names do not establish that they reproduce TypeSafe’s RL method; Appendix D compares the eight inspected industry and community sources and the controls needed to measure their benefits. The productive systems question is which combination of training, probability readout, and serving design improves the application’s measured decisions.

Beyond the ranking-decision gap itself, the decision view determines when the reporting-capability gap matters: whether a more uneven success-rate profile is an improvement depends on the thresholds the deployment applies (Section 3.2). An ordering, a reusable magnitude, and a learned action each imply a different comparator and test. An answer-or-abstain interface also bundles three separable objects—a stated probability, a rule that turns costs into a cutoff, and the executed choice—and the evidence above shows that they can change independently. A probability field in an interface therefore does not show whether training achieved an ordering, a reusable magnitude, or a learned action; Section 6 turns these distinctions into comparator choices and an evaluation contract.

## 6. When does RL add value, and how should it be shown?

The three gaps raise a central practical question: what RL contributes beyond post-hoc calibration or direct supervision, and what evidence would show that contribution. The answer has two parts: the comparator an RL claim must beat, which depends on the intended intervention and the available gradients (Section 6.1), and an evaluation that establishes what improved, under which information and resource constraints, and for which decision (Sections 6.2–6.5). Because calibration-aware training may change answers, confidence reports, parsing success, and the subset of queries a system accepts, no single aggregate metric can identify which change produced a gain; the evaluation contract below separates these effects while still reporting the native system’s end-to-end performance.

### 6.1. Choosing the comparator

Supervised estimation, prompting, and post-hoc fitting provide substantive alternatives to policy learning, and whichever alternative matches the intended intervention is the comparator an RL claim must beat. Early verbalized-confidence SFT learns confidence words or numbers from empirical subtask accuracies [88]. Kapoor et al. compare supervised probes and language-formulated correctness classifiers on fixed generated answers [66]. Probes of reasoning-model hidden states extend supervised probing to intermediate answers within a trace [112], and EliCal fits a pre-generation capability head with the generator frozen [63]. Tian et al. elicit confidence from existing RLHF models through prompts, with optional labeled temperature fitting [89]; Xiong et al. systematically compare prompting, sampling, and aggregation choices [96]. These approaches separate the cost of obtaining a useful confidence signal from the cost of changing the answering policy. ConfTuner, described in Section 4.3, is a particularly direct training comparator: because it fits the expected report loss exactly, comparing against it tests whether sampled confidence actions add anything beyond supervised reporting [57].

For answer-or-abstain behavior, prompts that state the payoff are a comparator that changes no weights. I-CALM combines elicited confidence, announced payoffs, and normative instructions [68, pp. 4–6]. Relative to abstaining by verbal-confidence rank at the same coverage on PopQA, the informedness of its abstentions (Youden’s  $J$ ) rose for two of three models, from 0.328 to 0.380 and from 0.275 to 0.490, but fell for the third, from 0.482 to 0.444 [68, Table 2].

An external predictor can also leave generation unchanged. Generalized Correctness Models learn from labeled histories across generators, using the question, response, and generator identity [132]. Their information and label requirements differ from those of an internal confidence head. At the post-hoc end, temperature scaling fits held-out outcomes [62], [133], while Dual-Align fits a temperature using a pre-trained model’s final distribution and layer signals as an unlabeled reference [134]. Self-consistency distillation is a label-free alternative when offline sampling is affordable: a regressor from a frozen model’s response embedding to the frequency of the modal answer among 100 unlabeled samples lowered average  $ECE_2$ , the binned  $\ell_2$  calibration error, from 0.22 for token probabilities to 0.09 across 36 model–dataset pairs [64, Fig. 2]. Labeled Platt scaling still had lower  $ECE_2$  in four of the five reported model and dataset groups, whereas the label-free predictor had a lower Brier score in four and tied in the fifth [64, Table 1]. Choosing among these comparators depends on access to logits, model internals, target-domain labels, and a suitable reference model.

Label-free sharpening is a further comparator for capability claims: entropy-minimization fine-tuning and RL reached average math scores of 40.2–42.9, against 42.1 and 43.9 for GRPO and RLOO trained with labels on the same 35K math prompts, although all runs were early-stopped on labeled validation accuracy [58, Table 2, pp. 22–23]. The same one-parameter temperature family that post-hoc calibration fits to outcomes can instead be tuned at each decoding step to lower entropy without labels [58, p. 7].

For class probabilities, answer-only calibration-aware fine-tuning (CFT) and its regularized variant provide supervised comparators [135]. Label smoothing changes the SFT target over the vocabulary [108]; CATTO augments DPO with a confidence-related token surrogate [136]. These methods address probability scale through different training signals. CATTO’s surrogate is linear in confidence for a fixed soft target, so its empirical classification results should be compared without importing a strict-propriety guarantee. BinaryPPO represents a neighboring approach that treats binary labels as actions [125]; its reported classification results motivate a matched supervised baseline, while sign inconsistencies in its printed objective (Appendix B) and limited probability-quality evaluation place it outside the established calibration evidence.

Table 5 collects these starting points. They form a comparison design, rather than a ranking of algorithms, and they reduce to one rule, the *comparator rule*. Post-hoc fitting is the comparator to beat when predictions can remain fixed.

**Table 5. Starting comparators for an RL claim.** The comparator follows from the intended intervention and the gradients available. RL is a distinct choice only when the behavior in the third column must adapt through outcome feedback; the rows form a comparison design, not a ranking of algorithms.

| Intended intervention                                                 | Justified starting comparator                                                                                                                              | What makes RL a distinct choice?                                                                          |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>A. Reports and estimators: the answer policy can stay fixed</b>    |                                                                                                                                                            |                                                                                                           |
| Recalibrate frozen scores                                             | Held-out fitting [62], [133]; reference-model or self-consistency fitting where labels are unavailable [64], [134]                                         | Changing the generator or the information acquired, beyond remapping a score                              |
| Assess fixed answers                                                  | Prompted or sampled reports [89], [96]; supervised estimation [66], [88]; exact expected-score training [57]                                               | Exploring an assessment process whose consequences require sampled feedback [29]                          |
| Add a correctness or capability predictor without changing generation | Historical cross-model prediction [132]; supervised capability heads with the generator frozen [63]                                                        | Adapting the answer policy or learning coupled behavior, rather than only a separate estimator            |
| <b>B. Answers and predictions: the policy itself must change</b>      |                                                                                                                                                            |                                                                                                           |
| Learn calibrated class outputs                                        | Answer-only CFT [135]; label smoothing [108]; class-logit supervision [113]                                                                                | Updating sampled reasoning from classification feedback [7], [36]                                         |
| Improve answers and report their reliability                          | Matched supervised answer and confidence training; preference regularization [136]; label-free sharpening for capability claims [58]                       | Updating sampled reasoning and answer choices from verifier or calibrated reward-model feedback [6], [30] |
| Improve a differentiable numerical prediction after reasoning         | Regression-aware supervised training                                                                                                                       | Learning which reasoning trajectories to sample while retaining prediction gradients [41]                 |
| Learn a rollout-induced predictive distribution                       | Distributional supervised training                                                                                                                         | Changing the generation distribution through sample-set feedback [42]                                     |
| <b>C. Decisions: a score is cut, spent, or acted on</b>               |                                                                                                                                                            |                                                                                                           |
| Optimize ordering or acceptance at one fixed cost                     | A rank or action objective with matched supervision; prompts that state the payoff, or elicited confidence followed by a separate decision step [68], [69] | Improving sequential choices from feedback, without requiring probability identification [39], [43]       |
| Certify a selective decision at a stated risk                         | Validation-selected thresholds with finite-sample risk bounds on frozen scores [68], [71]                                                                  | Changing which items the model can handle or what it outputs, rather than where an existing score is cut  |
| Allocate or stop test-time computation                                | Training-free token-statistic gates [74], [75]; success or correctness estimators fitted with the generator fixed [72], [73], [87], [112]                  | Changing the reasoning policy together with its stopping or budget signal [76], [77]                      |
| Adapt information acquisition to per-query costs                      | Recalibrated or supervised priors supplied, with the stated costs and discounted-reward objective, to a prompted agent [78]                                | Learning the exploration policy from discounted outcome reward while conditioning on those priors [78]    |

Direct supervision is the comparator to beat when the desired report or statistic has a tractable differentiable loss. RL becomes a distinct intervention only when reasoning, retrieval, answering, abstention, or information acquisition must adapt through outcome feedback; when a differentiable loss is also available, hybrid methods can retain both gradient routes (Section 4.3). Calibrate-Then-Act divides the work in this way: its probabilities come from isotonic recalibration or a supervised predictor, and GRPO then learns an exploration policy conditioned on them. Relative to RL trained without the priors, conditioning on them raised discounted reward from 0.259 to 0.268, with significant paired gains in two of four cost settings [78, Tables 4, 6]. The strongest evidence for a distinct RL contribution therefore identifies the behavior being changed, supplies the relevant supervised or post-hoc baseline, and matches access, labels, and compute.

## 6.2. Separate reporting quality from policy improvement

Two complementary evaluations are needed. A **fixed-answer evaluation** compares reporters on a common collection of questions, answers, histories, and labels. It measures how well each reporter assesses the same outcomes. A **native-policy evaluation** lets each trained system generate its own answers and confidence and then measures predictive quality and decision utility under its intended deployment procedure. The first isolates reporting quality; the second credits the benefit of generating better answers. Neither substitutes for the other: attributing a Brier gain to the report requires fixed outcomes, and hence a constant base rate, which only the fixed-answer evaluation provides (Section 2.3).

Where interfaces permit, a crossed diagnostic adds two intermediate comparisons. Let  $\pi_0, \pi_1$  be the original and updated answer policies and  $R_0, R_1$  their confidence reporters. Generate answer pools  $A_i$ , with their histories  $H_i$ , from each  $\pi_i$ , and evaluate

$$C_{ij} = R_j(X, A_i, H_i), \quad i, j \in \{0, 1\},$$

against labels for the same  $A_i$ . Comparing reporters within a pool tests reporting changes on fixed answers; comparing pools reveals dependence on the answer distribution. The diagonal entries should also be compared with native operation, because an imposed reporting interface may itself change behavior.

The diagnostic is interpretable when the reporters have compatible inputs. A reporter may require unavailable hidden states, a particular reasoning format, or information that another policy never produces. Crossed results should identify those compatibility restrictions and any out-of-distribution use. Outcome feedback and reference-answer access should be matched, and reported uncertainty should be measured before the decision it informs.

## 6.3. A common evaluation contract

The contract in Table 6 makes these comparisons operational. Its purpose is to align evidence with the intended claim, not to require every task to optimize the same metric. Its implemented-update row carries the objective-update gap of Section 4 into evaluation: two runs with the same reward can differ in centering, standardization, token aggregation, KL placement, or rollout count.

Holdout design should account for every selection step: fit transformations and choose checkpoints or thresholds without consulting the final evaluation labels; if a benchmark’s test partition is subdivided for calibration, identify the reserved subset and exclude it from final reporting; and use nested validation or a separate selection split for more complex tuning. How each example was actually used matters more than a partition’s name.

Resource matching also requires more than equal annotation totals. Preference-distribution learning in [82] compares 50,000 annotation calls allocated as 5,000 inputs with ten labels each for SFT and 50,000 inputs with one label each for RL. This is an informative comparison of sparse and dense supervision with tenfold different input coverage, but it does not isolate the optimizer under identical inputs. Likewise, a transferred calibrator may use no target labels while still requiring substantial labeled source data [137].

Decision evaluation should include two operating conditions. First, apply a probability threshold determined by declared costs, testing the reported numerical meaning directly. Second, select a threshold on independent validation data to test the achievable usefulness of the score for that deployment. Present the resulting coverage and error costs on test data. A ranking method may perform well in the second condition despite lacking an interpretable probability scale. Attaining a target accuracy in one small accepted subset is empirical evidence, rather than a distribution-free certificate [40, Table 3]. A finite-sample bound belongs in threshold selection whenever a guarantee is claimed. In the selective-judging study of Section 5.2, point-estimate thresholds chosen on the same labeled data without a bound met the agreement target in only 47.5–79.0% of splits across three judges [71, Table 2].

**Table 6. A common contract for interpretable evaluation.** Each component specifies what a comparison must hold fixed or report so that a gain can be attributed to the reporter, the answer policy, or the decision rule. The supporting evidence is discussed in the sections named in the last column.

| Component                      | Required comparison or record                                                                                                                                                                                                                                                                                                                                                                                              | Failure it guards against                                                                                       |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Target and information         | Specify the event, label rule and grader, report timing, and evidence visible to the reporter; retain these fields for each example                                                                                                                                                                                                                                                                                        | Comparing numbers that estimate different events or use different information (Section 2.1)                     |
| Elicitation and evidence use   | Record the prompt, role framing, response scale and polarity, and whether confidence is elicited with or after the answer; report threshold-decision changes under equivalent re-elicitation, and confidence changes when the evidence a report is said to use is masked or perturbed                                                                                                                                      | Decisions that depend on wording; confidence that ignores the reasoning it is attributed to (Sections 2.1, 6.3) |
| Answer and reporter changes    | Report fixed-answer and native-policy results, including the estimated success-rate profile under the named decoder; add compatible crossed evaluations when diagnosing their interaction                                                                                                                                                                                                                                  | Crediting the reporter with a change in the answer distribution (Sections 3.2, 6.2)                             |
| Implemented update             | Report the advantage estimator (centering, standardization, leave-one-out baselines), token or sequence aggregation, the anchor, placement, and coefficient of any KL term, clipping activity, active gradient routes, rollout count, and validity filtering; vary one of them at a time on common samples when attributing a gain                                                                                         | Crediting the objective with an effect of its implementation (Sections 4.1–4.4, 7.2)                            |
| Supervision and selection      | Record labeled inputs, labels per input, calibration data, and checkpoint and threshold selection data; reserve an untouched final test population                                                                                                                                                                                                                                                                         | Test-set selection and unequal label budgets (Sections 6.3–6.4)                                                 |
| Inference resources            | Count reasoning tokens, candidate samples, retries, tool calls, auxiliary judge calls, and calibration rollouts; compare fixed-budget performance or explicit cost–quality curves                                                                                                                                                                                                                                          | Gains bought with uncharged computation or information (Section 5.4)                                            |
| Output validity and abstention | Report counts of attempted, not-attempted, and valid outputs, together with parsing and fallback rules and the utility of failures; where inputs may be unanswerable, label answerability and report abstention recall and precision separately from accuracy on answerable inputs; for agents, log executed actions separately from verbal outputs; show valid-only statistics as a separately labeled conditional result | Denominator shifts that turn refusal or parse failure into apparent gains (Section 6.3)                         |
| Prediction and decisions       | Pair the target’s proper loss and calibration diagnostics with ranking, coverage, and utility at declared operating points; report utility at a cost-derived threshold and at a threshold selected on independent validation data; bound the risk of any threshold selected to support a guarantee                                                                                                                         | Metrics that do not match the decision the score serves (Sections 5.1–5.3)                                      |
| Uncertainty of estimates       | Give sample counts, accepted and error counts, paired or clustered uncertainty intervals, and seed variability where training randomness is material                                                                                                                                                                                                                                                                       | Reading differences within sampling or seed noise as gains (Section 6.5)                                        |

The elicitation protocol belongs in this contract because it can change the report even when answers are fixed. The ownership-framing study holds candidate answers fixed while changing whether they appear as assistant output or user-provided content, and the resulting confidence changes identify a contextual confound; it also scores all answer options rather than only the top answer [17]. Semantically equivalent wording also matters. In a small training-free study, two equivalent re-elicitations of P(True)-style confidence for 99 fixed TriviaQA answers changed scores by 0.043–0.084 on average and moved 4.0–9.1% of answers across a 0.8 threshold; because the study is small (99 questions and eight or nine errors per model), the authors present its statistical intervals as descriptive [138, Table 2, p. 4]. Token likelihood, numerical confidence, and linguistic uncertainty extracted by another model differ in information access, parsing failures, and inference cost [139]. A reproducible comparison therefore retains the answer text, role framing and template, elicitation wording and order, the scored population (selected answer or all candidates), parse-failure outcomes, and any auxiliary judge calls. Beyond these records, whether a report actually uses the evidence attributed to it requires a separate test, such as masking that evidence, because outcome-level calibration cannot reveal it (Section 2.1) [84].

Refusal further changes the evaluated population. Reasoning-model comparisons that calculate calibration only on attempted answers should also report all-query coverage and a comparison on the questions that every compared model attempted [13]. Refusal on explicitly unanswerable questions supplies decision evidence distinct from probability calibration [14]. AbstentionBench makes the population explicit by labeling whether abstention is appropriate for each input and reporting abstention recall separately from accuracy on answerable inputs; because its judge detects more abstention when reasoning traces are included, an evaluation must also state which part of the output it treats as the decision [15, p. 5, Fig. 8a].

The population and the interface also determine what a refusal means. Where every question is answerable, not attempting is lost coverage rather than a correct abstention. SimpleQA therefore reports accuracy over all questions separately from accuracy on attempted ones, and similar F-scores can conceal different attempt policies: 35.0 with 35.0% of questions not attempted versus 38.4 with 1.0% [83, Table 3]. Where declining is correct because the outcome is unknowable, a matched answerable arm keeps blanket refusal from scoring well [111, pp. 4–5]. Agent benchmarks add the executed action to the record: scoring committed tool calls separately from verbal refusals showed that 2.6% of qualifying should-abstain runs, 115 in all, acted while claiming to abstain, 98 of them through a critical tool call [79, pp. 3–5, 8]. A forced best guess after each abstention can restore a correctness label for withheld items, on the assumption that the guess stands in for the answer the model would otherwise have given [68, pp. 5–6]. The RLVR measurement position paper makes a related argument: attempted-answer counts, shared-set stated confidence, and matched inference budgets can change how post-training comparisons are interpreted. Its heterogeneous checkpoint and budget comparisons call for controlled measurement rather than support a universal direction of calibration change [140].

## 6.4. What within-study comparisons establish

The examples in Table 7 illustrate different claims; their numbers should not be combined into a cross-study leaderboard.

Each row supports a narrower conclusion than a headline might suggest. Although the first study describes RL/DPO-induced miscalibration, its reported mean ECE after GRPO or DPO improves on initialization but remains worse than after SFT, so the supported conclusion is comparator-specific [12]. DPO in that comparison learns from fixed preference pairs without sampling from the policy [11, p. 5], so under the comparator rule of Section 6.1 it is closer to direct supervision than to learning from outcomes. EpiCaR explicitly discusses discrimination versus absolute calibration, and RLSR is designed around selective ranking; their results call for evaluation against each method’s intended objective. The preference example goes further: its metrics disagree because they score different targets, not because they summarize the same probability error differently.

The EliCal, AbstentionBench, and trajectory rows are narrower still. EliCal’s temperature-scaling comparators judge an appended greedy answer and use more labels, so that contrast also differs in information and supervision [63, p. 21]. AbstentionBench compares released checkpoints whose reasoning training is distillation or supervised fine-tuning; its one explicit RL stage, Tülu 3’s PPO-RLVR, changed average abstention recall by only 0.01–0.02 [15, Table 4]. The trajectory study places outcome metrics reproduced from the original evaluation beside locally computed grounding scores; its negative 4B post-training score is not distinguishable from zero; and its common-population recomputation shows the grounding sign reversal only for VL-Calibration at 8B [84, pp. 14–15, 19].

**Table 7. Within-study evidence that improvements are property- or comparator-specific.** Each row contrasts properties measured in one study; numbers are not comparable across rows, and no cross-study ranking is implied.

| Within-study comparison                                                                                            | Observed result                                                                                                                   | Supported interpretation                                                                                  |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>A. Probability quality: which property improved?</b>                                                            |                                                                                                                                   |                                                                                                           |
| Qwen3-4B training comparison [12, Tables 1–2]                                                                      | Mean ECE is 0.163 initially, 0.135 after GRPO, 0.117 after DPO, and 0.034 after SFT                                               | GRPO and DPO outperform initialization on this metric but remain worse than the supervised comparator     |
| EpiCaR transfer, Qwen3-8B on GSM8K [141, Table 3]                                                                  | AUROC rises 0.654→0.722 while raw ECE rises 0.215→0.364                                                                           | Better error ordering does not establish better probability scaling                                       |
| RLSR versus RLCR, HotpotQA-trained Qwen2.5-7B, in-domain aggregate [40, Table 1]                                   | AURC improves 0.51→0.44 and accuracy at 10% coverage 53.5%→66.0%, while ECE rises 0.05→0.24                                       | A selective-ranking objective can improve its intended behavior without improving calibration             |
| Gemma-2-9B preference evaluation [82, Table 1]                                                                     | RL-Brier has lower soft-target MSE than temperature scaling, 0.0764 versus 0.0839, but higher reported ECE, 0.0879 versus 0.0827  | Soft preference proportions and tie-excluded majority labels are different evaluation targets             |
| RLVP-Anneal versus XGBoost on six single tasks [36, Table 1]                                                       | AUROC improves on 6/6 tasks; ECE improves on 2/6. Unweighted mean ECE, calculated from the reported rows, is 0.1005 versus 0.0743 | A useful ranking gain is not uniform calibration dominance                                                |
| EliCal with 1k labels versus the Thermometer temperature-scaling baseline, Qwen2.5-7B, in-domain [63, Tables 2, 6] | ECE is 0.05 versus 0.06, while AUROC is 84.36 versus 59.48                                                                        | Similar calibration error can hide a large difference in discrimination                                   |
| <b>B. Behavior: what the probability field does not show</b>                                                       |                                                                                                                                   |                                                                                                           |
| AbstentionBench, reasoning-tuned versus instruction-tuned counterparts [15, Table 4]                               | Average accuracy rises 0.78→0.81 and 0.75→0.80, while abstention recall falls 0.66→0.46 and 0.71→0.43                             | Higher accuracy on answerable inputs does not establish appropriate abstention on unanswerable ones       |
| VL-Calibration versus its Qwen3-VL-8B base, ten benchmarks [84, Tables 2, 5]                                       | ECE falls 0.402→0.081 and AUROC rises 0.561→0.768, while trajectory grounding falls 0.079→−0.018 (95% CI −0.029 to −0.005)        | Better outcome-level calibration does not establish that confidence depends on the model’s own trajectory |

Four further cases show why the contract records selection, population, and the metric behind each claim. The ACL version of decision-token calibration adds calibration-weight and KL sensitivity analyses in which in-domain and transferred ECE can move in opposite directions, yet it still selects its best checkpoint on the test set [7]. EliCal likewise selects checkpoints for all trained methods on the in-domain test set [63, p. 18]. EpiCaR reserves 500 examples from the original test split for temperature fitting and excludes them from final reporting, which, as described, is a legitimate holdout rather than test-set selection; however, one passage specifies fitting by minimizing ECE and another by minimizing negative log-likelihood (NLL) [141]. RLVP’s “up to 55%” teacher-surpass ratio and its unseen-task wins concern AUROC rather than calibration, and its six-task ECE values do not support the paper’s best-average-ECE statement [36]. Appendices E.2, E.3, and D.2 give the details for the decision-token, EpiCaR, and RLVP cases; each qualification retains the positive result while stating its evidential scope.

The apparently conflicting observations of Section 1 fit the same pattern. The GPT-4 and reasoning-model results read different probabilities, answer-option probabilities versus discretized verbal confidence [8], [9]; the abstention results differ in population, answerable versus unanswerable inputs and attempted versus all questions [13], [14], [15]; and the factual-QA results differ in pipeline, supervised fine-tuning followed by RL versus RL alone [16]. Once the event, information, and population of each claim are stated, none of these contrasts is a contradiction.

## 6.5. Metrics, implementation, and uncertainty

Metric names are insufficient identifiers. State the label, binning and weighting, report transformation, sample retention, and aggregation level. Among the recent protocols reviewed here, ECE is computed against graded judge ratings [35], over twelve equal-mass bins [64], or over ten bins [32], [63]. Label rules alone can also move measured performance: correcting exact-match false negatives, and dropping one disputed question, in one small audit raised measured accuracy from 0.86 to 0.909 and from 0.88 to 0.919 [138, p. 3]. The audited papers and released evaluators include per-example absolute error labeled “ECE,” differing hard-versus-graded correctness targets, and bin weighting that differs between a paper and its linked metric dependency [37], [99], [142]. Appendix D.1 records these source-specific details and distinguishes inspected releases from historical experiment runs.

Finally, uncertainty estimates should respect the unit of dependence. Multiple trajectories from one question are not independent test questions; repeated annotations may also share an annotator or task. Use paired or clustered resampling accordingly, report the number of rare errors and accepted examples, and keep label uncertainty separate from training-seed variability. Significance tests over inference seeds alone, as in one recent comparison, do not capture training variability [32, p. 14]. These practices turn a metric improvement into an interpretable result: a better reporter on fixed outcomes, a better native policy, or a better decision under specified costs. The next section asks which comparisons are still missing for attributing such results to a mechanism.

## 7. Research agenda: open questions about mechanisms and decisions

An improved confidence metric admits at least four explanations: better reporting, a changed answer policy, an implemented update that optimizes a different objective, and better operating-point selection. The three gaps indicate which comparisons separate them. Author-reported studies provide evidence within their own protocols, and the analytic constructions show what is possible under explicit assumptions. Neither shows how often a mechanism explains results for other trained models, and this survey reports no independently trained LLM results. The four questions below, one for each gap and one for adaptive computation, state the evidence that would distinguish the explanations.

### 7.1. When does joint scoring change reporting or capability allocation?

Section 3 showed that exact reward optimization over a coupled policy family can prefer lower mean accuracy even with truthful reports, whereas a feasible pointwise-dominating policy aligns the objectives when truthful reports are attainable and the calibrated value is nondecreasing. RLCR and reward-family comparisons provide existing evidence about joint training [30], [31], and published rewards whose calibrated value is not monotone offer natural test cases (Section 3.3) [32], [35]. Identifying which decomposition term explains a particular outcome, however, requires more than aggregate ECE and accuracy.

The informative comparisons distinguish objective preference from restricted reporting capacity and finite optimization. Experiments with known success probabilities can vary reward curvature and accuracy weight across coupled and independently adjustable policy families; attainable truthful reports then isolate allocation, and a pointwise-dominating policy serves as a negative control. For LLM evidence, common input weights, decoders, and verifiers, together with uncertainty estimates for conditional success profiles, make attribution more credible. A correctness-only RL control on the same decoder isolates the sharpening that correctness rewards produce on their own [86]; compatible crossed reporters on fixed answer pools can identify reporting changes separately from native-policy gains; and selective utility shows whether a changed success-rate profile benefits the intended decision.

Label-free certainty rewards pose the capability-side version of this question. Their reported gains depend on initial policy entropy and training duration [51, Fig. 3, Tables 4–5], and one study reporting such gains early-stops on labeled validation accuracy [58, p. 23]. Informative evidence would report full training trajectories, a label-free stopping rule, and outcome-based calibration of any confidence read from the trained policy.

### 7.2. Which effects belong to the score, and which to the update?

A population scoring objective, its finite-sample estimate, and its implemented gradient need not have the same properties. For  $K \geq 2$ , replacing the conventional pairwise factor  $1/(2K^2)$  with  $1/[2K(K-1)]$  yields an unbiased estimate of population CRPS under iid ensemble draws, finite first moments, and independence from the realized outcome

conditional on the input [27]. The Gaussian construction of Section 4.4 shows that the conventional score can favor insufficient spread even before neural optimization enters. This analysis does not establish that the mechanism explains DAR’s reported results: DAR adds leave-one-out credit, normalization, and valid-sample handling [42]. Fairness of the scalar score does not certify any transformed gradient, and dependent generation and parse filtering need analysis beyond the independent-draw reference.

Author-reported normalization ablations [18] and REAL’s direct-gradient ablation [41] already motivate examining update choices, but their conclusions remain specific to the evaluated targets and configurations. Differences between published formulas and inspected launchers, moreover, cannot identify historical run settings. Attribution would compare raw scores, rollout credits, transformed advantages, and gradients on common samples. Varying  $K$  and the pairwise denominator under explicit sampling budgets, with sampling, validity handling, and advantage transformations held fixed, separates estimator effects from additional computation. Comparing centering with standardization distinguishes baseline subtraction from random rescaling, and for parameter-dependent rewards, the exact two-route gradient, a stopped-gradient reward, and a stabilized estimator should be compared as distinct conditions.

### 7.3. When do reusable probability magnitudes add decision value?

The four-context construction of Section 5.2 fixes answers and outcomes. It shows that a strictly increasing transformation can preserve ordering and attainable selection frontiers while changing utility at a transported threshold, and that reselecting the threshold recovers the accepted set without recalibrating the probabilities. Existing RLSR and TruthRL comparisons supply evidence for selective ranking and action learning at their respective operating conditions [40], [43]; they do not establish a universal preference among probability reporting, selective ranking, and direct action learning.

Beyond the two-threshold evaluation of Section 6.3, which separates numerical probability meaning from attainable deployment performance on common answers, the unresolved comparison concerns transfer across costs, contexts, and available labels. Native-policy evaluation should accompany that comparison, so that any benefit from changing answers is also measured. Evaluating transfer to withheld costs without reusing evaluation labels could establish when calibrated, informative reports reduce the need for additional supervision, when a rank score suffices after threshold fitting, or when direct action learning remains preferable at a fixed utility. Contextual costs and abstention-induced missing labels must enter that comparison explicitly, because they change the information needed for a valid decision [24].

A related comparison concerns consistency between stated belief and executed choice. Announcing payoffs does not make a model apply the implied threshold (Section 5.1), and sweeping stated payoffs over fixed tasks is one design for measuring this inconsistency [69, pp. 2–4]. The studies reviewed in Sections 5.1 and 5.3 do not report a direct measure of belief–gate consistency, the difference between a model’s own act-or-abstain choices and those that an external threshold would make from its stated probabilities [68], [69], [111]. The nearest checks are one-sided: in one agent study, requiring stated probabilities at least 15 points from 50 would have cut commitment from 54.0% to 7.5%, and a threshold on the stated edge, at its best value, separated answerable from unknowable questions more accurately than the model’s own act-or-decline choices did (83.7% versus 67.9%) [111, pp. 5, 12].

### 7.4. Which uncertainty estimate supports an adaptive decision?

The question is which probability target remains useful at the moment each adaptive intervention is chosen. Confidence in a completed answer and the expected benefit of another action concern different events and information sets. Section 5.4 distinguishes response confidence from decoder-specific capability confidence, and Table 4 shows that adaptive-computation decisions differ in what they need from an estimate: an ordering within a problem, a validated threshold, or a calibrated magnitude. The capability study provides evidence for matching the predictor to a repeated-success target [72]; its reported pass@64 simulation does not by itself validate a deployed selector. Even under independent trials, finding at least one correct candidate differs from recognizing and selecting it. The decoder can also differ between target and evaluation: EliCal’s target is the success rate of temperature-1 samples, whereas its evaluation scores the greedy answer [63, Eq. 4, pp. 7, 18].

Two comparisons are missing from the reviewed evidence. A stopping decision compares the current candidate’s success probability with that of the answer continued reasoning would deliver, yet no reviewed study estimates both at

the same state: early-exit gates estimate the former [75], [112] and continuation estimators the latter [73], [87]. Escalation compares current confidence with the expected gain from a stronger resource, yet cascaded judges escalate on the current judge’s confidence in its own verdict and report cost only after the fact [71, pp. 5, 8–9].

An informative adaptive-computation comparison would retain pre-action states, available information, chosen actions, observed costs, candidate outcomes, and the selection rule. Recording whether a trigger is visible from the query, from the tool inventory, or only after tool observations is one way to document the information available at each decision [79, p. 24]. Comparing training-free gates [74], [75], supervised value predictors [73], [87], [112], capability estimates [72], and RL controllers [76], [77] under the same information and budget could distinguish better prediction from additional information or computation.

Calibrate-Then-Act compares RL with and without supplied priors on the same training data [78, p. 8], but its priors are fitted on data from the evaluation distribution [78, pp. 5–6], so whether supplied magnitudes transfer remains untested. Evaluating on a held-out decoder, policy, or budget regime would test the scope of transfer. Where information sets are nested and the outcome remains fixed, the conditional-expectation identity (A.2) of Appendix A.1 provides a coherence check. Where retrieval, routing, or continued generation changes the outcome distribution, the resulting policy and its cost must be evaluated directly.

Multi-turn interaction makes the information set of Section 2.1 concrete. A follow-up that supplies a counterexample or a source changes the evidence about the original answer; one that only repeats doubt changes it only if, in the deployment population, challenges are more frequent after wrong answers than after correct ones. A report conditioned on the declared information should move in the first case, and in the second only by as much as that difference in frequency warrants; where the outcome is fixed and the information sets are nested, identity (A.2) constrains the average movement. The multi-turn evaluation in this collection does not separate these cases: Firm or Fickle challenges only answers that were initially correct, with doubt or misinformation [85, pp. 3–4], so it measures capitulation but not warranted revision, and CARG’s gain was measured in that setting [85, p. 9]. An informative design would cross the type of follow-up (repeated doubt, misleading claim, correct evidence) with initial correctness and record the stated probability and the keep-or-revise action at each turn, so that warranted revision and the belief–gate consistency of Section 7.3 can both be measured across turns.

Across the four questions, the missing evidence takes a common form: comparisons that change one mechanism while holding fixed the event, information, and population of the probability claim.

## 8. Conclusion

Calibration is a property of a claim, not of a model. A reported probability means something only relative to an event, the information behind the report, and the population or policy over which it is evaluated, and calibration-aware RL can change all three while it trains. The survey has traced such claims through three gaps. The reporting–capability gap arises because proper scores make truthful reporting optimal, but once the policy selects its own answers, the same reward places value on how success is distributed across inputs (Section 3). The objective–update gap arises because group transformations, gradient routes, and finite-ensemble scoring can make the implemented update depart from the stated objective (Section 4). The ranking–decision gap arises because useful ranking and reusable probability magnitudes solve overlapping but different problems (Section 5).

The practical consequence is the comparator rule of Section 6.1. A calibration gain is evidence of a distinct RL contribution only if it beats the comparator appropriate to the intended intervention under matched access, labels, and compute: post-hoc fitting wherever the predictions could have remained fixed, and direct supervision wherever the desired report had a tractable differentiable loss. RL’s distinct role lies where the behavior being trained, not just the number it reports, has to change in response to outcomes. A claimed gain should then be checked by an evaluation that isolates reporting, the answer policy, the implemented update, and operating-point selection (Sections 6.2–6.5).

For method papers, this means stating which gap a reported gain addresses and against which comparator it was measured; for benchmarks, stating the event, population, and payoff that define the score; and for deployed interfaces, establishing that a probability field estimates the event the consuming decision needs. The central opportunity is to connect improvements at individual stages to evidence of reliable decisions, with enough detail to explain both when the connection holds and which comparison is needed when it does not.

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## Appendix A. Mathematical analysis and exact examples

This appendix derives the identities and computes the exact examples that the main text cites, using classical conditional expectation and proper scoring rules [26], [27]; none is claimed as a new statistical theorem. Constructed distributions are distinguished from empirical LLM results throughout. Each statement specifies its reward sign, admissible reports, information set, and the distribution over which expectations are taken.

### A.1. Reporting error and available information

This subsection decomposes the Brier risk of a report according to the reporter’s information and relates ideal reports under nested information. It formalizes the conditioning requirements of Sections 2.1 and 2.3 and supplies the identity behind the coherence check proposed in Section 7.4. Let  $Z \in \{0, 1\}$ , let  $\mathcal{F}$  be a sigma-field describing the reporter’s information, and write  $p_{\mathcal{F}} = \mathbb{E}[Z \mid \mathcal{F}]$ . For any  $\mathcal{F}$ -measurable report  $C \in [0, 1]$ ,

$$\mathbb{E}(Z - C)^2 = \mathbb{E}[p_{\mathcal{F}}(1 - p_{\mathcal{F}})] + \mathbb{E}(C - p_{\mathcal{F}})^2. \quad (\text{A.1})$$

Expand  $Z - C = (Z - p_{\mathcal{F}}) + (p_{\mathcal{F}} - C)$ . The cross term vanishes conditional on  $\mathcal{F}$ . The first term measures the uncertainty that remains under that information; the second is the report’s excess Brier risk relative to the conditional probability, a population quantity rather than an empirical ECE estimator.

For nested information  $\mathcal{F} \subseteq \mathcal{H}$ , with the same outcome and joint distribution, the tower property gives

$$\mathbb{E}[p_{\mathcal{H}} \mid \mathcal{F}] = p_{\mathcal{F}}, \quad \mathbb{E}(Z - p_{\mathcal{F}})^2 - \mathbb{E}(Z - p_{\mathcal{H}})^2 = \mathbb{E}(p_{\mathcal{H}} - p_{\mathcal{F}})^2. \quad (\text{A.2})$$

This is a coherence condition for ideal reports before and after receiving information. A pre-generation capability estimate and a post-generation report can be compared this way only if they refer to the same realized correctness outcome under the same generation policy. A policy change, a revised verifier, or a different population alters the outcome or its joint distribution, so the identity no longer applies.

Score calibration,  $\mathbb{E}[Z \mid C] = C$ , is weaker than  $C = p_{\mathcal{F}}$ . A constant base-rate report can be calibrated while ignoring predictive information. Two calibrated scalar scores therefore do not automatically satisfy a tower relationship. Likewise, uncertainty arising from computational limitations should not be conflated with residual randomness under complete semantic conditioning. If the conditioning includes a complete prompt, complete answer, and deterministic correctness rule, the corresponding conditional label probability is 0 or 1. Intermediate probabilities require a specified information restriction, latent or stochastic labels, or an action denoting a stochastic answer procedure.

### A.2. Bayes reward and the joint-objective decomposition

This subsection derives the decomposition of a joint correctness-plus-score objective stated in Section 3.2, specializes it to RLCR’s reward, and computes the two-context example behind Figure 4. For a proper binary score  $S(c, z)$  in reward orientation, define

$$L(c, p) = pS(c, 1) + (1 - p)S(c, 0), \quad G(p) = L(p, p), \quad D_G(p, c) = G(p) - L(c, p) \geq 0.$$

Strict propriety makes the zero-regret report unique within the stated admissible domain. Under the regular Savage representation, this regret takes the Bregman form  $D_G(p, c) = G(p) - G(c) - g(c)(p - c)$ , where  $g(c)$  is an appropriate subgradient [26]. The decomposition below needs only the definition of  $D_G$ , not this representation.

For one fixed policy  $\pi$ , let  $p_{\pi}$  be its conditional probability of success under the declared reporting information, and let  $c_{\pi}$  be a report measurable with respect to that information. Consider

$$R = \alpha Z + \kappa + \beta S(c_{\pi}, Z), \quad \beta > 0,$$

with fixed finite weights and well-defined expectations. Writing  $\mu_{\pi} = \mathbb{E}p_{\pi}$  gives

$$J(\pi, c_\pi) = \alpha\mu_\pi + \kappa + \beta G(\mu_\pi) + \beta\{\mathbb{E}G(p_\pi) - G(\mu_\pi)\} - \beta\mathbb{E}D_G(p_\pi, c_\pi). \quad (\text{A.3})$$

The identity holds for any admissible report, truthful or not. Its proof conditions the expected score on the reporting information and then adds and subtracts  $G(\mu_\pi)$ . Its three varying components are mean capability, the Jensen gap of the success-probability profile, and reporting regret. The Jensen gap is generally not variance; variance alone does not order expectations of every convex function.

For correctness plus negative binary Brier,  $\alpha = \beta = 1$ ,  $\kappa = 0$ ,  $G(p) = p^2 - p$ , and  $D_G(p, c) = (p - c)^2$ . Thus

$$J = \mathbb{E}[p_\pi^2] - \mathbb{E}(p_\pi - c_\pi)^2 = \mu_\pi^2 + \text{Var}(p_\pi) - \mathbb{E}(p_\pi - c_\pi)^2. \quad (\text{A.4})$$

This identity specializes the objective used in RLCR [30]. Under truthful reporting, its conditional value is  $p^2$ , which increases in  $p$ . Such pointwise preference does not settle comparisons among policies whose achievable success rates are coupled across inputs.

To isolate this issue, give two contexts equal weight and allow two candidate policies with independently attainable truthful reports (Table A.1).

**Table A.1. Two policies that separate joint reward, mean accuracy, and selective utility.** Two equally weighted contexts; truthful reports are attainable under both policies. Utility scores correct, wrong, and rejected answers as +1, −1, and 0 at threshold 0.5. Bold marks the better policy under each criterion.

| Policy | Conditional success rates | Mean accuracy | Optimized joint reward | Utility at +1/−1/0 |
|--------|---------------------------|---------------|------------------------|--------------------|
| A      | 0.6, 0.6                  | <b>0.600</b>  | 0.36000                | 0.200              |
| B      | 0.95, 0.2                 | 0.575         | <b>0.47125</b>         | <b>0.450</b>       |

The joint reward prefers B despite its lower mean accuracy, and no reporting error or optimizer failure is involved. B also has higher selective utility when correct, wrong, and rejected answers earn +1, −1, 0: with threshold 0.5, it accepts only its first context. The example shows that joint reward, mean accuracy, and selective utility are different objectives, not that the joint reward’s preference for B is universally undesirable. It compares true conditional success rates across contexts, not randomized extreme reports for one fixed event.

If reports are also constrained to  $\mathcal{C}_\pi$ , the ideal feasible value is

$$J_{\text{feasible}}^*(\pi) = \alpha\mu_\pi + \kappa + \beta\mathbb{E}G(p_\pi) - \beta \inf_{c \in \mathcal{C}_\pi} \mathbb{E}D_G(p_\pi, c).$$

Because the final term can depend on the answer policy, it cannot be discarded as a common constant, and the feasible value need not inherit a simple convex shape.

### A.3. Curvature, capability constraints, and accuracy weights

This subsection proves the four results that Section 3.2 uses: strict convexity of the calibrated value, a reward–accuracy reversal within a restricted policy family, conditions that exclude such reversals, and a bound on the accuracy lost under a finite accuracy weight. Throughout,  $\text{Acc}(\pi) = \mathbb{E}p_\pi$  denotes mean accuracy, and  $J^*(\pi)$  denotes the report-optimized value of Eq. (A.3), which equals  $\kappa + \mathbb{E}V(p_\pi)$  when truthful reports are feasible. Classical strict propriety implies strict convexity of the Bayes reward on a nondegenerate convex probability domain [26]. A short proof avoids differentiability. For  $u \neq v$ ,  $t \in (0, 1)$ , and  $m = tu + (1 - t)v$ ,

$$G(m) = L(m, m) = tL(m, u) + (1 - t)L(m, v) < tG(u) + (1 - t)G(v).$$

The inequality is strict because strict propriety applies at both  $u$  and  $v$ , where the report  $m$  differs from the truthful report. Consequently the calibrated value  $V(p) = \alpha p + \beta G(p)$ , with the constant  $\kappa$  omitted as in Section 3.2, re-

mains strictly convex for any fixed finite affine accuracy bonus and  $\beta > 0$ . A positive slope, as in pointwise best-effort conditions [31], can coexist with that curvature.

There is therefore no universal guarantee that expected reward preserves mean-accuracy rankings over every coupled feasible policy family. Choose interior  $u < v$ , let  $m = (u + v)/2$ , and define the strict Jensen gap  $\Delta = [V(u) + V(v)]/2 - V(m) > 0$ . Because a convex function is continuous in the interior of its domain, some sufficiently small  $\epsilon > 0$  satisfies  $V(m + \epsilon) < V(m) + \Delta$ . Two equal-weight profiles  $A = (m + \epsilon, m + \epsilon)$  and  $B = (u, v)$  then satisfy

$$\text{Acc}(A) > \text{Acc}(B), \quad J^*(A) < J^*(B). \quad (\text{A.5})$$

A family containing only A and B therefore exhibits the reversal. The reversal is an elementary consequence of established score geometry, not an empirical explanation of any particular reported accuracy drop.

The quantifier matters: some families permit reversal; not every family does. If probabilities take only fixed endpoints 0 and 1, then  $\mathbb{E}V(p) = V(0) + [V(1) - V(0)]\mathbb{E}p$ , which preserves accuracy order when  $V(1) > V(0)$ . The same affine argument applies to any fixed two-value support with positive secant slope. If each context can attain its highest success probability independently of the others, or one feasible policy dominates pointwise, a nondecreasing  $V$  permits joint optimization of reward and accuracy. For fixed external events the issue does not arise, because the policy does not change the success profile. Weak propriety, nonconvex probability domains, report constraints, and richer non-affine mechanisms require their own analyses.

A larger accuracy weight nonetheless bounds the accuracy that exact optimization can sacrifice. Suppose  $G$  is bounded, truthful reports are feasible, and exact optimizers exist over a common policy family. Let  $\pi_\alpha$  maximize  $\alpha \text{Acc}(\pi) + \beta \mathbb{E}G(p_\pi)$  for  $\alpha > 0$ , and let  $\pi_{\text{acc}}$  maximize accuracy. Optimality implies

$$0 \leq \text{Acc}(\pi_{\text{acc}}) - \text{Acc}(\pi_\alpha) \leq \frac{\beta \text{osc}(G)}{\alpha}, \quad \text{osc}(G) = \sup G - \inf G. \quad (\text{A.6})$$

Subtract the two objective values and bound the difference in expected  $G$  by its oscillation. For negative binary Brier on  $[0, 1]$ ,  $\text{osc}(G) = 1/4$ . If the most accurate policy in a fixed finite family leads every other policy by a positive gap, a sufficiently large weight therefore selects it; because the gap can be arbitrarily small, no single finite weight does so for every family. The bound concerns exact optima and does not guarantee that finite neural optimization attains them.

#### A.4. Fixed-cost decision regret

This subsection bounds the regret of thresholding a report at the cost-derived threshold of Section 5.1 in terms of the report’s excess Brier risk, and separates that bound from what score calibration alone guarantees. Hold the joint distribution of answers and outcomes fixed, let  $p = \mathbb{E}[Z \mid \mathcal{F}]$ , and let  $c$  be  $\mathcal{F}$ -measurable. Accepting a correct answer earns  $b > 0$ , accepting a wrong one loses  $\ell > 0$ , and rejection earns zero. The expected gain from acceptance is  $(b + \ell)(p - \tau)$ , where  $\tau = \ell/(b + \ell)$ . Let  $D_p = \mathbf{1}\{p \geq \tau\}$  and  $D_c = \mathbf{1}\{c \geq \tau\}$ . The regret of thresholding the report instead of the conditional probability is

$$\mathcal{R}(D_c) = (b + \ell)\mathbb{E}[|p - \tau|\mathbf{1}\{D_c \neq D_p\}] \leq (b + \ell)\mathbb{E}|p - c| \leq (b + \ell)\sqrt{\mathbb{E}(p - c)^2}. \quad (\text{A.7})$$

When the two rules disagree,  $\tau$  lies between  $p$  and  $c$ , which gives the first inequality; Cauchy–Schwarz gives the second. Because the report and the conditional probability use the same information, the squared term is the excess Brier risk in Eq. (A.1). The result is not an ECE bound or a distribution-shift guarantee.

With only score calibration,  $\mathbb{E}[Z \mid C] = C$ , thresholding is optimal among policies observing that score under these fixed costs. It need not match a policy observing additional information or context-dependent costs. This distinction explains how an uninformative constant report can be calibrated, with thresholding optimal for anyone who observes only that report, yet support worse decisions than a report that separates cases.

## A.5. Finite-ensemble CRPS

This subsection derives the finite-ensemble bias of the empirical CRPS and the Gaussian example that Section 4.4 and Figure 6 summarize. Let  $X_1, \dots, X_K$  be independent draws from  $F$ , with finite first moments, and let  $y$  be a fixed outcome. The same argument applies conditionally on an input when the ensemble members are independent of its realized outcome. For the empirical forecast distribution,

$$C_K = \frac{1}{K} \sum_i |X_i - y| - \frac{1}{2K^2} \sum_{i,j} |X_i - X_j|.$$

The diagonal distances are zero, and each of the  $K(K-1)$  off-diagonal pairs has the expected distance between two independent draws. Therefore

$$\mathbb{E}C_K = \text{CRPS}(F, y) + \frac{\mathbb{E}|X - X'|}{2K}. \quad (\text{A.8})$$

Replacing  $K^2$  by  $K(K-1)$  in the pair term makes the score unbiased for population CRPS when  $K > 1$  under these sampling conditions. Ferro calls the relevant property fairness for ensemble forecasts [27]; it concerns the generating distribution, rather than demographic fairness. A proper score for the empirical distribution and a fair score for its generating distribution are different requirements.

For a transparent analytic example, let truth be  $Y \sim N(0, 1)$  and the predictive generator be  $N(0, \sigma^2)$ , with independent ensemble draws and outcome. Averaging the conventional empirical CRPS over ensemble draws and outcome gives the expected score and its minimizer:

$$L_K(\sigma) = \sqrt{\frac{2}{\pi}} \sqrt{1 + \sigma^2} - \frac{1 - 1/K}{\sqrt{\pi}} \sigma, \quad \sigma_K^* = \frac{1 - 1/K}{\sqrt{2 - (1 - 1/K)^2}}. \quad (\text{A.9})$$

The optima at  $K = 2, 4, 8$  are approximately 0.378, 0.626, and 0.788, matching the classical example in [27] (PDF p. 5, Fig. 2), whereas the fair score has optimum 1. This is an analytic distributional illustration, not an LLM experiment. An implementation with dependent sampling, changing valid-sample counts, or additional reward transformations does not inherit an unbiased policy-gradient claim from Eq. (A.8). In particular, DAR’s leave-one-out credit and subsequent normalization require separate analysis [42]; Appendix C.5 documents the released implementation.

## A.6. Exact decision data and verification

This subsection reports the exact population metrics behind the decision example of Section 5.2 and Figure 7 (Table A.2), and lists the numerical checks for this appendix. The example uses four equally probable contexts with success rates (0.1, 0.4, 0.6, 0.9). Truthful reports equal these rates; sharpened reports are (0.001, 0.1, 0.9, 0.999); the constant report is 0.5. Outcomes are integrated exactly rather than drawn as a finite sample.

**Table A.2. Exact population metrics for the decision example of Section 5.2.** Four equally probable contexts with success rates (0.1, 0.4, 0.6, 0.9). Truthful and sharpened reports share an ordering and hence an AUROC, but a fixed threshold of 0.8 accepts different sets under them, with opposite-signed utility when correct, wrong, and rejected answers score +1, −4, and 0.

| Report    | Population Brier | Population L1 reliability | AUROC | Coverage at 0.8 | Utility at +1/−4/0 |
|-----------|------------------|---------------------------|-------|-----------------|--------------------|
| Truthful  | 0.1650000        | 0                         | 0.825 | 0.25            | 0.125              |
| Sharpened | 0.2149005        | 0.1995                    | 0.825 | 0.50            | −0.125             |
| Constant  | 0.2500000        | 0                         | 0.500 | 0               | 0                  |

The Brier decomposition [90] separates an uncertainty term of 0.25, common to all three reports, from resolution and squared reliability error. Truthful and sharpened reports both have resolution 0.085, because sharpening is a strictly increasing, hence one-to-one, map that preserves the information in the truthful score. Their squared reliability errors are 0 and 0.0499005, respectively, whereas the constant report has zero resolution and zero squared reliability error.

These squared terms differ from the L1 reliability in Table A.2, the expected absolute gap between a report and the success rate given that report (Section 5.2).

The truthful and sharpened scores have the same ordering and tie structure and hence the same attainable threshold frontiers on this population. Reselecting a sharpened-score threshold strictly above 0.9 and no greater than 0.999 recovers the truthful acceptance set at 0.8. Equal AUROC alone, without the ordering condition, would not establish equal frontiers.

The calculation scripts supplied with the survey (Appendix F) verify Eqs. (A.1), (A.2), and (A.7) and the Brier algebra on 100 exact rational tables, and Eq. (A.3) on another 100 tables. They also enumerate all outcomes of discrete independent ensembles at  $K = 2, \dots, 5$  to check Eq. (A.8), and they check the analytic stationary points in Eq. (A.9). These checks validate the constructed examples and algebra. They do not reproduce any paper’s model results.

## A.7. Exponentiating a score changes the population incentive

This subsection shows that exponentiating a proper score can change which report maximizes expected reward, using the RLVP reward discussed in Section 3.4. It separates the teacher-matching incentive, the scoring property at the hard-label endpoint, and the limits of both results for finite training. RLVP’s probability reward is  $\exp\{-(1-\lambda)D_{\text{KL}}(e_Y\|p) + \lambda D_{\text{KL}}(q\|p)\}$ , with  $\lambda$  annealed from 1 to 0 [36] (PDF p. 3, Eq. 2). The following is our analysis of that formula. Fix a teacher vector  $q$ , a realized label  $y$ , and  $\lambda \in [0, 1]$ , and define  $m = (1 - \lambda)e_y + \lambda q$ . On compatible supports,

$$\begin{aligned} (1 - \lambda)D_{\text{KL}}(e_y\|p) + \lambda D_{\text{KL}}(q\|p) &= D_{\text{KL}}(m\|p) + C, \\ C &= (1 - \lambda)D_{\text{KL}}(e_y\|m) + \lambda D_{\text{KL}}(q\|m) \geq 0. \end{aligned} \tag{A.10}$$

Thus the isolated probability component for this fixed sample is maximized at  $p = m$ , but its maximum is  $\exp(-C)$ , generally below one. Replacing the weighted KL sum by  $D_{\text{KL}}(m\|p)$  alone drops a factor that can vary across outcomes and affect the composite reward. At the soft endpoint the maximizer equals the teacher vector, and fidelity to the teacher does not establish fidelity to the true label distribution.

At the hard endpoint,  $\exp[-D_{\text{KL}}(e_y\|p)] = p_y$ . For an external label distribution  $r$  and a report issued before observing the label, the ungated expected component is  $r^\top p$ . With  $r = (0.7, 0.3)$ , truthful reporting gives 0.58, whereas the strictly positive report  $(0.999, 0.001)$  gives 0.6996. Both choose the same class, so an added argmax-correctness reward and a constant valid-format reward do not remove this difference. A proper log score instead rewards  $\log p_y$ , not  $p_y$ . More generally, a strictly increasing transformation of a proper score need not preserve the maximizer of its expectation over outcomes.

This counterexample concerns the displayed component on an unrestricted report simplex. The paper’s teacher-correct gate changes the samples to which the component applies, and how that filtering is implemented remains unresolved. The counterexample is also not a prediction that finite RLVP training converges to a concentrated report: the curriculum, parameter sharing, optimizer, and stopping rule all matter. Nor does the paper claim a propriety theorem for this reward; its explicit proper-scoring statement concerns the evaluation Brier loss. An accompanying RLVP calculation check verifies Eq. (A.10), the endpoint reduction, and arithmetic transcribed from the paper’s result tables, without reproducing a model run.

## Appendix B. Comparative method inventory

This inventory compares 77 learning, estimation, elicitation, post-hoc, and boundary protocols, followed in its final subsection by four analyses with controlled learning experiments. Its 81 records count protocols and analyses, not distinct RL algorithms. Variants from one paper share a record, whereas Table 3 compares learning families through selected examples. Other theoretical work, diagnostic studies, and deployment descriptions cited in the survey serve as evidence and are not reclassified here as prediction methods.

Each record groups labeled fields into three columns: method and prediction object, supervision and update, and use and qualifications. The fields keep the target separate from the reporting interface, and feedback, objective, learning rule, and downstream use separate from one another. Descriptions refer to each paper’s stated protocol unless a public-code qualification is explicit. Source locators below use **physical PDF pages** of the archived version; the

decision-token calibration paper uses its ACL version, and the confidence-elicitation study its ICLR v2. Fixed-commit implementation checks and evaluation qualifications appear in Appendices C–D. The structured companion is `research/method_comparison_v0.4.csv`.

CE denotes cross-entropy; BCE, binary cross-entropy; KL, Kullback–Leibler divergence; SFT, supervised fine-tuning; FT, fine-tuning; RM, reward model; PRM, process reward model; IAS, instance-adaptive scaling; LoRA, low-rank adaptation; LOO, leave-one-out; MSE, mean squared error; NLL, negative log-likelihood; EM, exact match (in EM-FT and EM-RL, entropy minimization); IDK, “I don’t know”; MCQ, multiple-choice question; OOD, out of distribution; RAG, retrieval-augmented generation; BNPO, beta-normalization policy optimization; GPG, group policy gradient; and std, within-group reward standard deviation. PPO and GRPO name policy-optimization families rather than fully specifying losses or reductions. A numeric report, a class-token probability, and an action probability retain different statistical meanings even when they are trained with the same optimizer.

## B.1. Fixed outcomes and fixed-answer reporting

| Method and prediction object                                                                                                                                                                                                                                  | Supervision and update                                                                                                                                                                                                                                                                               | Use and qualifications                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Proper-score forecasting [28]</b><br>RL / hybrid protocol; pp. 3–5, Eqs. 1–7<br><i>Target</i> : External binary event<br><i>Output</i> : Reasoning + numeric probability                                                                                   | <i>Feedback / objective</i> : Observed event outcome; Log/Brier/spherical/Beta proper rewards<br><i>Update</i> : Group mean, no std, no KL                                                                                                                                                           | <i>Use</i> : Forecasting<br><i>Qualifications</i> : Fixed-step comparison; token budgets and seed rankings differ                                                                                                   |
| <b>Verifiable forecast rewards [116]</b><br>RL / hybrid protocol; pp. 4–6, 12, Eq. 1<br><i>Target</i> : Conditional game win probability<br><i>Output</i> : Direct numeric span or reasoning + span                                                           | <i>Feedback / objective</i> : Historical state-bucket empirical rate; Squared error to teacher rate<br><i>Update</i> : GRPO with reward scaling; direct-report KL 0.01; masked-report KL 0 and final-span credit                                                                                     | <i>Use</i> : Forecasting<br><i>Qualifications</i> : Empirical state-bucket teacher (built from training outcomes with hierarchical shrinkage) and temporal splits determine the target                              |
| <b>Outcome-based forecasting RL [101]</b><br>RL / hybrid protocol; pp. 2–4, 7<br><i>Target</i> : External future binary event<br><i>Output</i> : Reasoning + probability                                                                                      | <i>Feedback / objective</i> : Market/resolved future outcomes; Negative Brier plus format constraints<br><i>Update</i> : ReMax vs GRPO vs mean-only GRPO                                                                                                                                             | <i>Use</i> : Forecasting / simulated trading<br><i>Qualifications</i> : Temporal checks and ensembles; not live profit evidence                                                                                     |
| <b>Future-as-Label [5]</b><br>RL / hybrid protocol; pp. 2–6; p. 5, Table 1<br><i>Target</i> : Future external binary event, given news before the forecast date<br><i>Output</i> : Reasoning + parsed probability clipped to [0.001, 0.999]                   | <i>Feedback / objective</i> : Outcomes resolved by a frozen LLM from later news; Log score<br><i>Update</i> : Group-mean centering over four trajectories, no standard-deviation division; offline historical outcomes                                                                               | <i>Use</i> : Forecasting<br><i>Qualifications</i> : No validation split or early stopping; resolver errors can contaminate labels; ensembling was not uniformly better                                              |
| <b>Market-probability forecasting RL [102]</b><br>RL / hybrid protocol; pp. 9–10, Eqs. 2.1–2.3; pp. 15–17; p. 21, Table 5.1; p. 28<br><i>Target</i> : Binary event probability as priced by a prediction market<br><i>Output</i> : Reasoning + 0–100 forecast | <i>Feedback / objective</i> : Polymarket prices as soft labels (resolved outcomes in an unsuccessful variant); Format plus negative cross-entropy capped at $-5$ , or negative squared error<br><i>Update</i> : GRPO with group-standardized advantages; full fine-tuning; KL coefficient unreported | <i>Use</i> : Forecasting<br><i>Qualifications</i> : Evaluated against market prices, not resolved outcomes; price and context timing unverified; single-author thesis                                               |
| <b>Rewarding Doubt [29]</b><br>RL / hybrid protocol; pp. 4–6, 15–16<br><i>Target</i> : Fixed-answer correctness<br><i>Output</i> : Answer-conditioned confidence number                                                                                       | <i>Feedback / objective</i> : Correctness labels for fixed answers; Clipped log-score confidence reward<br><i>Update</i> : PPO updates confidence generation; answer held as input                                                                                                                   | <i>Use</i> : Answer assessment<br><i>Qualifications</i> : Discrete reports, clipping and shared weights delimit the ideal score argument; only answer-assessment trajectories are sampled for the confidence update |

## B.2. Joint reports, expert feedback and communication

| Method and prediction object                                                                                                                                                                                                                                              | Supervision and update                                                                                                                                                                                                                                                                                                                 | Use and qualifications                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RLCR [30]</b><br>RL / hybrid protocol; pp. 4, 16–18; code qualification in Appendix C<br><i>Target:</i> Chosen-answer correctness<br><i>Output:</i> Answer + uncertainty analysis + number                                                                             | <i>Feedback / objective:</i> Binary verifier;<br>$z - (c - z)^2$<br><i>Update:</i> Group mean without std; no KL; BNPO                                                                                                                                                                                                                 | <i>Use:</i> Answering/abstention<br><i>Qualifications:</i> Pointwise reporting incentive; joint capability separate; released recipe also includes format and small report-dependent bonuses                        |
| <b>Multi-RLCR [103]</b><br>RL / hybrid protocol; pp. 3–4, 19, its Appendix B<br><i>Target:</i> Candidate-specific correctness marginals<br><i>Output:</i> Multiple answers + numeric confidences                                                                          | <i>Feedback / objective:</i> Answer-set verifier; Sum correctness minus mean Brier<br><i>Update:</i> Group mean without std; sequence-level update                                                                                                                                                                                     | <i>Use:</i> Answer sets<br><i>Qualifications:</i> Marginals do not define joint set probability                                                                                                                     |
| <b>OpenForecaster [33]</b><br>RL / hybrid protocol; pp. 4, 8–9, 17, 23–24<br><i>Target:</i> Correctness of a self-chosen open-ended forecast<br><i>Output:</i> Reasoning + free-form answer + probability                                                                 | <i>Feedback / objective:</i> Outcomes from later news, judged by an LLM; Correctness plus $z - (q - z)^2$ for free-form questions; negative squared error for binary questions<br><i>Update:</i> GRPO with group-mean centering, no standard-deviation division; KL 0.005; asymmetric clipping                                         | <i>Use:</i> Forecasting<br><i>Qualifications:</i> Joint answer choice and reporting; the added correctness term counters an “Unknown” answering strategy; test questions filtered for resolvability                 |
| <b>VL-Calibration [32]</b><br>RL / hybrid protocol; pp. 4–5, Eqs. 2–12; p. 12, Table 5<br><i>Target:</i> Answer correctness (holistic report); visual-certainty proxy (visual report)<br><i>Output:</i> Visual and reasoning 0–10 confidences combined by a harmonic mean | <i>Feedback / objective:</i> Answer verifier; batch-standardized sigmoid of log image-masking KL minus log token entropy; Accuracy $- 2 \times$ holistic Brier loss $- 0.4 \times$ visual squared error<br><i>Update:</i> GRPO with group-normalized advantages, clipping and KL; negative visual-token advantages rescaled by 0.9–1.1 | <i>Use:</i> Multimodal reasoning<br><i>Qualifications:</i> The visual report has batch-relative, not probability, semantics; weight 2 on the Brier term breaks best effort below $p = 1/4$ ; no post-hoc comparator |
| <b>DCPO [100]</b><br>RL / hybrid protocol; p. 6, Eqs. 20–24; p. 17<br><i>Target:</i> Hybrid prompt/response correctness<br><i>Output:</i> Post-answer numeric confidence                                                                                                  | <i>Feedback / objective:</i> Group mean mixed with per-rollout correctness; Correctness plus negative absolute error to a mixed group/instance target<br><i>Update:</i> Separate standardized advantages and token masks; no KL                                                                                                        | <i>Use:</i> Reasoning confidence<br><i>Qualifications:</i> Implemented absolute error and stated proper-score theory have different incentives                                                                      |
| <b>CoCA [59]</b><br>RL / hybrid protocol; p. 4, Eqs. 6–11; pp. 7–8<br><i>Target:</i> Current-policy prompt solvability<br><i>Output:</i> Pre-answer numeric confidence                                                                                                    | <i>Feedback / objective:</i> Current rollout success rate; Separate correctness and squared-error rewards<br><i>Update:</i> Segmented std-normalized advantages; no KL                                                                                                                                                                 | <i>Use:</i> Early confidence and prospective routing<br><i>Qualifications:</i> Pre-report conditions later generation; demonstrated report cost differs from routing utility                                        |
| <b>CALIBER [60]</b><br>RL / hybrid protocol; pp. 5–8, 18<br><i>Target:</i> Pre-answer policy solvability and post-answer correctness<br><i>Output:</i> Pre- and post-answer numeric confidences                                                                           | <i>Feedback / objective:</i> Group mean pre; instance outcome post; Two squared-error rewards + accuracy/format<br><i>Update:</i> No std/KL; warmup then joint                                                                                                                                                                         | <i>Use:</i> Reasoning confidence<br><i>Qualifications:</i> In-distribution evaluation split selects checkpoints; warmup addresses weak variation in the pre-answer training signal                                  |

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## B.2. Joint reports, expert feedback and communication (continued)

| Method and prediction object                                                                                                                                                                                                                                                                         | Supervision and update                                                                                                                                                                                                                                                                                                                                                                 | Use and qualifications                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Calibration-aware decision-token RL [7]</b><br>RL / hybrid protocol; ACL pp. 7–8, Eq. 4; p. 13<br><i>Target:</i> Final candidate-token probabilities<br><i>Output:</i> Reasoning + decision-token logits                                                                                          | <i>Feedback / objective:</i> Gold class and sampled-answer correctness; RL accuracy; local CE one-hot/uniform<br><i>Update:</i> Zero decision-token RL advantage plus local CE; shared parameters; no KL                                                                                                                                                                               | <i>Use:</i> Classification/moderation<br><i>Qualifications:</i> Wrong-response uniform target is a heuristic; test-set checkpoint selection                                                                                                                  |
| <b>SaySelf [99]</b><br>RL / hybrid protocol; pp. 3–5, Eq. 2; p. 6, Eq. 4<br><i>Target:</i> Chosen-answer correctness<br><i>Output:</i> Confidence + reflection                                                                                                                                       | <i>Feedback / objective:</i> Cluster-frequency SFT; correctness feedback on scales that differ between the paper and the current evaluator; $1 - 2(c - z)^2$<br><i>Update:</i> PPO after SFT                                                                                                                                                                                           | <i>Use:</i> Selective retrieval<br><i>Qualifications:</i> Published ECE is a per-example absolute error; current evaluator confirms no bins and uses graded correctness                                                                                      |
| <b>PPO-M / PPO-C [6]</b><br>RL / hybrid protocol; pp. 5–6, Eqs. 2–3; pp. 9, 21–22<br><i>Target:</i> Confidence relative to response quality<br><i>Output:</i> Answer + confidence number                                                                                                             | <i>Feedback / objective:</i> Preference pairs and learned reward-model quality; M: confidence-conditioned RM ranking; C: relative-quality reward modulation<br><i>Update:</i> RM adaptation then PPO, or online reward transformation; CDPO extension                                                                                                                                  | <i>Use:</i> RLHF confidence<br><i>Qualifications:</i> Reward-model quality is a proxy; PPO-C is linear in reported confidence and measures quality relative to a moving reward baseline                                                                      |
| <b>UARM [45]</b><br>RL / hybrid protocol; pp. 4–7, Eqs. 6–11, 20–22; p. 8, Table 1<br><i>Target:</i> Scalar preference-proxy reward label for a prompt–response pair<br><i>Output:</i> Reward quantiles; median reward and conformal interval                                                        | <i>Feedback / objective:</i> Pointwise helpfulness, overall-score or severity labels; Pinball loss, then a split-conformal interval fitted on a held-out 20% of the training split<br><i>Update:</i> GRPO with clipping and KL; unweighted group mean; signal standard deviation after subtracting mean interval noise; per-sample weight $\sigma_s^2 / (\sigma_s^2 + \sigma_{n,i}^2)$ | <i>Use:</i> RLHF reward reliability<br><i>Qualifications:</i> Reported evidence is uncertainty-ranked reward prediction, without policy-training results; coverage presumes exchangeability with reward-model data; weighted advantages need not be zero-sum |
| <b>RLMF [34]</b><br>RL / hybrid protocol; pp. 5, 28, 34–36<br><i>Target:</i> Sentence confidence relative to sampling consistency<br><i>Output:</i> Sentence scores and linguistic uncertainty                                                                                                       | <i>Feedback / objective:</i> Sampled-answer consistency plus correctness judge; Squared proxy alignment, factuality and format rewards<br><i>Update:</i> GRPO without std; selectively scales positive faithfulness advantages                                                                                                                                                         | <i>Use:</i> Uncertainty expression<br><i>Qualifications:</i> Proxy consistency and factual correctness are separate evaluation targets                                                                                                                       |
| <b>Semantic-level reward (CSR) [38]</b><br>RL / hybrid protocol; pp. 4, 12–14; p. 6, Table 1<br><i>Target:</i> Per-question accuracy, estimated from semantic agreement among $K = 8$ samples<br><i>Output:</i> Natural-language answers; confidence $\exp(-H)$ of the semantic-cluster distribution | <i>Feedback / objective:</i> LLM-judged correctness and pairwise semantic equivalence; Correctness plus a cross-entropy agreement term weighted from 0.1 to 0.2<br><i>Update:</i> GRPO with LoRA and KL 0.1; standard-deviation and clipping settings unstated                                                                                                                         | <i>Use:</i> Question answering<br><i>Qualifications:</i> The reported number is an inverse effective count of answer classes, not a correctness probability; accuracy can fall as ranking improves; single seed                                              |
| <b>LoVeC [35]</b><br>RL / hybrid protocol; pp. 4–6, Eqs. 1–8; p. 15, Eq. 9 and reward code<br><i>Target:</i> Graded judge veracity of each generated sentence<br><i>Output:</i> Sentence text with 0–10 confidence tags                                                                              | <i>Feedback / objective:</i> GPT-4o or frozen-policy fact-checks with retrieved evidence; Clipped, normalized, power-stretched log score plus a small correctness bonus (GRPO); pairs differing only in scores (DPO)<br><i>Update:</i> GRPO with group standard deviation, token mean and KL to a frozen copy; DPO after an SFT format warm start; LoRA                                | <i>Use:</i> Long-form factuality<br><i>Qualifications:</i> Graded labels identify a mean judge rating; the free-form reward is not best-effort; no post-hoc or exact expected-score comparator                                                               |

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## B.2. Joint reports, expert feedback and communication (continued)

| Method and prediction object                                                                                                                                                                                                            | Supervision and update                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Use and qualifications                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Linguistic calibration</b> [117]<br>RL / hybrid protocol; pp. 8–9, 48–52<br><i>Target</i> : Reader-induced answer belief<br><i>Output</i> : Long-form linguistic uncertainty                                                         | <i>Feedback / objective</i> : Reader probability model / QA answers; Reader log score + normalization penalty<br><i>Update</i> : PPO with KL and learned surrogates                                                                                                                                                                                                                                                                                                                                    | <i>Use</i> : Reader decisions<br><i>Qualifications</i> : Depends on reader model and scored question                                                                                                                            |
| <b>LACIE</b> [37]<br>preference-optimization protocol; pp. 4–7, 13–14; p. 7, Table 1<br><i>Target</i> : Listener acceptance consistent with answer correctness<br><i>Output</i> : Answer text with implicit and explicit certainty      | <i>Feedback / objective</i> : Exact-match correctness and a fixed Mistral-7B listener’s acceptance probability, split at the training median; Preferences ranking correct-accepted and wrong-rejected above correct-rejected, above wrong-accepted<br><i>Update</i> : DPO with QLoRA; checkpoints selected on listener precision and recall                                                                                                                                                            | <i>Use</i> : Reader decisions<br><i>Qualifications</i> : Risk-ordered preferences are not a proper score; abstention rises, and the unweighted ECE excludes abstentions                                                         |
| <b>RLVP</b> [36]<br>RL / hybrid protocol; pp. 2–6, Eqs. 1–4; p. 10, Table 4<br><i>Target</i> : Full class-label distribution intended to approximate $P(Y   x)$<br><i>Output</i> : Generated chain of thought + JSON probability vector | <i>Feedback / objective</i> : Observed class and same-training-split XGBoost soft probabilities; teacher-correct filtering; Format and argmax correctness rewards plus $\exp\{-(1 - \alpha)D_y - \alpha D_t\}$ , with $D_y = \text{KL}(e_y \  p)$ , $D_t = \text{KL}(q \  p)$ , one-hot label $e_y$ , teacher $q$ , and teacher weight $\alpha : 1 \rightarrow 0$<br><i>Update</i> : Paper: GRPO without reference-policy KL; 8 rollouts, temperature 0.7; public repository unavailable at inspection | <i>Use</i> : Multiclass tabular and mixed text/tabular classification<br><i>Qualifications</i> : Exponentiated KL is not a proper-score guarantee; full-vector downstream evaluation needed; teacher-filtering scope unresolved |
| <b>Verbalized Probability Distribution</b> [80]<br>RL / hybrid protocol; pp. 2–4, 14<br><i>Target</i> : Chosen-answer confidence within reported vector<br><i>Output</i> : Full verbalized candidate vector                             | <i>Feedback / objective</i> : Correctness of the finally selected answer; separate validity penalty; Chosen-answer $z - (c - z)^2$<br><i>Update</i> : GRPO with mean-only group centering                                                                                                                                                                                                                                                                                                              | <i>Use</i> : MCQ confidence<br><i>Qualifications</i> : Full-vector output; selected-answer scalar supervision                                                                                                                   |
| <b>Judging with Confidence</b> [82]<br>RL / hybrid protocol; pp. 4, 18–20<br><i>Target</i> : Preference distribution<br><i>Output</i> : Autorater numeric probability                                                                   | <i>Feedback / objective</i> : Persona-conditioned judge votes; Brier/log rewards for distribution matching<br><i>Update</i> : GRPO; exact normalization incompletely specified                                                                                                                                                                                                                                                                                                                         | <i>Use</i> : LLM-as-a-judge<br><i>Qualifications</i> : Main annotators are synthetic personas; soft-preference and majority-label metrics differ                                                                                |

### B.3. Ranking, process and agent feedback

| Method and prediction object                                                                                                                                                       | Supervision and update                                                                                                                                                                                            | Use and qualifications                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CAPO [39]</b><br>RL / hybrid protocol; pp. 3–6, Eqs. 5, 16–20<br><i>Target:</i> Correct-vs-incorrect response ranking<br><i>Output:</i> Mean response log probability           | <i>Feedback / objective:</i> Math verifier; Pairwise logistic AUC surrogate<br><i>Update:</i> Custom advantage; no std/KL; response mask                                                                          | <i>Use:</i> Sample aggregation<br><i>Qualifications:</i> Ranking score is not a correctness probability                                                         |
| <b>CONQORD [98]</b><br>RL / hybrid protocol; pp. 4–5, Eqs. 1–5; pp. 7–8<br><i>Target:</i> Relative confidence vs reward quality<br><i>Output:</i> Answer + numeric confidence      | <i>Feedback / objective:</i> Preference-trained quality reward; Quality plus confidence-quality pair differences<br><i>Update:</i> PPO with KL                                                                    | <i>Use:</i> RAG triggering<br><i>Qualifications:</i> Ranking/covariance objective is not proper probability elicitation                                         |
| <b>RLSR [40]</b><br>RL / hybrid protocol; pp. 4–6, Eqs. 2.13–2.15; pp. 7, 18<br><i>Target:</i> Confidence ranking for risk-coverage<br><i>Output:</i> Answer + confidence          | <i>Feedback / objective:</i> Binary correctness; Harmonic rank-weighted error / AURC<br><i>Update:</i> Group-centered advantage, no std/KL                                                                        | <i>Use:</i> Selective prediction<br><i>Qualifications:</i> Ranking can improve while ECE worsens; current released batch size differs from the paper            |
| <b>RLCM [77]</b><br>RL / hybrid protocol; pp. 4–5, 19<br><i>Target:</i> Success of an answer forced from a reasoning prefix<br><i>Output:</i> Hidden-state probe probability       | <i>Feedback / objective:</i> Monte Carlo forced-answer success by prefix; Probe BCE plus trajectory correctness and confidence-margin reward<br><i>Update:</i> Probe-only supervised gradient; GRPO policy update | <i>Use:</i> Early exit and answer aggregation<br><i>Qualifications:</i> Prefix success differs from step validity; margin rewards do not identify probabilities |
| <b>DualStake [142]</b><br>RL / hybrid protocol; pp. 2–5; pp. 13–14<br><i>Target:</i> Evidence and final-answer confidence<br><i>Output:</i> Two confidence numbers in search trace | <i>Feedback / objective:</i> Answer-token F1 during training; evaluation conventions differ; Correctness plus linear confidence stakes<br><i>Update:</i> GRPO with warmup weights                                 | <i>Use:</i> Search agents<br><i>Qualifications:</i> Clipped linear stake does not identify interior probabilities; metric targets differ                        |
| <b>C3RL / CAS [76]</b><br>RL / hybrid protocol; pp. 3–4, 13, 15<br><i>Target:</i> Correctness confidence / stopping score<br><i>Output:</i> Confidence 1–10                        | <i>Feedback / objective:</i> Verifier and base-model reference groups; Thresholded confidence and reference rewards<br><i>Update:</i> VeRL clipped RL; KL 0.001                                                   | <i>Use:</i> Adaptive sample count<br><i>Qualifications:</i> Threshold reward not proper; OOD hyperparameter selection                                           |
| <b>RefGRPO [120]</b><br>behavior / boundary protocol; pp. 5–6, 15<br><i>Target:</i> Binary post-feedback correctness assessment<br><i>Output:</i> Reflection bit                   | <i>Feedback / objective:</i> Execution-grounded task outcomes; Outcome + decaying reflection-match bonus<br><i>Update:</i> Std-normalized GRPO+; no KL                                                            | <i>Use:</i> Self-verifier / selection<br><i>Qualifications:</i> Binary self-assessment accuracy differs from continuous-probability calibration                 |
| <b>CURE [104]</b><br>RL / hybrid protocol; pp. 4–6, 13–14<br><i>Target:</i> Atomic-claim correctness<br><i>Output:</i> Claim + numeric confidence + reasoning                      | <i>Feedback / objective:</i> Retrieved verification + teacher corrections; DPO calibration then masked factuality RL<br><i>Update:</i> SFT/format RL, calibration DPO, then claim-token factuality RL             | <i>Use:</i> Claim filtering<br><i>Qualifications:</i> Claim precision, recall and confidence quality are different endpoints                                    |

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## B.3. Ranking, process and agent feedback (continued)

| Method and prediction object                                                                                                                                                                  | Supervision and update                                                                                                                                                                                          | Use and qualifications                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CAR / MSCR [105]</b><br>RL / hybrid protocol; p. 5; pp. 7–8<br><i>Target:</i> Final tool-agent answer correctness<br><i>Output:</i> Numeric confidence after tool trace                    | <i>Feedback / objective:</i> EM outcome; Weighted Brier or margin-separated reward<br><i>Update:</i> Search-R1/GRPO; incomplete hyperparameters                                                                 | <i>Use:</i> Search/code agents<br><i>Qualifications:</i> Unequal MSCR weights change elicited probability                                           |
| <b>RAC [107]</b><br>RL / hybrid protocol; pp. 3–6, 16<br><i>Target:</i> Relative correctness / evidence confidence<br><i>Output:</i> Numeric multimodal confidence                            | <i>Feedback / objective:</i> Correctness and corruption comparisons; Ranking hinge + clean/corrupt directional hinge<br><i>Update:</i> GRPO reward shaping; paired branches                                     | <i>Use:</i> Multimodal reasoning<br><i>Qualifications:</i> Directional constraints do not identify absolute probabilities                           |
| <b>Entropy-target RL [119]</b><br>RL / hybrid protocol; pp. 3, 7–8<br><i>Target:</i> Question-level error proxy<br><i>Output:</i> Separate reflection + numeric uncertainty                   | <i>Feedback / objective:</i> Sampling entropy mapped through held-out Platt fitting; Tolerance-clipped absolute error to teacher<br><i>Update:</i> GRPO uncertainty adapter; answer generation fixed            | <i>Use:</i> Amortized uncertainty<br><i>Qualifications:</i> Question-level proxy does not automatically become answer-conditioned error probability |
| <b>C<sup>2</sup>GSPG [56]</b><br>RL / hybrid protocol; pp. 4–6, Eqs. 5–10; pp. 15–18<br><i>Target:</i> Sequence-likelihood confidence<br><i>Output:</i> Geometric-mean trajectory probability | <i>Feedback / objective:</i> Verifier; Sequence policy objective plus BCE on geometric-mean token likelihood<br><i>Update:</i> Centered reward divided by one minus old confidence; direct BCE; conflict gating | <i>Use:</i> Reasoning<br><i>Qualifications:</i> Sequence likelihood is a trained proxy, not inherently a correctness posterior                      |

## B.4. Actions and selective decisions

| Method and prediction object                                                                                                                                                                                                | Supervision and update                                                                                                                                                                                                                           | Use and qualifications                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Behaviorally calibrated RL [44]</b><br>RL / hybrid protocol; pp. 5–6, Eqs. 3–4; pp. 25–26<br><i>Target:</i> Correctness confidence for threshold decisions<br><i>Output:</i> Answer probability or outcome-value readout | <i>Feedback / objective:</i> Final correctness and threshold-dependent utilities; Threshold-integrated Brier or clipped log-based score<br><i>Update:</i> PPO confidence/value variants; Confidence-CE uses GRPO; post-hoc acceptance thresholds | <i>Use:</i> Selective answering<br><i>Qualifications:</i> Threshold-weighting convention determines the score; claim probabilities require separate supervision |
| <b>TruthRL [43]</b><br>behavior / boundary protocol; pp. 3–4, 16<br><i>Target:</i> Answer/abstain action<br><i>Output:</i> Answer or refusal                                                                                | <i>Feedback / objective:</i> Task correctness and explicit abstention utility; Ternary correct/abstain/wrong reward<br><i>Update:</i> GRPO on discrete action behavior                                                                           | <i>Use:</i> Selective answering<br><i>Qualifications:</i> Action probabilities are not explicit correctness reports                                             |
| <b>UCPO [110]</b><br>behavior / boundary protocol; p. 4, Eqs. 3–5; pp. 13–14<br><i>Target:</i> Uncertainty action preference<br><i>Output:</i> Answer or explicit uncertain behavior                                        | <i>Feedback / objective:</i> Right/wrong/uncertain outcome groups; Dynamic uncertainty advantage<br><i>Update:</i> Ternary decoupling and rollout filtering                                                                                      | <i>Use:</i> Selective answering<br><i>Qualifications:</i> Behavioral metrics; no numeric-probability calibration evidence                                       |

Continued on next page

## B.4. Actions and selective decisions (continued)

| Method and prediction object                                                                                                                                                                                                                        | Supervision and update                                                                                                                                                                                                                                | Use and qualifications                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Intellectual humility</b> [106]<br>behavior / boundary protocol; pp. 2–3, 5<br><i>Target</i> : Answer-vs-abstain action<br><i>Output</i> : Answer or IDK                                                                                         | <i>Feedback / objective</i> : Task verifier; Correct +1; wrong –1; tuned abstain reward<br><i>Update</i> : GRPO after optional abstention SFT                                                                                                         | <i>Use</i> : Abstention<br><i>Qualifications</i> : Direct behavioral evidence; coverage and raw accuracy must accompany accepted-answer precision                                           |
| <b>BinaryPPO</b> [125]<br>behavior / boundary protocol; p. 3, Eqs. 3–9; pp. 4–5<br><i>Target</i> : Binary label-action probability<br><i>Output</i> : Two-action policy probabilities                                                               | <i>Feedback / objective</i> : Static binary class labels; Confidence-dependent shaping plus value, supervised and entropy terms<br><i>Update</i> : PPO-style contextual action learning; printed signs unresolved                                     | <i>Use</i> : Binary classification<br><i>Qualifications</i> : Standard probability-quality evaluation is missing; retained as a boundary protocol                                           |
| <b>GLiClass</b> [81]<br>behavior / boundary protocol; pp. 3–6; p. 8, Table 2<br><i>Target</i> : Match between an input text and each candidate label<br><i>Output</i> : Independent sigmoid label scores from an encoder                            | <i>Feedback / objective</i> : Multilabel classification data; Configurable weighted rewards on sampled or thresholded label choices, with value, KL, and entropy terms<br><i>Update</i> : Pretraining, a PPO stage (clip 0.2), and later adaptation   | <i>Use</i> : Open-label classification<br><i>Qualifications</i> : No calibration metric or RL-stage ablation; sigmoid scores across candidate sets need not be comparable                   |
| <b>Calibrate-Then-Act</b> [78]<br>behavior / boundary protocol; pp. 3–6; p. 8, Table 4; p. 17, Table 6; p. 22<br><i>Target</i> : Explore-or-commit action under stated costs<br><i>Output</i> : Actions conditioned on supplied prior probabilities | <i>Feedback / objective</i> : Discounted task success; priors from isotonic-recalibrated verbal confidence (QA) or a supervised format predictor (FileReading)<br><i>Update</i> : Prompting, or GRPO (group 8, KL 0.01) with the priors in the prompt | <i>Use</i> : Cost-aware exploration<br><i>Qualifications</i> : Probabilities come from post-hoc or supervised fitting, not RL; priors fitted on the evaluation distribution; one base model |

## B.5. Numerical prediction

| Method and prediction object                                                                                                                                                                                                                                 | Supervision and update                                                                                                                                                                                                                      | Use and qualifications                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DAR</b> [42]<br>RL / hybrid protocol; pp. 3–4, 14–17; code qualification in Appendix C<br><i>Target</i> : Numerical predictive distribution<br><i>Output</i> : Multiple generated numeric samples                                                         | <i>Feedback / objective</i> : Regression targets; Empirical CRPS with K-squared pair denominator and leave-one-out contribution rewards<br><i>Update</i> : GRPO centering/std after LOO; KL loss 0.001; valid-sample filtering and fallback | <i>Use</i> : Regression<br><i>Qualifications</i> : Finite-set score, LOO credit and normalized update require separate interpretation                                                                                         |
| <b>REAL</b> [41]<br>RL / hybrid protocol; pp. 4–5, Eqs. 4–7; p. 8, Table 9; code qualification in Appendix C<br><i>Target</i> : Numeric predictive mean and correlation with ratings<br><i>Output</i> : Numerical token probabilities summarized as a scalar | <i>Feedback / objective</i> : Human ratings; Paper: squared mean error plus target log likelihood<br><i>Update</i> : Paper: LOO/std/advantage clamp and weighted direct derivative; current GPG launcher has direct coefficient 0 and KL 0  | <i>Use</i> : LLM-as-a-judge<br><i>Qualifications</i> : Paper and public launcher differ; mean/correlation evidence is not distributional calibration; the paper’s direct-gradient weight is distinct from reference-policy KL |

## B.6. Supervised, elicitation and post-hoc comparators

| Method and prediction object                                                                                                                                                                                                                                                                                                          | Supervision and update                                                                                                                                                                                                     | Use and qualifications                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ConfTuner [57]</b><br>non-RL comparator; p. 4, Eq. 1; pp. 23, 26<br><i>Target</i> : Semantic value of confidence number<br><i>Output</i> : Distribution over confidence-number tokens                                                                                                                                              | <i>Feedback / objective</i> : Fixed-answer correctness; Exact expected squared error across numeric tokens<br><i>Update</i> : Direct backpropagation of the exact expected loss; optional supervised language loss         | <i>Use</i> : Confidence reporting<br><i>Qualifications</i> : Probability of reporting 0.8 is not correctness probability 0.8                                                                                               |
| <b>Verbalized-confidence SFT [88]</b><br>non-RL comparator; pp. 4–6, 16–18<br><i>Target</i> : Fixed-answer correctness/subtask rate<br><i>Output</i> : Confidence words or percentages                                                                                                                                                | <i>Feedback / objective</i> : Empirical subtask accuracies; Supervised confidence-token likelihood<br><i>Update</i> : Supervised fine-tuning; fixed-answer protocol                                                        | <i>Use</i> : Reporting<br><i>Qualifications</i> : Subtask rate labels and decoding determine the report; RL is future work                                                                                                 |
| <b>Supervised correctness estimation [66]</b><br>non-RL comparator; pp. 4–7, 10; its Appendix C.1<br><i>Target</i> : Correctness of a fixed generated answer<br><i>Output</i> : Probe or Yes/No-style language classifier                                                                                                             | <i>Feedback / objective</i> : Labeled fixed answers; Binary classification with optional representation regularization<br><i>Update</i> : Frozen-feature probe or LoRA estimator; held-out temperature fitting             | <i>Use</i> : Answer assessment<br><i>Qualifications</i> : Generator and estimator are separate model copies                                                                                                                |
| <b>Just Ask [89]</b><br>elicitation comparator; pp. 2–5, 8<br><i>Target</i> : Reported correctness of generated answers<br><i>Output</i> : Numeric/linguistic confidence prompts                                                                                                                                                      | <i>Feedback / objective</i> : Existing model; labels only for optional calibration; Prompted elicitation; optional NLL temperature fitting<br><i>Update</i> : No LLM update; prompt/decoding choice and post-hoc fitting   | <i>Use</i> : Answer assessment<br><i>Qualifications</i> : Some methods change the selected answer; temperature-scaled results use calibration labels                                                                       |
| <b>Confidence elicitation study [96]</b><br>elicitation comparator; ICLR v2 pp. 3–6<br><i>Target</i> : Reported answer confidence and sampling-based uncertainty<br><i>Output</i> : Black-box prompts and aggregated samples                                                                                                          | <i>Feedback / objective</i> : Existing generator; correctness labels for evaluation; Comparison of prompting, sampling and aggregation protocols<br><i>Update</i> : No policy update; inference-time confidence extraction | <i>Use</i> : Answer assessment and selection<br><i>Qualifications</i> : Calibration error and error ranking vary with elicitation protocol                                                                                 |
| <b>Semantic entropy [67], [97]</b><br>elicitation comparator; Kuhn et al. pp. 4–7, Eqs. 2–4; Farquhar et al. pp. 1–5, 7–9, Eq. 5<br><i>Target</i> : No declared event; dispersion of sampled answers over meaning classes<br><i>Output</i> : Entropy of meaning-class probabilities or, in the discrete variant, of class frequencies | <i>Feedback / objective</i> : No outcome labels; about ten samples per question clustered by bidirectional entailment<br><i>Update</i> : No training; model fixed                                                          | <i>Use</i> : Error detection and rejection<br><i>Qualifications</i> : Evaluated by AUROC (the later study also by rejection accuracy), not calibration; does not detect consistent errors; depends on the entailment judge |
| <b>CFT / RCFT [135]</b><br>non-RL comparator; pp. 5–6, Eq. 6; p. 7, Table 2<br><i>Target</i> : Candidate-label probability<br><i>Output</i> : Candidate logits                                                                                                                                                                        | <i>Feedback / objective</i> : Gold classification label; Answer-only CE (CFT); prompt/answer loss plus ECE surrogate (RCFT)<br><i>Update</i> : Supervised fine-tuning                                                      | <i>Use</i> : Classification<br><i>Qualifications</i> : Main CFT gains use no explicit calibration regularizer; ECE and accuracy trade off                                                                                  |

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## B.6. Supervised, elicitation and post-hoc comparators (continued)

| Method and prediction object                                                                                                                                                                                                                              | Supervision and update                                                                                                                                                                                                                                                                     | Use and qualifications                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CATTO [136]</b><br>preference-optimization comparator;<br>pp. 4–5, Eqs. 6–7; pp. 24–26<br><i>Target:</i> Token confidence, evaluated at final labels<br><i>Output:</i> Token/class probabilities                                                       | <i>Feedback / objective:</i> Preferred/rejected examples and probability-margin proxy; DPO plus confidence-linear surrogate regularization<br><i>Update:</i> Differentiable preference optimization                                                                                        | <i>Use:</i> Classification and candidate selection<br><i>Qualifications:</i> Fixed-target linear surrogate is not strictly proper; comparison changes can affect decoding; DPO changes token probabilities without online rollout-reward learning |
| <b>Label-smoothed SFT [108]</b><br>non-RL comparator; pp. 2, 5–6, 8, 20<br><i>Target:</i> Token distribution, evaluated on class confidence<br><i>Output:</i> Vocabulary and label-token probabilities                                                    | <i>Feedback / objective:</i> Instruction demonstrations; Cross-entropy to one-hot/uniform mixture<br><i>Update:</i> Supervised instruction tuning; efficient smoothed-CE kernel                                                                                                            | <i>Use:</i> Class confidence<br><i>Qualifications:</i> SFT evidence does not establish persistence through later RL                                                                                                                               |
| <b>CaOPD [61]</b><br>non-RL comparator; pp. 3–6, 28, 34–36<br><i>Target:</i> Current student’s marginal success rate on the question<br><i>Output:</i> Reasoning + answer + numeric confidence                                                            | <i>Feedback / objective:</i> Verifier pass rate over $K = 8$ student rollouts replaces the confidence in student and teacher contexts; Token-level reverse-KL on-policy distillation<br><i>Update:</i> Self-distillation from a privileged teacher (EMA rate 0.05); no proper-score reward | <i>Use:</i> Distilled reasoning with confidence<br><i>Qualifications:</i> Out-of-distribution calibration gains need not improve ranking; shared parameters couple reasoning and confidence positions                                             |
| <b>Generalized Correctness Models [132]</b><br>non-RL comparator; pp. 3–4, 22<br><i>Target:</i> Correctness conditional on response and generator identity<br><i>Output:</i> External Yes/No predictor                                                    | <i>Feedback / objective:</i> Labeled histories from multiple generators; Correctness-token cross-entropy; optional spline calibration<br><i>Update:</i> Supervised LoRA; loss only on correctness tokens                                                                                   | <i>Use:</i> Cross-model filtering<br><i>Qualifications:</i> Leaves generation fixed; target-domain calibration labels and history freshness matter                                                                                                |
| <b>P(IK) training [65]</b><br>non-RL comparator; pp. 3–5, 14, 18; pp. 24–31<br><i>Target:</i> Probability that a temperature-1 sample answers correctly, before answering<br><i>Output:</i> Binary value head read at the last question token             | <i>Feedback / objective:</i> Correctness of 30 sampled answers per question; Cross-entropy on each sample’s correctness label<br><i>Update:</i> Supervised fine-tuning of the language model together with the value head                                                                  | <i>Use:</i> Self-knowledge and answer gating<br><i>Qualifications:</i> Out-of-distribution transfer can be under- or overconfident; the target depends on the model, prompt, and sampling temperature                                             |
| <b>EliCal [63]</b><br>non-RL comparator; pp. 3–6, Eqs. 1–11; p. 18<br><i>Target:</i> Pre-generation success rate of temperature-1 samples<br><i>Output:</i> Linear-head score on the last question token                                                  | <i>Feedback / objective:</i> Self-consistency on about 560k unlabeled questions, then 20-sample accuracy on 1k or more labeled questions; Mean squared error<br><i>Update:</i> Two-stage supervised LoRA plus linear head; generator frozen                                                | <i>Use:</i> Honesty and answer gating<br><i>Qualifications:</i> Evaluated on greedy-answer correctness; test-set checkpoint selection; temperature-scaling baselines use post-answer judgments                                                    |
| <b>Single-generation self-consistency calibrator [64]</b><br>non-RL comparator; pp. 4–7, Eqs. 1–4; p. 24, Code Box 1<br><i>Target:</i> Correctness of a single sampled final answer<br><i>Output:</i> Ridge-plus-isotonic score from a response embedding | <i>Feedback / objective:</i> Modal-answer frequency among 100 unlabeled samples; Ridge regression, then an isotonic fit to the proxy<br><i>Update:</i> No LLM update; post-hoc regressor on the model’s own or an external model’s embeddings                                              | <i>Use:</i> Answer gating and selective prediction<br><i>Qualifications:</i> Can inherit consistent-error failures of the proxy; labeled Platt scaling keeps lower ECE in four of five groups                                                     |

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## B.6. Supervised, elicitation and post-hoc comparators (continued)

| Method and prediction object                                                                                                                                                                  | Supervision and update                                                                                                                                                                                 | Use and qualifications                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dual-Align [134]</b><br>non-RL comparator; pp. 3–5, Eqs. 6–11; pp. 11, 13<br><i>Target</i> : Class confidence after post-training<br><i>Output</i> : Temperature-rescaled final logits     | <i>Feedback / objective</i> : Pre-trained-model distributions and layer signals; Distribution KL plus layer-entropy alignment<br><i>Update</i> : Unlabeled scalar-temperature fitting; generator fixed | <i>Use</i> : Post-hoc class calibration<br><i>Qualifications</i> : Reference-model fidelity is an assumption; theoretical bound requires qualification |
| <b>Temperature scaling [62]</b><br>non-RL comparator; pp. 4, 6<br><i>Target</i> : Class-label confidence from fixed logits<br><i>Output</i> : Temperature-rescaled class probabilities        | <i>Feedback / objective</i> : Held-out labeled outcomes; Negative log likelihood<br><i>Update</i> : Fit one positive temperature; generator fixed                                                      | <i>Use</i> : Post-hoc thresholding<br><i>Qualifications</i> : Preserves class argmax; distribution shift still requires evaluation                     |
| <b>SALSA [113]</b><br>non-RL comparator; p. 2, Eq. 1; p. 10, Algorithm 1<br><i>Target</i> : Structured classification labels<br><i>Output</i> : Single-pass structured classification readout | <i>Feedback / objective</i> : Supervised labels; Cross-entropy on renormalized class-token logits<br><i>Update</i> : Supervised LoRA; no autoregressive label rollout                                  | <i>Use</i> : Typed decisions<br><i>Qualifications</i> : Interface comparator; source does not measure calibration                                      |

## B.7. Test-time computation: estimators and gates

These records estimate success or correctness to allocate, extend, or stop test-time computation with the generator fixed, or apply training-free gates to token statistics; self-certainty selection and CARG use token statistics to choose among candidates or to condition later dialogue turns. They are the comparators for RL controllers of the same decisions (Table 4).

| Method and prediction object                                                                                                                                                                                               | Supervision and update                                                                                                                                                                                     | Use and qualifications                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Capability probes [72]</b><br>non-RL comparator; pp. 3–4, 18–19, 21, 23<br><i>Target</i> : Success under a specified decoding policy<br><i>Output</i> : Query-only probe                                                | <i>Feedback / objective</i> : Repeated sampled-answer correctness; Supervised success-probability fitting<br><i>Update</i> : BCE-trained probe; answer policy fixed                                        | <i>Use</i> : Compute allocation<br><i>Qualifications</i> : Targets policy-level capability, not a realized answer’s correctness                                                                                |
| <b>PRM calibration and IAS [73]</b><br>non-RL comparator; pp. 3–7, 9, 15–21, 31<br><i>Target</i> : Continuation success under a specified generator<br><i>Output</i> : Quantiles of prefix success probability             | <i>Feedback / objective</i> : Rollout frequencies (8 continuations per prefix); Weighted quantile (pinball) loss<br><i>Update</i> : LoRA on PRM layers and a quantile head; generator fixed                | <i>Use</i> : Sample-count allocation<br><i>Qualifications</i> : The guarantee assumes a selector that recognizes success; requires per-generator recalibration; calibration rollouts not charged to the budget |
| <b>Distributional PRM [87]</b><br>non-RL comparator; pp. 3–5, 7–9, 13, 15<br><i>Target</i> : Predictive distribution of prefix success probability<br><i>Output</i> : Conditional quantile function from PRM hidden states | <i>Feedback / objective</i> : Rollout frequencies; Conditional optimal-transport dual objective with calibration-area early stopping<br><i>Update</i> : Offline input-convex map; PRM and generator frozen | <i>Use</i> : Sample-count allocation<br><i>Qualifications</i> : Gains depend on the scoring PRM; quantile read-out underspecified                                                                              |

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## B.7. Test-time computation: estimators and gates (continued)

| Method and prediction object                                                                                                                                                                                                                                                                                                         | Supervision and update                                                                                                                                                                                       | Use and qualifications                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Intermediate-answer probe</b> [112]<br>non-RL comparator; pp. 3–8, 14–16<br><i>Target</i> : Correctness of an intermediate answer in a reasoning chunk<br><i>Output</i> : Hidden-state probe probability                                                                                                                          | <i>Feedback / objective</i> : LLM-judged answer correctness; Class-weighted binary cross-entropy<br><i>Update</i> : Probe only; reasoning model frozen                                                       | <i>Use</i> : Early exit<br><i>Qualifications</i> : Class weighting is not proper unless the weight is 1; calibration degrades out of distribution; truncation simulated on completed traces                                                                  |
| <b>DEER</b> [75]<br>non-RL comparator (inference-time); pp. 4–5, 18, 23<br><i>Target</i> : Implied correctness of an induced trial answer<br><i>Output</i> : Geometric-mean token probability                                                                                                                                        | <i>Feedback / objective</i> : No outcome labels; Fixed threshold 0.95<br><i>Update</i> : No training; exit at reasoning transitions                                                                          | <i>Use</i> : Early exit<br><i>Qualifications</i> : Sequence-likelihood score; calibration assumed in the analysis, not measured                                                                                                                              |
| <b>Self-certainty selection</b> [70]<br>non-RL comparator (inference-time); p. 4, Eqs. 4–9; p. 6, Eq. 11; p. 8, Table 1; p. 14, Fig. 10<br><i>Target</i> : No declared event; within-question candidate quality<br><i>Output</i> : Mean $D_{\text{KL}}(U \  p_\theta)$ over response positions, with $U$ uniform over the vocabulary | <i>Feedback / objective</i> : No outcome labels; Borda voting weights candidates by self-certainty rank<br><i>Update</i> : No training; requires full logits                                                 | <i>Use</i> : Best-of- $N$ selection<br><i>Qualifications</i> : Unbounded score whose level falls with difficulty; alone it trails self-consistency on MATH at $N = 64$ ; no calibration metric                                                               |
| <b>DeepConf</b> [74]<br>non-RL comparator (inference-time); pp. 3–7, 17, 19, 24<br><i>Target</i> : No declared event; trace-quality score<br><i>Output</i> : Windowed token-confidence statistics                                                                                                                                    | <i>Feedback / objective</i> : No outcome labels; Percentiles of same-problem warm-up traces<br><i>Update</i> : No training; filtering, weighted voting, trace termination, consensus stop                    | <i>Use</i> : Parallel sampling<br><i>Qualifications</i> : Uses within-problem rankings and vote share; no calibration evaluation                                                                                                                             |
| <b>CARG</b> [85]<br>non-RL comparator (inference-time); pp. 4–5, §3.4.1; pp. 8–9, §5; p. 16, Table 6<br><i>Target</i> : No declared event; confidence in the stated answer at each turn<br><i>Output</i> : Geometric-mean token probability of “The correct answer: X”                                                               | <i>Feedback / objective</i> : No outcome labels; confidence appended to each turn of the dialogue history<br><i>Update</i> : No training; the model conditions on its earlier confidences, with no threshold | <i>Use</i> : Keeping correct answers under follow-up challenges<br><i>Qualifications</i> : Confidence measured for one model and not evaluated for calibration; only initially correct answers are challenged, and no follow-up supplies correct information |

## B.8. Intrinsic certainty and entropy as capability signals (boundary protocols)

These records use statistics of the policy’s own distribution as rewards, or regulate the policy’s entropy during correctness-reward training. They are retained because Section 3.6 compares those statistics with reported probabilities; none defines a probability target.

| Method and prediction object                                                                                                                                                                                                    | Supervision and update                                                                                                                                                                                                                                                                                                                  | Use and qualifications                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INTUITOR [47]</b><br>capability / boundary protocol; pp. 4–6, 8, 14, 20–21<br><i>Target:</i> Self-certainty of the policy’s token distributions<br><i>Output:</i> Reasoning + answer                                         | <i>Feedback / objective:</i> No gold labels; Mean $D_{\text{KL}}(U \  p_{\theta})$ over response tokens, scored online by the current policy<br><i>Update:</i> GRPO with group standardization, clipping and KL to the initial policy; learning rates and KL differ between text and appendix                                           | <i>Use:</i> Label-free capability adaptation<br><i>Qualifications:</i> Separates correct from incorrect responses without being a correctness probability; failures under weak priors, noisy inputs and high learning rates                                                                             |
| <b>RENT [48]</b><br>capability / boundary protocol; p. 4, §3.3; p. 15, Table 5<br><i>Target:</i> Low next-token entropy along the sampled response<br><i>Output:</i> Reasoning + answer                                         | <i>Feedback / objective:</i> No gold labels; Mean negative token entropy<br><i>Update:</i> GRPO with 5 samples per prompt; KL loss 0.001; clip 0.2; standard-deviation division not stated                                                                                                                                              | <i>Use:</i> Label-free capability adaptation<br><i>Qualifications:</i> Training and evaluation share problems except GSM8K; token-selection rule of the main runs unstated; calibration asserted from a correlation, not measured                                                                       |
| <b>EM-FT / EM-RL [58]</b><br>capability / boundary protocol; pp. 3–5, Eqs. 1–4, Table 1; p. 23<br><i>Target:</i> Low token- or trajectory-level entropy of sampled outputs<br><i>Output:</i> Reasoning + answer or code         | <i>Feedback / objective:</i> No gold labels in the objective; Minimized summed token entropy (FT, token RL) or sequence log-likelihood (sequence RL)<br><i>Update:</i> FT: direct entropy gradient, one sample; RL: score-function reward, RLOO baseline, $N = 4$ , KL 0.001                                                            | <i>Use:</i> Math and code post-training<br><i>Qualifications:</i> Early-stopped on labeled validation accuracy; weak for Llama-3.1-8B; no calibration evaluated; EM-INF is decoding-time only                                                                                                           |
| <b>TTRL [49]</b><br>capability / boundary protocol; pp. 4–5, Eqs. 1–3; p. 7; p. 22, Eq. 4<br><i>Target:</i> Agreement with the policy’s own majority answer<br><i>Output:</i> Reasoning + extracted final answer                | <i>Feedback / objective:</i> No gold labels; majority vote over 64 samples; Binary match reward on 32 downsampled rollouts<br><i>Update:</i> GRPO per benchmark; KL and standard-deviation settings not stated                                                                                                                          | <i>Use:</i> Label-free capability adaptation<br><i>Qualifications:</i> Trains on the evaluation questions; reward accuracy relies on scattered wrong answers; entropy and response length fall; no calibration evaluated                                                                                |
| <b>EMPO [50]</b><br>capability / boundary protocol; pp. 4–5, Eqs. 3–7; p. 7, Table 1; p. 8, Table 2<br><i>Target:</i> Agreement of an output’s answer with the policy’s other samples<br><i>Output:</i> Reasoning + answer      | <i>Feedback / objective:</i> No gold labels; reward equals the frequency of the output’s semantic answer cluster among $G$ samples, which lowers semantic entropy<br><i>Update:</i> GRPO-style group advantages with clipping and no KL; trains only on prompts whose entropy lies between two thresholds whose values are not reported | <i>Use:</i> Label-free capability adaptation<br><i>Qualifications:</i> Free-text clustering uses a supervised verifier model; below gold-label GRPO on MMLU-Pro at all three scales and on GPQA at 7B and 14B; authors note possible convergence to a confident wrong cluster; no calibration evaluated |
| <b>Positive-Advantage Reweighting [118]</b><br>capability / boundary protocol; pp. 4, 7–8, Eq. 5; p. 13<br><i>Target:</i> Task correctness, with the generation policy’s entropy regulated<br><i>Output:</i> Reasoning + answer | <i>Feedback / objective:</i> Verifier correctness; Positive-advantage token losses scaled by $\lambda \in [0, 1]$ , scheduled or adjusted toward an entropy target<br><i>Update:</i> GRPO with group standardization, token-level loss, and no KL                                                                                       | <i>Use:</i> Math reasoning<br><i>Qualifications:</i> Likelihood separation is not probability calibration; in-domain gains came with an out-of-domain cost; checkpoints selected on evaluation sets                                                                                                     |

## B.9. Analyses and controlled mechanism studies

These records develop incentive or estimator analyses and report controlled learning experiments. They are separated from the main landscape to avoid treating a theorem, a diagnostic comparison, and a deployable predictor as the same type of contribution.

| Method and prediction object                                                                                                                                                                                                                                         | Supervision and update                                                                                                                                                                                  | Use and qualifications                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reward-family analysis [31]</b><br>analysis / controlled protocol; pp. 3–4, Theorem 1; pp. 31–34<br><i>Target:</i> Joint chosen-answer correctness<br><i>Output:</i> Reasoning + integer confidence                                                               | <i>Feedback / objective:</i> Binary correctness; Properness + monotone optimal-answer reward<br><i>Update:</i> DrGRPO; no std; format/length rewards                                                    | <i>Use:</i> Decision utility<br><i>Qualifications:</i> Static reporting/answer incentives and finite optimization are distinct                                                                                                                       |
| <b>Abstention-collapse analysis and report repair [109]</b><br>analysis / controlled protocol; pp. 2–7, 20–21<br><i>Target:</i> Success probability and answer gate<br><i>Output:</i> Discrete gate vs continuous report head                                        | <i>Feedback / objective:</i> Binary outcome; Error penalties vs Brier + accuracy<br><i>Update:</i> Gate-policy analysis; forced answers plus direct Brier report-head learning                          | <i>Use:</i> Abstention<br><i>Qualifications:</i> Collapse law is conditional on gate/readout assumptions; repair uses a continuous head                                                                                                              |
| <b>Group-update calibration analysis [18]</b><br>analysis / controlled protocol; pp. 3–6; pp. 10–11, its Appendix A.1<br><i>Target:</i> Stochastic external event probability<br><i>Output:</i> Numeric probability token/report                                     | <i>Feedback / objective:</i> Observed stochastic outcome; Log proper score<br><i>Update:</i> Compare std-normalized GRPO vs no-std/PPO/RLOO                                                             | <i>Use:</i> Forecasting<br><i>Qualifications:</i> Outcome-weighting result holds in a specified stochastic setup; public-code paths differ                                                                                                           |
| <b>Internal-feedback analysis [51]</b><br>analysis / controlled protocol; pp. 5–7, Eqs. 2–4, Props. 1–3; pp. 16–18; p. 19, Tables 4–5<br><i>Target:</i> Self-certainty, token entropy or length-normalized log-likelihood<br><i>Output:</i> Reasoning + boxed answer | <i>Feedback / objective:</i> No gold labels; Three intrinsic rewards on MATH training prompts<br><i>Update:</i> VerL GRPO, $G = 8$ , 128 prompts per update, KL 0.005; tabular-softmax entropy analysis | <i>Use:</i> Label-free reasoning<br><i>Qualifications:</i> Single greedy evaluations; early gains largely format-related; later decline or collapse except for a math-specialized base model; overconfidence is behavioral, not a calibration metric |

## Appendix C. Implementation evidence and version boundaries

This audit distinguishes a paper’s mathematical formulation, its reported experimental settings, and an accessible public execution path. We inspected the registered PDFs and statically read official repositories discovered through paper or author-project links on 19 September 2026. We did not execute training or reproduce reported metrics. A checked commit is an immutable source reference, not proof of the configuration used for a published checkpoint. Paper-code differences below therefore identify evidence boundaries; they do not establish that the reported experiments were conducted incorrectly.

The accompanying claim ledger records individual page and code locators. This appendix retains the differences most relevant to interpreting objectives, gradients, and comparison settings: Table C.1 summarizes support boundaries for representative protocols, and C.1–C.6 examine the implementation evidence for RLCR, the reward-family experiments of Tan et al., stochastic-outcome GRPO, REAL, DAR, and ConfTuner. All PDF page references are physical pages.

**Representative support boundaries.** Table 3 compares learning families side by side. Table C.1 instead records, for selected protocols, what the evaluated design supports and which inferences require additional evidence, with source locators for implementation-oriented reading.

**Table C.1. Representative support boundaries.** For selected protocols, what the evaluated design supports and which inferences require additional evidence. Locators give physical PDF pages of the inspected version.

| Representative protocol                                  | What the evaluated design supports                                                                                  | Inference requiring additional evidence                                                      |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Temperature scaling [62]<br>PDF pp. 4, 6                 | Fitting a scalar correction with fixed logits; positive temperature preserves the argmax                            | Improved generation or arbitrary distribution-shift reliability                              |
| ConfTuner [57]<br>PDF pp. 4–5                            | Direct supervised learning over a confidence-numeral support                                                        | Necessity of policy gradients or identification of the numeral distribution with correctness |
| Decision-token calibration [7]<br>ACL PDF pp. 7–8, 14–15 | A masked GRPO/CE intervention with calibration, ranking, and sensitivity results                                    | A proper-score guarantee for the entire shared-parameter update                              |
| RLCR [30]<br>PDF pp. 4–5, 16–18                          | Conditional reporting incentives in a joint answer/confidence objective                                             | Global maximum mean accuracy within a constrained policy family                              |
| RLVP [36]<br>PDF pp. 3–7                                 | Full verbalized class vectors trained with an expert-guided reward curriculum; ranking and calibration measurements | Propriety of an exponentiated score or calibration transfer inferred from AUC alone          |
| Group-update analysis [18]<br>PDF pp. 4–6, 10–11         | Changed incentives from reward standardization in specified stochastic-outcome settings                             | The same failure for every score, probability object, or GRPO application                    |
| RLSR [40]<br>PDF pp. 4–7                                 | Learning useful confidence ordering for selective prediction                                                        | Numerical probability semantics or certified risk at a new threshold                         |
| Preference-distribution RL [82]<br>PDF pp. 4–7           | Fidelity to a declared soft annotator target                                                                        | Human-population validity or equivalence to majority-label calibration                       |
| REAL [41]<br>PDF pp. 4–5, 8                              | A mean/likelihood objective with sampled reasoning and a possible direct-gradient component                         | Correlation as probability calibration or equivalence of paper and released configurations   |
| DAR [42]<br>PDF pp. 4, 17                                | Empirical CRPS, leave-one-out credit, and a group-transformed update                                                | Automatic preservation of population-score incentives through finite ensembles               |

### C.1. RLCR: the principal recipe agrees, but the reward has additional terms

The RLCR paper reports group centering without reward-standard-deviation division, no KL penalty, and BNPO aggregation over active tokens in a local batch [30], pp. 5, 18. The public `configuration` sets `beta=0` and `scale_rewards=False`. Its local `arguments definition` defaults to BNPO; the `loss branch` implements that aggregation.

The implemented `Brier reward` is  $1 - (z - c)^2$ , with zero for invalid formatting. The HotpotQA recipe weights the format, accuracy, and Brier rewards by 0.5 each and assigns weights of  $10^{-6}$  to two confidence-related bonuses. For valid formatting, its main component is a positive affine transformation of correctness-plus-Brier. The total reward is nevertheless not exactly the bare formula: report-dependent bonuses and formatting rules remain. The current recipe also uses a warmup ratio of 0.05 rather than the paper’s 0.1. These qualifications preserve the main mechanistic comparison while avoiding an exact-equivalence claim unsupported by the configuration.

### C.2. Reward-family experiments: missing implementation evidence remains missing

Tan et al. report Dr. GRPO without reward standardization and with equal token weighting, together with additional format, length, and valid-confidence rewards [31], pp. 31–34. Their theoretical reward family must therefore be distinguished from the complete experimental reward. For example, valid confidence contributes an additional unit reward, while length and XML-format checks contribute further bonuses.

The paper links an [anonymous code release](#), but this audit received HTTP 403 and could not inspect it. The failed access is not evidence that code was never released. We therefore leave the actual KL, clipping, and masking behavior unresolved rather than inferring it from generic Unsloth defaults. Likewise, the paper’s best-effort property concerns its stated expected-reward problem, not the stability of every implementation in that reward family.

### C.3. Stochastic-outcome GRPO: a controlled language-model contrast and a synthetic-code qualification

The public [GRPO](#) and [no-standard-deviation](#) launchers retain the same main LM settings while switching the normalization flag [18]. Both use four rollouts, learning rate  $10^{-6}$ , and a KL-loss coefficient of 0.001; disabling their KL-in-reward flag does not remove this separate loss term.

The repository pins a specific upstream verl version, whose [corresponding advantage implementation](#) confirms the centering-versus-standardization distinction. This supports the paper’s LM estimator comparison, although reproducing the reported values would require executing the complete protocol, including checkpoint selection.

A separate issue affects the released synthetic notebook. It [constructs a clipped surrogate and then overwrites the loss](#) with an unclipped ratio expression before backpropagation. Thus the current notebook does not establish that the reported synthetic clipping ablations used that clipping path. Run-specific sources would resolve the difference. The notebook’s [RLOO baseline](#) also divides the sum of other rewards by  $G$ , rather than  $G - 1$ . This scaled baseline is not the conventional leave-one-out mean, but it remains independent of the scored sample’s own action under the appropriate sampling conditions, so the departure from the conventional definition does not by itself establish score-gradient bias.

### C.4. REAL: follow the specialized actor, not generic GRPO flags

REAL’s paper distinguishes the auxiliary likelihood coefficient  $\lambda$ , the direct prediction-gradient weight  $\beta$ , and stabilization of the trajectory term [41], pp. 4–5. REAL’s [public launcher](#) sets the direct-gradient coefficient and dedicated REAL KL coefficient to zero. The specialized actor [gates its KL calculation](#) on that dedicated coefficient. Generic GRPO KL flags elsewhere in the script therefore do not demonstrate an active REAL KL penalty.

The implemented trajectory advantage also differs from the paper’s combined negative-MSE-plus-log-likelihood expression. The actor separately applies leave-one-out baselines, standardization, and clipping to [negative MSE](#) and [target-token probability](#), then combines them. Probability and log probability are not interchangeable after these transformations. The [GPG loss](#) uses  $-\log \pi$  times advantage, without PPO ratio clipping. Advantage clipping is a different operation.

The REAL branch [zeros answer-token advantages](#), with direct prediction supervision controlled by the separate coefficient. Because parameters are shared, zeroing these advantages does not freeze the predictions. The enabled remove-padding path [weights selected digit probabilities from a full-vocabulary softmax](#), without renormalizing within the digit set. Dedicated partial-group filtering also persists independently of generic filtering flags.

These findings motivate precise reporting of both the paper’s estimator and the public recipe. They do not justify claiming that the paper’s  $\beta = 0.01$  ablation or released checkpoints were trained with the launcher’s zero default. The reported Pearson/RMSE trade-off remains author-reported evidence, rather than an independently reconstructed run.

### C.5. DAR: empirical scoring, valid samples, and transformed credit

DAR’s [reward manager](#) agrees with the paper’s  $K^2$  empirical-CRPS denominator and  $(K - 1)^2$  leave-one-out denominator [42], p. 4. Neither denominator implements the fair-score  $K(K - 1)$  adjustment for independent ensemble members [27].

The [GRPO implementation](#) subsequently centers and standardizes contribution rewards. Furthermore, [validity handling](#) makes the scoring ensemble size depend on successful parsing, with pointwise fallbacks for exceptional groups. Invalid outputs receive a separate worst-reward-minus-ten penalty. Thus the iid fixed- $K$  identity in Eq. (A.8) is a diagnostic for the score construction, not an exact characterization of the complete training update.

The [KBSS launcher](#) agrees with the paper on twelve rollouts and a KL-loss coefficient of 0.001, but specifies a minibatch size of 128 rather than the paper’s 16 [42], p. 17. We report both values rather than treating either as the configuration that produced the results.

## C.6. ConfTuner: a directly differentiated comparator

ConfTuner’s [training computation](#) restricts logits to confidence tokens, normalizes them, and exactly sums their probability-weighted squared errors [57]. Because it backpropagates this expected loss directly, it is a supervised comparator, not policy-gradient RL over confidence reports.

When its optional supervised regularizer is enabled, the code adds that loss with coefficient one; the [example recipes](#) enable it for Llama and disable it for Qwen. The regularizer should not be described as explicit reference-policy KL. These examples establish an available implementation, not a run-by-run reconstruction of every reported setting.

## Appendix D. Decision and system evidence audit

This appendix records distinctions that affect numerical comparison and reproducibility. Paper locators use physical PDF pages. Public repositories were inspected on 19 September 2026; links below fix the examined commits where available. We distinguish the published protocol, the inspected release, and any unresolved correspondence between them. Static inspection establishes what the released code specifies, not what a reported run executed. The observations qualify individual claims rather than rank entire papers.

### D.1. Metrics with different names, targets, and implementations

**SaySelf.** The paper defines its “ECE” as the mean per-example absolute error between confidence and a binary correctness indicator, without binning [99] (p. 6, Eq. 4). The public [evaluation code](#) likewise averages `abs(confidence-correctness)/10`, which confirms that the inspected release applies no binning. This quantity should retain an explicit definition when quoted; silently pooling it with conventional binned ECE would change the estimand.

A further distinction concerns labels. The same evaluator obtains a GPT correctness grade, reports mean grade divided by ten as accuracy, and thresholds grades at seven for AUROC. Its [default grading prompt](#) also conditions the grade on the presence of a chain of thought. These graded labels differ from the paper’s binary-indicator notation. Without a historical run manifest, the release does not establish that every table used this evaluator. The reported HotpotQA comparison remains interpretable with qualifications: adding RL lowered the quantity labeled “ECE” from 0.3704 to 0.3558 and reported accuracy from 0.3708 to 0.3585 [99] (p. 7, Tables 1–2).

**LACIE.** The paper specifies nine-bin, unweighted ECE and excludes abstentions [37] (p. 6). Its [evaluator](#) constructs `ECEMetric(n_bins=9)`. However, the author-linked [metric dependency](#) defaults to count weighting. The inspected documentation does not pin that dependency to the paper run. The appropriate record is therefore “paper specifies unweighted ECE; current linked dependency defaults to weighted ECE,” with the historical implementation unresolved.

The population also matters. The evaluator’s `skip_none` option removes examples where either compared model gives no answer. This is a particular paired filtering rule, not an automatic property of every invocation. In the paper, Mistral-base abstention rose from 0.80% to 25.27%, while ECE fell from 0.15 to 0.12 and recall, the share of correct answers the listener accepted, fell from 0.76 to 0.55 [37] (p. 7, Table 1). A deployment comparison should preserve the unanswered cases and their costs alongside any conditional calibration result.

**DualStake.** Its appendix defines ECE bin accuracy using mean answer F1, whereas the accuracy column and Brier formula use exact-match correctness [142] (pp. 13–14). Evidence confidence and answer confidence are also elicited at different trajectory positions. These targets and prefixes should remain separate in a comparison matrix even when both scores are supervised using downstream answer quality.

The [public metric wrapper](#) defaults to twenty bins and passes an `is_correct` array into all three metric functions; the paper specifies ten bins. The imported `statistics/confidence_metrics.py` is absent from the inspected public tree. Accordingly, the paper’s F1/EM distinction is source-supported, while the underlying metric implementation and its connection to published runs remain unresolved. The reported eight-dataset averages for

DualStake are ECE 0.178, AUROC 0.712, and Brier 0.220; temperature scaling has a lower ECE (0.151) but a worse AUROC (0.538) and Brier score (0.249) [142] (p. 6, Table 1). These are qualified within-paper comparisons, not harmonized calibration measurements.

**Just Ask for Calibration.** Its “AUC” denotes the area under the selective accuracy–coverage curve, not AUROC; the ECE description specifies squared bin error [89] (pp. 2–3). No attributable implementation was located to resolve the squared-versus-absolute wording. Its temperature-scaled results use labels: one of five folds fits the temperature and the remaining four evaluate it. Learned linguistic confidence mappings instead use four folds for fitting and one for evaluation; the combined procedure separates mapping, temperature, and evaluation folds [89] (p. 8). Each procedure thus relies on labeled data, allocated differently. On GPT-4 TriviaQA, label probabilities and verbalized one-stage top-one reports give raw ECE of 0.078 and 0.024, respectively, but temperature-scaled Brier scores of 0.077 and 0.084 [89] (p. 3, Table 2), so neither dominates under every metric.

## D.2. What the objective and released artifact identify

RLSR’s paper states an effective batch of 1,536 completions and removal of reward-standard-deviation normalization [40] (p. 7). The inspected *HotpotQA configuration* agrees on `scale_rewards: false` but implies an effective batch of 2,048 completions (four devices, batch eight, and accumulation sixty-four). This difference belongs in artifact provenance; it does not invalidate the reported selective-prediction comparison. A reproducibility claim needs the configuration and dependency versions for the particular run, rather than the repository name alone.

For two richer output interfaces, a comparison must also record which object the reward scores. Verbalized Probability Distribution emits a candidate vector, chooses its maximum, and rewards the selected answer’s scalar confidence, with a separate probability-sum penalty [80] (pp. 2–3). Its full-vector Brier improvement is empirical evidence, not full-vector supervision; that score ranges from zero to two, unlike its main binary Brier metric [80] (p. 14, Table 6). In Judging with Confidence, the training event is an individual preference draw, the soft MSE target is a preference proportion, and the ECE/Brier target is a binarized majority after excluding ties [82] (pp. 18–19). The main comparison also matches annotation count while giving RL ten times as many distinct inputs as dense-label SFT [82] (pp. 5, 20–21). Reporting these choices supports sensible method selection without attributing all differences to the named optimizer.

**RLVP: a full-vector reward curriculum.** This audit uses the published WWW 2026 version [36]. The policy generates chain-of-thought text and a JSON class vector; the displayed reward sums format-validity and argmax-correctness terms and the exponential of a negative weighted sum of label-to-report and expert-to-report KL divergences (PDF pp. 2–3, Eqs. 1–4). The expert weight decays linearly from one to zero. The expert term is a concrete source of denser comparisons between probability reports, but the endpoint component is  $p_Y$ , not a proper log score; Appendix A.7 gives the precise distinction. The reward specifies no semantic process supervision for chain-of-thought faithfulness.

The expert is XGBoost fitted to each task’s full training split, then used to predict that same split. The paper describes stratified 80/20 task splits and teacher-correct filtering (PDF pp. 3–4). RLVP’s Section 2.3 restricts calculation of the probability component to teacher-correct samples, whereas its Section 3.1.1 describes retaining such training samples; it is unclear whether the other examples are dropped from all losses or only from this component. The description does not establish test-label leakage, but it also does not establish cross-fitted teacher probabilities or rule out teacher-selection bias. The nominal optimizer is GRPO without reference-policy KL, with eight rollouts, temperature 0.7, effective batch 1,024, learning rate  $2 \times 10^{-6}$ , and three joint-training or six single-task epochs (PDF pp. 6, 10). The label-distribution KL in the reward is distinct from the omitted reference-policy KL.

The numerical evidence is metric-specific. In the paper’s Table 1, RLVP-Anneal beats XGBoost on AUC in all six single tasks but has lower ECE only on Customer Churn (0.0726 versus 0.1681) and Uber Analysis (0.0153 versus 0.0431); the other four ECE comparisons favor XGBoost. Equal-task averages calculated from that table are AUC 0.9294 versus 0.8124 and ECE 0.1005 versus 0.0743 (PDF p. 5), so these rows do not support the paper’s accompanying statement that RLVP has the best average ECE. The paper defines macro one-vs-rest AUC, summed multiclass Brier, and a ten-bin weighted absolute confidence/accuracy ECE (PDF pp. 5–6, Eqs. 5–7); the ECE formula is not a full-vector calibration test.

Joint training uses Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct on 169 UniPredict training splits. The reported up-to-55% Qwen and 50% Llama teacher-surpass ratios concern AUC on those tasks’ test splits; the mean gain in its Figure 3 is averaged only over the tasks RLVP wins (PDF pp. 6–7). On eight unseen AMLB tasks, its Table 2 compares RLVP with

zero or 32 in-context examples to XGBoost trained/tuned on the same 32 examples. RLVP wins four and five of the eight tasks, respectively; adding examples raised its AUC on three tasks and lowered it on five. The table reports no held-out ECE or Brier score, so it does not support the claim of calibration transfer to unstructured text. The protocol does not fully specify demonstration exclusion from evaluation, tuning partitions, seed dispersion, or all text preprocessing.

The displayed expression for the adapted RLCR baseline adds a Brier term, which leaves reward orientation and scalar/vector notation unresolved (PDF p. 5); this ambiguity does not establish an implementation error. Likewise, format rules name tags, JSON parsing, and a  $10^{-5}$  sum tolerance without fully specifying invalid-output fallback or all component constraints. The [paper-linked repository](#) and its public GitHub API endpoint returned HTTP 404 at inspection on 19 September 2026. No attributable code was therefore available to this audit, although the repository may become available later. The source paper and our arithmetic, rather than a verified implementation or reproduced run, support the comparisons above. RLVP’s Appendix A.3 reports approximately 48 hours on 32 A100-80GB GPUs for a three-epoch joint run; this resource scope should accompany comparisons with task-specific tabular models.

### D.3. Eight industry and community sources and their evidential roles

These sources provide useful engineering evidence but differ in disclosure level and statistical target. Table D.1 cites their original pages or repositories and describes each source as inspected on its access date.

**Table D.1. Industry and community sources and the evidence each supplies.** Each source is cited to its original page or repository and described as inspected on its access date; rows separate what a source discloses from what its evidence supports.

| Source                                                  | Disclosed approach or interface                                                                                                                     | What the evidence supports                                                                                                     |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Mantic technical post [143]</a>             | Brier-reward forecasting with prepared research context, reasoning, and distribution tools; centered advantages without standard-deviation division | A documented industrial training case with external event outcomes; information gathering and training contribute jointly      |
| <a href="#">Jev introduction [1]</a>                    | Parallel typed decisions and probability outputs; workflow evaluation against strong-model reference probabilities                                  | Product interface and reported system comparisons; reference-model agreement is not independently labeled correctness          |
| <a href="#">TypeSafe primer [4]</a>                     | RLCD expanded as Reinforcement Learning for Calibrated Decisions; calibration described through group frequencies                                   | Official terminology and intended statistical contract; no complete reproducible training algorithm                            |
| <a href="#">TypeSafe confidence documentation [127]</a> | Option/level probabilities plus a distribution-shape confidence statistic                                                                           | Distinct API fields whose semantics must be checked before choosing a threshold                                                |
| <a href="#">harshat heg/Qwen-2.5-1B-RLCD [128]</a>      | Shared prefill, candidate-logit readout and programmatic assembly using Qwen2.5-1.5B-Instruct-4bit                                                  | An inference-engine proposal; normalized candidate scores and the artifact name do not establish calibration-aware RL training |
| <a href="#">rorshopping/jev-on-a-laptop [129]</a>       | Unofficial stock-model experiments with parallel finite-field decisions                                                                             | A local interface comparator; its own documentation distinguishes raw candidate softmax from calibrated probabilities          |
| <a href="#">kiconcuo/jevfire [130]</a>                  | Existing LM-head scoring of verified single-token labels, shared-prefix reuse, and JSON assembly without retraining                                 | Serving and schema evidence; independent field scoring differs from general sequential structured generation                   |
| <a href="#">AlexWortega/openjev [131]</a>               | Last-token sequence classification trained by cross-entropy over entailment, contradiction, and neutral labels                                      | A supervised-training comparator, distinct from inference wrappers and the undisclosed official RL recipe                      |

The community artifacts therefore implement at least two different interventions: changing inference around existing weights and training a classifier. Neither establishes identity with Jev’s proprietary method. Similarly, structural validity prevents unsupported output forms but does not ensure that an allowed answer is correct. The useful comparison separates model training, probability readout, and output serialization.

## D.4. Controls for an industrial decision benchmark

A benchmark should first fix the semantic task: candidate sets, per-field labels, abstention behavior, and whether fields are independent or condition on preceding decisions. Otherwise a parallel field scorer and a sequential generator may solve different problems. Candidate verbalizers, shared token prefixes, fallback continuation, and invalid-output handling affect both computation and the resulting scores. Preserve the same input evidence and reasoning allowance where the comparison permits; state the differences where it does not.

Serving measurements should disclose weights, precision, hardware, engine, batching, concurrency, context length, and cache state. Separate prefill, scoring, assembly, and end-to-end latency; distinguish cold model loading from fresh-prefix requests on warm kernels. Compare finite-choice scoring with constrained-output baselines, not only with unrestricted prose. JEVfire’s source itself separates cache conditions and notes that engine scheduling can require multiple batches; its game demonstration includes an explicit physics guard [130]. Controller rules, retrieval, and verification should therefore be ablated or held fixed when attributing performance to the model.

Statistical evaluation should use independent task outcomes, or explicitly identify a reference-model target. Report accuracy, proper loss, stated calibration diagnostics, coverage, and utility under declared costs; evaluate both cost-derived and validation-selected thresholds. Multiple decisions from one workflow are not automatically independent test cases. For subjective labels, preserve the annotator population and distinguish individual preferences from majority decisions. For forecasting, record information cutoffs, event grouping, and access to retrieval. Mantic reports a Metaculus baseline-score increase from 38.6 to 45.8, where the score rescales log probability rather than measuring ECE [143] (Results). That claim concerns the described forecasting system, not an isolated causal effect of the reward formula.

These controls make positive engineering results easier to interpret: reduced serialization work, better classification after supervised training, improved numerical probabilities, and better decisions can each be valuable. A comparison should establish which of these changed, what resources the change required, and whether the gain survives the application’s chosen costs and information constraints.

## Appendix E. Neighboring surveys and protocol qualifications

This appendix documents the scope comparisons and selected protocol details behind Sections 1.1, 2, and 6. The comparison concerns the question each source organizes, not a ranking of survey quality. Page locators refer to physical PDF pages starting at 1; publication and arXiv versions are distinguished where relevant.

### E.1. Scope of neighboring surveys

**Table E.1. Scope of neighboring surveys.** Coverage refers to the inspected version and to the question each source organizes; the comparison does not rank survey quality.

| Survey and inspected version                                                                                    | Established coverage relevant to this article                                                                                                             | More specific question pursued here                                                                     |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Geng et al., <i>A Survey of Confidence Estimation and Calibration in Large Language Models</i> [19]<br>arXiv v2 | Relative versus absolute confidence; generation and classification; estimation, calibration, selective generation, and human variation (PDF pp. 2–9)      | How a declared probability target interacts with a joint answer objective and implemented policy update |
| Pan et al., <i>Reward Modeling for Reinforcement Learning-Based LLM Reasoning</i> [20]<br>arXiv v2              | Reward architecture, granularity and semantics; PPO/DPO/GRPO/DAPO; uncertainty penalties, reward hacking, and downstream evaluation (PDF pp. 7–19, 37–43) | Whether a reporting incentive survives the transformations applied to its reward and gradient           |

*Continued on next page*

Table E.1 (continued)

| Survey and inspected version                                                                                                                          | Established coverage relevant to this article                                                                                                                                                                                                                                                | More specific question pursued here                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zhang et al., <i>A Survey of Reinforcement Learning for Large Reasoning Models</i> [21]<br>arXiv v3                                                   | Verifiable, generative, dense and unsupervised rewards and reward shaping; critic-based and critic-free optimization, including group-standardized advantages; KL, entropy and length regularization; sampling; sharpening-versus-discovery and normalization debates (PDF pp. 11–34, 37–38) | How a probability claim’s event, information, and population survive those reward, normalization, and sampling choices; confidence appears there as an intrinsic reward for correctness (PDF pp. 17–18), not as a reported probability evaluated for calibration |
| Zhang et al., <i>From Passive Metric to Active Signal</i> [22]<br>Findings of ACL 2026                                                                | Uncertainty-guided reasoning, agents and RL; local discrimination, intrinsic rewards, process supervision, and control-oriented evaluation (PDF pp. 3–9)                                                                                                                                     | Connecting those uses to explicit probability targets, objective properties and estimator assumptions                                                                                                                                                            |
| Shorinwa et al., <i>A Survey on Uncertainty Quantification of Large Language Models</i> [23]<br>arXiv v2                                              | Token, verbal, semantic and mechanistic approaches; RL-based confidence learning, proper scores, and uncertainty-guided assistance (PDF pp. 3–4, 12–14, 19–24)                                                                                                                               | Comparing particular scoring objectives and actual updates, rather than grouping methods by uncertainty-estimation channel                                                                                                                                       |
| Wen et al., <i>Know Your Limits</i> [24]<br>arXiv v3                                                                                                  | Query answerability, model knowledge and human values; supervised/preference alignment, inference-time abstention, and coverage–quality trade-offs (PDF pp. 2–11)                                                                                                                            | Distinguishing learned action preferences at specified costs from reusable probability reports                                                                                                                                                                   |
| Li et al., <i>Confidence as Control</i> [25]<br>Author-hosted public copy                                                                             | Decision states, units, signals, transformations and policies; training, inference, cascading, retrieval, risk and agentic control; semantics, composition and evaluation (PDF pp. 2–6, 22–24)                                                                                               | Method-level conditions connecting probability-learning incentives to implemented updates and downstream losses                                                                                                                                                  |
| Unpublished companion manuscript, <i>Beyond Natural-Language Interfaces: Decisions, Scores, and Predictive Distributions</i><br>Manuscript version V6 | Predictive-report semantics; supervision and readout paths; classification, scoring, numerical properties and distributions; transformations, consumer losses, access and execution costs                                                                                                    | Probability-learning incentives under joint answer objectives, practical reward and gradient transformations, and finite-ensemble scoring, connected to downstream losses                                                                                        |
| Li et al., <i>A Survey on the Honesty of Large Language Models</i> [144]<br>arXiv v1; TMLR publication metadata verified                              | Self-knowledge and self-expression; calibration, selective prediction, and SFT/RL methods for abstention and confidence (PDF pp. 3, 6–7, 10–14)                                                                                                                                              | Comparing the statistical target, reward incentive, update, and deployment loss within confidence learning                                                                                                                                                       |
| Xia et al., <i>A Survey of Uncertainty Estimation Methods</i> [145]<br>arXiv v2; Findings of ACL 2025 metadata                                        | Four inference-time estimation families; calibration training is explicitly outside its scope (PDF pp. 2–3, 6–8)                                                                                                                                                                             | Studying interventions that train or transform probability reports and the updates that implement them                                                                                                                                                           |
| Liu et al., <i>Uncertainty Quantification and Confidence Calibration</i> [146]<br>KDD 2025 copy                                                       | Input, reasoning, parameter and prediction uncertainty; computational cost; explicit ranking/calibration distinction (PDF pp. 2–5)                                                                                                                                                           | Comparing reward geometry, policy updates and finite-sample objectives at the method level                                                                                                                                                                       |
| Xie et al., <i>A Survey of Calibration Process for Black-Box LLMs</i> [147]<br>arXiv v1                                                               | Estimation-to-calibration pipeline; postprocessing, auxiliary models and API access (PDF pp. 2, 6–8)                                                                                                                                                                                         | Connecting external calibration alternatives to the reasons for changing a policy through RL                                                                                                                                                                     |
| Kang et al., <i>Uncertainty Quantification for Hallucination Detection</i> [148]<br>arXiv v1                                                          | Signal, sampling, access and training-dependence axes for uncertainty-based detection (PDF pp. 6–8, 14–15)                                                                                                                                                                                   | Separating detection ranking from probability learning and decision requirements under policy changes                                                                                                                                                            |

The comparison identifies substantial established coverage. Ranking versus calibration, confidence-guided control, supervised estimators, and RL-based abstention already appear across these surveys. The RL-for-reasoning survey adds

the algorithmic side of this coverage, including group-standardized advantages, the debate between debiasing and normalization, and pass@ $k$ -aligned objectives (PDF pp. 19–20, 24, 38); in the pages read and a full-text search, it does not treat proper scoring rules, calibration metrics, or abstention [21]. *Confidence as Control* additionally covers training-data selection, preference and policy optimization, reward-model uncertainty, and calibrated abstention (PDF pp. 4–6, Table 1), and its general control loop is closely related to the five-component reading aid of Figure 2. The distinction of the present synthesis therefore rests on the assumptions and evidence connecting stages, not on that diagram or the presence of an RL section. Where terminology differs between surveys, we retain the primary statistical definitions of Section 2; for example, AUROC measures discrimination, and a min–max transformation alone does not establish probability reliability.

Both *Confidence as Control* and the unpublished companion manuscript have overlapping authorship with this article. The latter was inspected in manuscript form and is included here for scope disclosure, outside the selected public literature collection. The shared material includes probability objects and conditioning in Section 2, proper-score and supervision comparisons in Sections 3, 4 and 6.1, and control and evaluation principles in Sections 5–7. The companion organizes reports, production paths and consumer requirements; this article follows particular learning incentives through implemented updates to decision losses. Neither companion excludes training, and the distinction does not require the three surveys to cover disjoint literatures. The multi-turn consistency study discussed in Sections 5.5 and 7.4 [85] also shares four authors with this article; it is described from its public arXiv text (v4), and its qualifications are recorded in Appendix B.7.

The present analysis combines existing theory and methods: proper-score reporting incentives, constrained joint-policy objectives, practical gradient transformations, and cost-specific decisions. Its contribution is the comparison of their premises and consequences, including criteria for distinguishing reporting changes from policy changes. The conditional-expectation and scoring identities used for this purpose are standard results, not new general statistical theorems.

The *Confidence as Control* source is a 39-page author-hosted copy, listed as an April 2026 preprint on the author’s homepage. Acceptance and a precise arXiv identifier were not verified. The substantive reading covered PDF pp. 1–6, 8, 11, 15–19, 22–25, 38–39; other main-body pages were structurally scanned. Its inclusion does not imply that every reference or every version was audited. Similarly, Table E.1 reports the coverage of the inspected versions; it is not a complete census of neighboring surveys.

The RL-for-reasoning survey was read in its 120-page arXiv v3 (October 2025). The substantive reading covered PDF pp. 1–34, 37–38, 55, 61–63 and 66; the remaining main-body pages were checked by full-text search only, and its reference list was not audited [21].

## E.2. Publication-version checks

**Decision-token calibration [7].** The originally read version is arXiv:2601.13284v1 (16 pages); the official Findings of ACL 2026 copy has 17 pages (printed pp. 12537–12553). The method remains a zero decision-token advantage combined with a cross-entropy target that is one-hot for correct generations and uniform over candidates for incorrect generations (physical PDF pp. 7–8, Eq. 4, both versions). This hybrid update does not inherit a full-trajectory proper-score theorem merely from its calibration purpose.

The explicit test-set checkpoint selection also remains: physical p. 13, Appendix B in arXiv v1 and Appendix A in the ACL copy. We did not independently quantify the effect of that selection. In its Table 4 (physical p. 8), Qwen3-4B CSQA accuracy is 81.84% for GRPO and 79.16% for the proposed method, so the results should not be summarized as preserving accuracy on every task.

The publication adds three useful evaluations (Table E.2).

In the calibration-weight analysis, increasing the weight from 0.001 to 0.005 for Qwen3-1.7B reduced CSQA ECE from 15.97 to 4.26 but raised OBQA ECE from 5.27 to 9.60 (its Table 8; ECE on a 0–100 scale).

In the ranking evaluation, the paper’s Table 7 reports AUROC on OpenAI Moderation of 77.888 for the proposed method versus 56.746 for GRPO with Qwen3-1.7B, on a 0–100 scale; SFT reaches 77.996. This is discrimination evidence, not measured utility at every error cost. The discussion extending the diagnosis to initially well-calibrated models is not new: it moves from arXiv p. 15 to ACL p. 16 and still leaves experimental validation to future work. Main-text claims that use the paper’s Tables 7–9 refer specifically to the ACL publication.

**Table E.2. Evaluations added in the ACL publication of decision-token calibration.** Locators are physical pages of the Findings of ACL 2026 copy.

| Added ACL evidence             | Exact physical locator       | Interpretation                                                                                 |
|--------------------------------|------------------------------|------------------------------------------------------------------------------------------------|
| Correct/incorrect ranking      | p. 14, Table 7               | All six reported size/task comparisons improve AUROC over GRPO; comparisons with SFT are mixed |
| Calibration-weight sensitivity | p. 15, Appendix B.3, Table 8 | The tested weights expose different in-domain and transferred calibration responses            |
| KL sensitivity                 | p. 15, Appendix B.4, Table 9 | Limited calibration gains from the tested KL coefficient on Qwen3-1.7B                         |

**Active-signal survey [22].** Both the inspected arXiv:2601.15690v2 and the official Findings of ACL 2026 copies have 20 pages; the latter occupies printed pp. 41525–41544. The checked publication passages retain local discrimination versus global calibration on physical p. 3, the RL comparison on pp. 6–8 and Table 3 on p. 7, and the control-evaluation agenda on p. 9. The limitation that no large-scale comparative experiment is provided also remains on p. 9. The checked text matches after removing whitespace and page-number lines; this bounded comparison does not certify that the two versions are identical in every respect.

### E.3. Selected protocol distinctions

Table 3 omits several details needed when implementing comparisons; Table E.3 records them.

**Table E.3. Protocol details that change interpretation.** Details omitted from Table 3 that a faithful comparison must reproduce or report.

| Protocol                                           | Detail that changes interpretation                                                                                                                                                                                           |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed-answer supervision [66]                      | Separate answering and confidence models, roughly 20,000 labeled source examples and a 2,000-example temperature holdout; joint online training is left open (PDF pp. 6, 10)                                                 |
| EpiCar [141]                                       | Correct trajectories train reasoning, while successes and failures train Yes/No self-evaluation; 500 reserved test examples are excluded from final reporting, but ECE/NLL fitting descriptions conflict (PDF pp. 4, 17, 21) |
| Direct actions and integrated utilities [43], [44] | Training on a correct/abstain/wrong payoff learns a specified action trade-off; integrating threshold-dependent bounded payoffs supplies a different reporting objective (PDF pp. 3–4; p. 6, Eq. 3)                          |
| Full-vector output [80]                            | Selected-answer confidence supplies the principal reward, while multiclass Brier improvements are empirical outcomes (PDF pp. 2–3, 14, Table 6)                                                                              |
| Capability estimates [72]                          | Repeated-success targets and budget calculations depend on the decoder, sampling assumptions, and the ability to recognize successful candidates (PDF pp. 3–4, 21, 23)                                                       |

The metric-implementation audit in Appendix D gives the detailed SaySelf, LACIE, and DualStake comparisons. These comparisons distinguish the published definition, the inspected public release, and the unverified historical run; no cross-paper harmonization is assumed.

## Appendix F. Source versions and reproducibility

The selected collection contains 140 academic records with acquired PDFs and reading notes over stated ranges, together with eight primary web sources. This total includes two foundational post-training papers added for the opening comparison, whose formal publication versions were checked on 21 September 2026. It also includes 35 records from a targeted extension that covered 2025–2026 work and added the primary sources of the named optimizers, and five records from a second extension that completes the lineage in Section 2.4. The method inventory in Appendix B contains

81 records. The collection combines calibration-learning methods, necessary comparators, theoretical foundations, diagnostic studies, neighboring surveys, and decision interfaces. Recording source identity and reading coverage does not amount to independent reproduction. Publication metadata identifies verified venues where available; Table F.1 identifies the source versions actually read.

The search was targeted to the survey scope and expanded through citation tracing and review-driven primary-source checks. Screening decisions and corpus-to-section coverage are retained with the research materials. The full text of Murphy (1973) could not be obtained; this candidate remains in the screening record but is excluded from the selected/read count and from substantive citations. This access limit does not imply that the paper is irrelevant.

The accompanying evidence register and revision reports retain physical source-page, equation, and table locators; the main narrative cites only selected locators. Page numbering starts at the first PDF page unless a printed-page convention is stated. Source model results are author-reported. Public-code observations refer to inspected versions rather than reconstructed historical experiment runs; no LLM training experiment was independently reproduced. The analytical examples and their numerical checks are reproducible from the supplied scripts.

The corpus was assembled on 19 September 2026, extended through targeted publication-metadata and Brier-foundation checks on 20 September and the foundational post-training checks on 21 September, extended again on 28 September through the targeted search described in Section 1.2, and extended on 1 October with the five lineage sources, whose citation links were checked in the full text of each later source (Table F.2). Bröcker’s analysis [90] is the read primary source for the Brier decomposition; the unavailable Murphy original is not counted as a read record.

## F.1. Versions used for substantive citations

**Table F.1. Source versions used for substantive citations.** Rows follow the reference numbering. A dash indicates that no qualification beyond the listed version applies.

| Ref. | Read version (arXiv identifier unless stated) | Version qualification                                                                  |
|------|-----------------------------------------------|----------------------------------------------------------------------------------------|
| [2]  | NeurIPS 2022 proceedings version              | Opening RLHF background; recorded methods and limitations pages read                   |
| [3]  | Nature 645:633–638 (2025), publisher PDF      | Opening RLVR background; recorded article and Methods pages read                       |
| [5]  | 2601.06336v2                                  | —                                                                                      |
| [6]  | 2410.09724v2                                  | —                                                                                      |
| [7]  | 2601.13284v1                                  | Findings of ACL 2026, July 2026                                                        |
| [8]  | 2303.08774v6                                  | —                                                                                      |
| [9]  | 2505.14489v2                                  | Formal-publication metadata verified; the listed arXiv version remains the read source |
| [10] | 2402.03300v3                                  | Technical report                                                                       |
| [11] | 2305.18290v3                                  | NeurIPS 2023 named on the read PDF                                                     |
| [12] | 2603.06604v1                                  | —                                                                                      |
| [13] | 2504.06564v3                                  | —                                                                                      |
| [14] | 2505.13988v1                                  | Formal-publication metadata verified; the listed arXiv version remains the read source |
| [15] | 2506.09038v1                                  | —                                                                                      |
| [16] | 2505.23646v1                                  | Marked as under review on the read PDF                                                 |
| [17] | 2606.03437v1                                  | —                                                                                      |
| [18] | 2508.11800v1                                  | —                                                                                      |
| [19] | 2311.08298v2                                  | —                                                                                      |
| [20] | 2602.09305v2                                  | —                                                                                      |
| [21] | 2509.08827v3                                  | Partial reading: PDF pp. 1–34, 37–38, 55, 61–63, 66                                    |
| [22] | 2601.15690v2                                  | Findings of ACL 2026, July 2026                                                        |
| [23] | 2412.05563v2                                  | —                                                                                      |
| [24] | 2407.18418v3                                  | —                                                                                      |

*Continued on next page*

Table F.1 (continued)

| Ref. | Read version (arXiv identifier unless stated)                                                              | Version qualification                                                                                         |
|------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| [25] | Author-hosted 39-page preprint, retrieved 19 September 2026; exact OpenReview revision relation unverified | Public author copy; publication status unverified; recorded sections read                                     |
| [26] | Published JASA article, 102(477), 359–378 (2007)                                                           | —                                                                                                             |
| [27] | Author-hosted QJRM article, 2013 online imprint; volume 140(683), 1917–1923 (2014)                         | 2013 online imprint; journal issue published 2014                                                             |
| [28] | 2608.28482v2                                                                                               | —                                                                                                             |
| [29] | 2503.02623v6                                                                                               | —                                                                                                             |
| [30] | 2507.16806v2                                                                                               | —                                                                                                             |
| [31] | 2607.04332v1                                                                                               | —                                                                                                             |
| [32] | 2604.09529v1                                                                                               | ACL 2026 Main in arXiv comments; no venue on the read PDF                                                     |
| [33] | 2512.25070v2                                                                                               | —                                                                                                             |
| [34] | 2606.32032v1                                                                                               | —                                                                                                             |
| [35] | 2505.23912v2                                                                                               | ACL 2026 Main in arXiv comments; no venue on the read PDF                                                     |
| [36] | WWW 2026 published version; pp. 7401–7411; DOI 10.1145/3774904.3792540                                     | —                                                                                                             |
| [37] | 2405.21028v2                                                                                               | —                                                                                                             |
| [38] | 2605.15588v2                                                                                               | —                                                                                                             |
| [39] | 2604.12632v1                                                                                               | —                                                                                                             |
| [40] | 2607.03528v1                                                                                               | —                                                                                                             |
| [41] | 2603.17145v2                                                                                               | —                                                                                                             |
| [42] | 2605.20740v1                                                                                               | —                                                                                                             |
| [43] | 2509.25760v2                                                                                               | —                                                                                                             |
| [44] | 2512.19920v3                                                                                               | ICLR 2026 non-archival workshop presentation verified; the listed arXiv version remains the read source       |
| [45] | 2606.19818v1                                                                                               | —                                                                                                             |
| [46] | 2606.09073v2                                                                                               | —                                                                                                             |
| [47] | 2505.19590v5                                                                                               | Formal-publication metadata verified; the listed arXiv version remains the read source                        |
| [48] | 2505.22660v4                                                                                               | —                                                                                                             |
| [49] | 2504.16084v3                                                                                               | —                                                                                                             |
| [50] | 2504.05812v3                                                                                               | Formal-publication metadata verified; the listed arXiv version, marked as a preprint, remains the read source |
| [51] | 2506.17219v2                                                                                               | —                                                                                                             |
| [52] | 2402.14740v2                                                                                               | —                                                                                                             |
| [53] | 1707.06347v2                                                                                               | —                                                                                                             |
| [54] | 2503.06639v4                                                                                               | —                                                                                                             |
| [55] | 2507.18071v2                                                                                               | —                                                                                                             |
| [56] | 2509.23129v2                                                                                               | —                                                                                                             |
| [57] | 2508.18847v2                                                                                               | Formal-publication metadata verified; the listed arXiv version remains the read source                        |
| [58] | 2505.15134v1                                                                                               | Marked as under review on the read PDF                                                                        |
| [59] | 2603.05881v2                                                                                               | —                                                                                                             |
| [60] | 2606.24281v1                                                                                               | —                                                                                                             |
| [61] | 2604.16830v1                                                                                               | —                                                                                                             |
| [62] | 1706.04599v2                                                                                               | —                                                                                                             |
| [63] | 2510.17509v3                                                                                               | ICLR 2026 header on the read PDF                                                                              |
| [64] | 2604.19444v1                                                                                               | —                                                                                                             |
| [65] | 2207.05221v4                                                                                               | —                                                                                                             |
| [66] | 2406.08391v3                                                                                               | —                                                                                                             |

Continued on next page

Table F.1 (continued)

| Ref.  | Read version (arXiv identifier unless stated) | Version qualification                                                                                                        |
|-------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| [67]  | 2302.09664v3                                  | Formal-publication metadata verified; the listed arXiv version remains the read source                                       |
| [68]  | 2604.03904v2                                  | —                                                                                                                            |
| [69]  | 2503.01332v2                                  | —                                                                                                                            |
| [70]  | 2502.18581v3                                  | Formal-publication metadata verified; the listed arXiv version remains the read source                                       |
| [71]  | 2407.18370v1                                  | —                                                                                                                            |
| [72]  | 2602.13540v1                                  | —                                                                                                                            |
| [73]  | 2506.09338v2                                  | NeurIPS 2025 named on the read PDF                                                                                           |
| [74]  | 2508.15260v1                                  | —                                                                                                                            |
| [75]  | 2504.15895v3                                  | —                                                                                                                            |
| [76]  | 2607.01612v1                                  | —                                                                                                                            |
| [77]  | 2604.23333v2                                  | —                                                                                                                            |
| [78]  | 2602.16699v3                                  | —                                                                                                                            |
| [79]  | 2607.10059v1                                  | —                                                                                                                            |
| [80]  | 2511.14275v2                                  | —                                                                                                                            |
| [81]  | 2508.07662v1                                  | —                                                                                                                            |
| [82]  | 2510.00263v1                                  | —                                                                                                                            |
| [83]  | 2411.04368v1                                  | —                                                                                                                            |
| [84]  | 2609.18453v1                                  | EMNLP 2026 Main in arXiv comments; no venue on the read PDF                                                                  |
| [85]  | 2503.22353v4                                  | Formal-publication metadata verified; the listed arXiv version remains the read source; shares four authors with this survey |
| [86]  | 2504.13837v5                                  | NeurIPS 2025 in the arXiv journal reference; no venue on the read PDF                                                        |
| [87]  | 2605.06785v2                                  | —                                                                                                                            |
| [88]  | 2205.14334v2                                  | —                                                                                                                            |
| [89]  | 2305.14975v2                                  | —                                                                                                                            |
| [90]  | 0806.0813v2                                   | Author arXiv v2; page anchors differ from 2009 journal pagination                                                            |
| [91]  | JRSSB 69(2):243–268 publisher version         | —                                                                                                                            |
| [92]  | NIPS 2017 proceedings version                 | —                                                                                                                            |
| [93]  | 1805.08206v4                                  | —                                                                                                                            |
| [94]  | 2107.07511v6                                  | —                                                                                                                            |
| [95]  | 2311.14648v3                                  | —                                                                                                                            |
| [96]  | 2306.13063v2                                  | —                                                                                                                            |
| [97]  | Nature 630, 625–630 (2024), journal PDF       | Main text, Methods and Extended Data read; Supplementary Information not read                                                |
| [98]  | 2404.17287v3                                  | —                                                                                                                            |
| [99]  | 2405.20974v3                                  | —                                                                                                                            |
| [100] | 2603.09117v3                                  | —                                                                                                                            |
| [101] | 2505.17989v4                                  | —                                                                                                                            |
| [102] | 2606.15917v1                                  | MSc thesis, University of Oxford (2025); not peer reviewed                                                                   |
| [103] | 2603.24844v1                                  | —                                                                                                                            |
| [104] | 2604.12046v1                                  | —                                                                                                                            |
| [105] | 2601.07264v1                                  | —                                                                                                                            |
| [106] | 2601.20126v1                                  | —                                                                                                                            |
| [107] | 2605.16999v1                                  | —                                                                                                                            |
| [108] | 2508.00264v2                                  | —                                                                                                                            |
| [109] | 2608.00301v1                                  | —                                                                                                                            |
| [110] | 2601.22648v2                                  | —                                                                                                                            |
| [111] | 2608.27167v2                                  | Single-author preprint                                                                                                       |

Continued on next page

Table F.1 (continued)

| Ref.  | Read version (arXiv identifier unless stated) | Version qualification                                                                  |
|-------|-----------------------------------------------|----------------------------------------------------------------------------------------|
| [112] | 2504.05419v1                                  | Marked as under review on the read PDF                                                 |
| [113] | 2510.22691v1                                  | —                                                                                      |
| [114] | 2410.10414v2                                  | —                                                                                      |
| [115] | 2503.03064v2                                  | Formal-publication metadata verified; the listed arXiv version remains the read source |
| [116] | 2607.00164v1                                  | —                                                                                      |
| [117] | 2404.00474v2                                  | —                                                                                      |
| [118] | 2511.05993v3                                  | —                                                                                      |
| [119] | 2603.06317v1                                  | —                                                                                      |
| [120] | 2606.14211v1                                  | —                                                                                      |
| [121] | 2503.14476v2                                  | Formal-publication metadata verified; the listed arXiv version remains the read source |
| [122] | 2503.20783v2                                  | —                                                                                      |
| [123] | 2510.21049v1                                  | —                                                                                      |
| [124] | 2509.04664v1                                  | —                                                                                      |
| [125] | 2602.02708v1                                  | —                                                                                      |
| [126] | 2405.21047v4                                  | Revised arXiv v4; differs from the 2024 proceedings version                            |
| [132] | 2509.24988v2                                  | —                                                                                      |
| [133] | 2410.06707v1                                  | —                                                                                      |
| [134] | 2601.04277v1                                  | —                                                                                      |
| [135] | 2505.01997v3                                  | —                                                                                      |
| [136] | 2601.23096v2                                  | —                                                                                      |
| [137] | 2403.08819v2                                  | —                                                                                      |
| [138] | 2609.20541v1                                  | Workshop paper; workshop not named in the PDF                                          |
| [139] | 2505.23854v1                                  | —                                                                                      |
| [140] | 2509.21882v3                                  | —                                                                                      |
| [141] | 2601.06786v1                                  | —                                                                                      |
| [142] | 2609.00935v1                                  | —                                                                                      |
| [144] | 2409.18786v1                                  | —                                                                                      |
| [145] | 2503.00172v2                                  | —                                                                                      |
| [146] | 2503.15850v2                                  | —                                                                                      |
| [147] | 2412.12767v1                                  | —                                                                                      |
| [148] | 2510.12040v1                                  | —                                                                                      |

## F.2. Documented links in the lineage figure

Table F.2 lists the relations behind Figure 3. A link is documented when the later source adopts the earlier method or statistic, uses it as a baseline or a stated contrast, names it as its closest related work or as the line it builds on, or, with shared authors, extends it; the locator is a physical page of the later source. Links marked *not drawn* are documented but omitted from the figure for legibility; the last group lists groupings that this survey draws and that the sources do not claim. A citation that only provides background is not counted as a link. Searches of the reference lists also established three absences: the self-certainty paper cites none of the label-free reward methods, CARG cites neither self-certainty nor DeepConf, and C3RL does not cite DeepConf.

**Table F.2. Documented links in the lineage of Figure 3.** Each row states the relation as the later source records it; locators are physical pages of the later source. Background citations are not counted as links.

| Later source                                               | Earlier source                                                             | Relation stated in the later source                                                                                                                                        | Locator               |
|------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Documented and drawn in Figure 3</b>                    |                                                                            |                                                                                                                                                                            |                       |
| Farquhar et al. [97]                                       | Kuhn et al. [67]                                                           | Shared authors; credits L. Kuhn with the initial formalization of semantic entropy and releases code in a repository of the same name, but does not cite the earlier paper | p. 11                 |
| CSR [38]                                                   | Farquhar et al. [97]                                                       | Adopts discrete semantic entropy as its confidence proxy                                                                                                                   | pp. 2–3, 13           |
| EMPO [50]                                                  | Kuhn et al. [67]                                                           | Chooses semantic entropy as its unsupervised objective                                                                                                                     | p. 4                  |
| SaySelf [99]                                               | Kadavath et al. [65]; Tian et al. [89]; Xiong et al. [96]; Lin et al. [88] | Baselines: prompting for correctness, direct prompting, self-consistency, and grouping-based calibration training                                                          | p. 6                  |
| CONQORD [98]                                               | Tian et al. [89]; Xiong et al. [96]                                        | Baselines: top- $k$ elicitation and chain-of-thought aggregation                                                                                                           | p. 5                  |
| Rewarding Doubt [29]                                       | Kadavath et al. [65]; Tian et al. [89]; Xiong et al. [96]                  | Baselines: surrogate token, top- $k$ , and chain-of-thought elicitation                                                                                                    | p. 6                  |
| ConfTuner [57]                                             | Tian et al. [89]; Xiong et al. [96]                                        | Prompt-based elicitation that it improves on; Xiong et al.’s ensemble as a baseline; two shared authors                                                                    | pp. 1, 5, 9           |
| RLCR [30]                                                  | SaySelf [99]; Rewarding Doubt [29]                                         | Named as the most closely related work and adapted as baselines                                                                                                            | pp. 11, 28            |
| Decision-token RL [7]                                      | RLCR [30]                                                                  | Cites RLCR as calibration-aware fine-tuning of verbalized confidence, a modality it distinguishes from its decision-token target                                           | pp. 1, 12             |
| DCPO [100]                                                 | RLCR [30]                                                                  | Baseline                                                                                                                                                                   | pp. 1–3               |
| CSR [38]                                                   | RLCR [30]                                                                  | Baseline                                                                                                                                                                   | p. 6                  |
| C3RL / CAS [76]                                            | RLCR [30]                                                                  | Baseline, described as building on SaySelf                                                                                                                                 | p. 5                  |
| INTUITOR [47]                                              | Self-certainty selection [70]                                              | Adopts the self-certainty metric; three shared authors                                                                                                                     | p. 4                  |
| DeepConf [74]                                              | Self-certainty selection [70]                                              | Adopts self-certainty, computed over its top- $k$ token confidence, as the global trace measure, then proposes local alternatives                                          | pp. 2–3               |
| RENT [48]                                                  | INTUITOR [47]; TTRL [49]                                                   | Compared as concurrent methods in its Table 2                                                                                                                              | pp. 7–8               |
| No Free Lunch [51]                                         | EMPO [50]; INTUITOR [47]                                                   | Describes EMPO as introducing RL from internal feedback and INTUITOR as building on it; evaluates the self-certainty reward                                                | pp. 2, 4–5            |
| <b>Documented, not drawn</b>                               |                                                                            |                                                                                                                                                                            |                       |
| EMPO [50]                                                  | Farquhar et al. [97]                                                       | Cited with Kuhn et al. for detecting hallucinations and for the clustering procedure                                                                                       | pp. 3, 5, 7           |
| CSR [38]                                                   | Rewarding Doubt [29]                                                       | Baseline                                                                                                                                                                   | p. 6                  |
| ConfTuner [57]                                             | SaySelf [99]                                                               | Baseline                                                                                                                                                                   | p. 5                  |
| C3RL / CAS [76]                                            | SaySelf [99]                                                               | Baseline                                                                                                                                                                   | p. 5                  |
| <b>Groupings drawn in this survey (dashed in Figure 3)</b> |                                                                            |                                                                                                                                                                            |                       |
| TTRL [49]                                                  | EMPO [50]                                                                  | Both reward agreement among sampled answers; EMPO’s cluster frequency grades TTRL’s majority indicator. Each lists the other as concurrent work                            | TTRL p. 14; EMPO p. 3 |
| CARG [85]                                                  | DeepConf [74]                                                              | Both act on token-probability confidence at inference, without training and without first calibrating it; neither cites the other                                          | —                     |

### F.3. Reproducibility and version boundaries

The source manifest records PDF hashes and actual reading ranges, while per-paper notes document evidence and limitations. The method matrix separates target, interface, feedback, objective, update, use, and qualification. Original manuscripts and figures remain preserved alongside the revised sources. The bibliography uses stable identifiers internally and conventional numeric references in publication exports.

RLVP’s paper-linked public repository was unavailable at inspection, so the method and table comparisons rest on its published PDF. As recorded in Appendix E.1, *Confidence as Control* received main-section reading with structural scanning of other pages, and the RL-for-reasoning survey a partial reading. These reading limits are stated explicitly rather than treated as full page-by-page coverage. The classical mathematical constructions demonstrate possible mechanisms; they do not measure the prevalence of those mechanisms in deployed LLMs.

### F.4. Web-source persistence

Web sources carry their access dates in the reference list, and Table D.1 describes each. During revision, we queried the Wayback Machine for existing snapshots, but its availability endpoint returned HTTP 429, so no unverified archive URLs are supplied. The failed lookups are recorded, and existing repository observations retain fixed-commit links where available. Mutable documentation and a frozen code commit provide different forms of reproducibility evidence.